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LIN et al.

(54) BATTERY CELL, BATTERY AND ELECTRIC APPARATUS

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(52) **U.S. Cl.**
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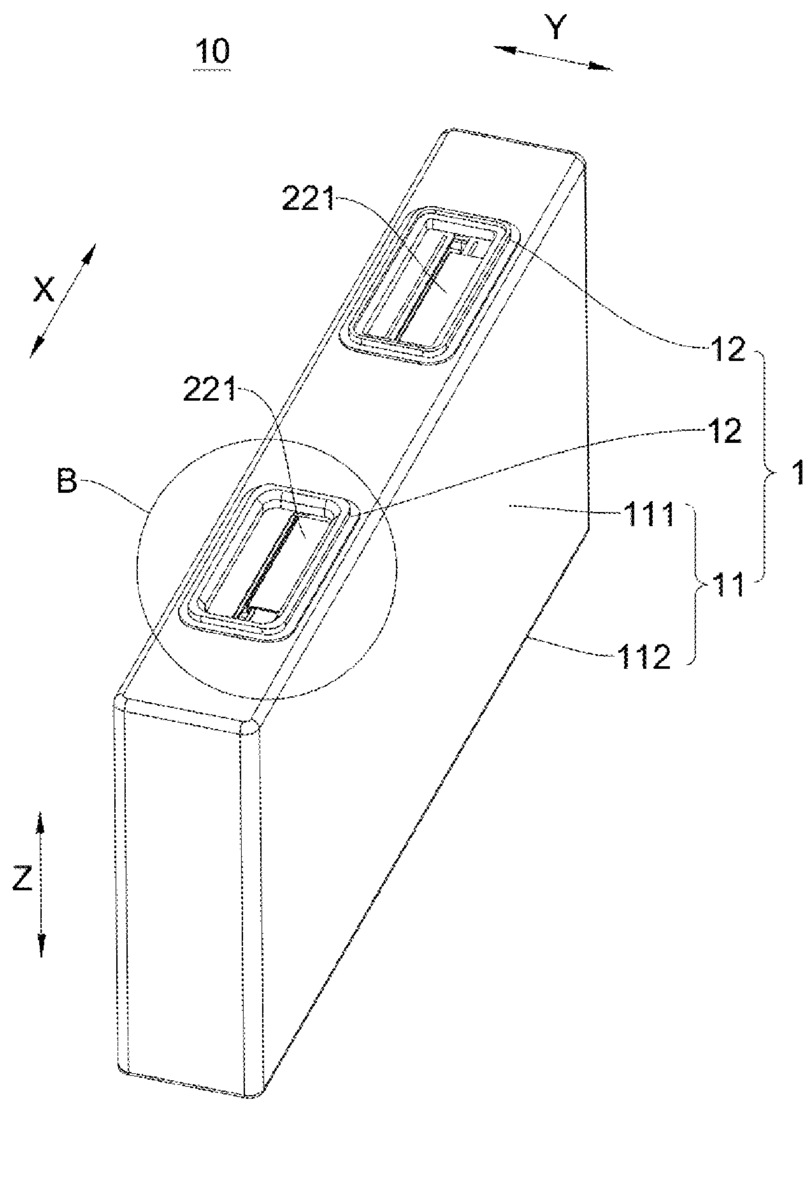
(57) **ABSTRACT**

A battery cell comprises: a casing assembly and a battery cell assembly, wherein the casing assembly comprises a casing and a first terminal arranged on the casing; and the battery cell assembly comprises an active-material coated portion and a conductive portion, the active-material coated portion being accommodated in the casing, the conductive portion being used for electrically connecting to the active-material coated portion and the first terminal, the first terminal being provided with an accommodating portion, and the conductive portion being at least partially accommodated in the accommodating portion.



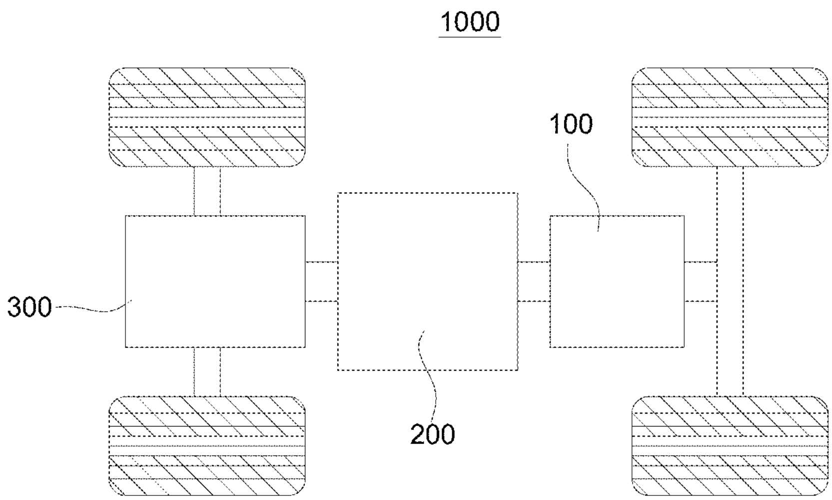


FIG. 1

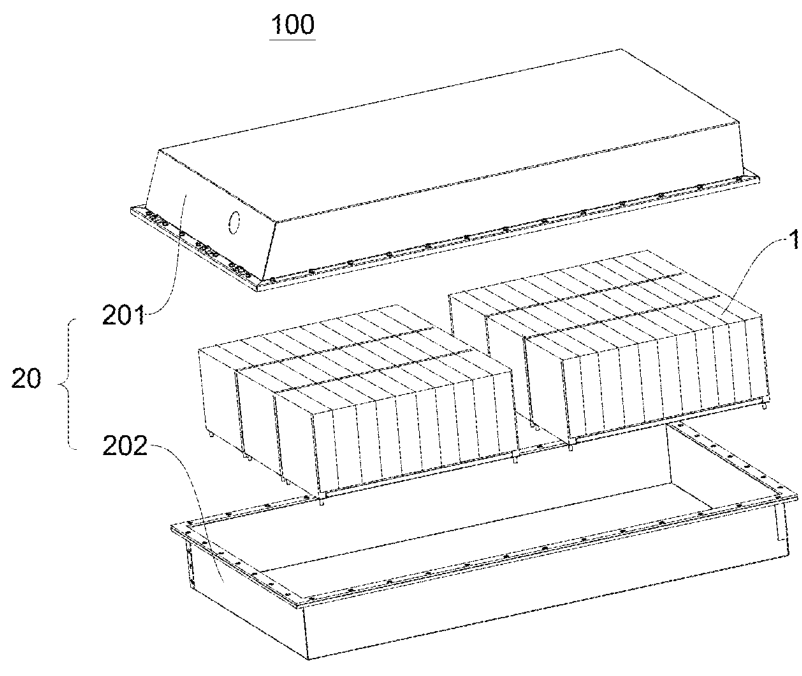


FIG. 2

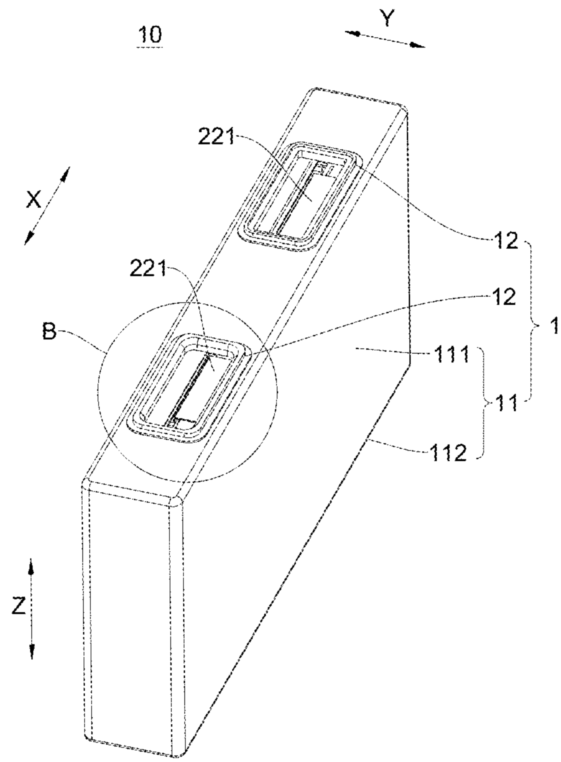
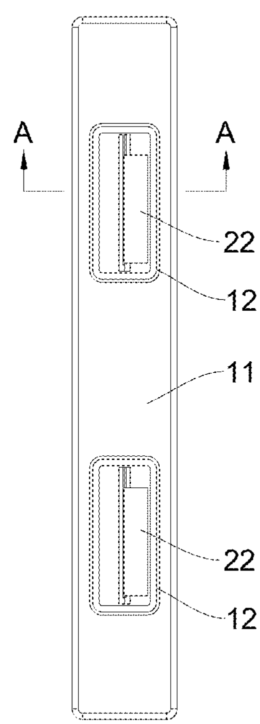


FIG. 3



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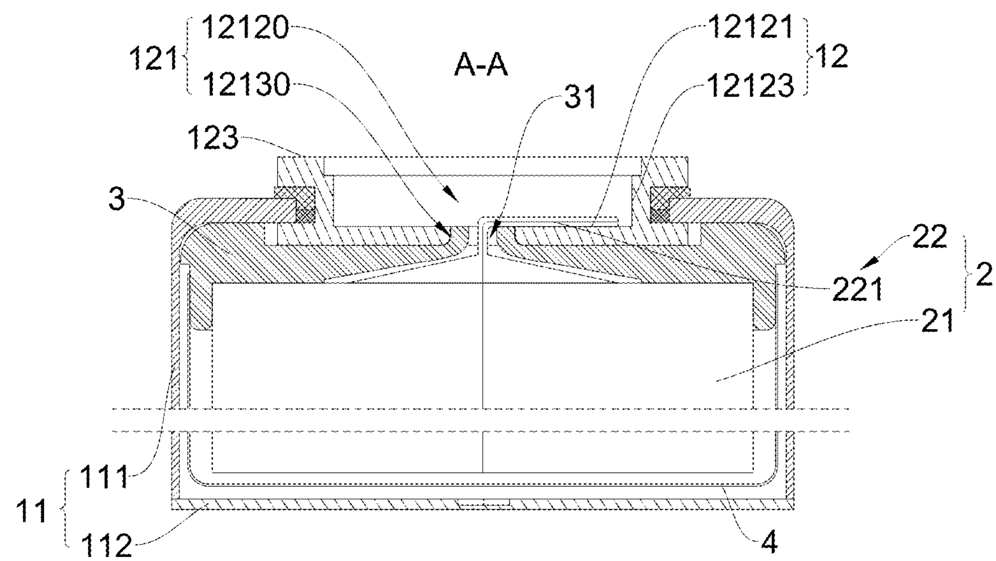
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FIG. 5

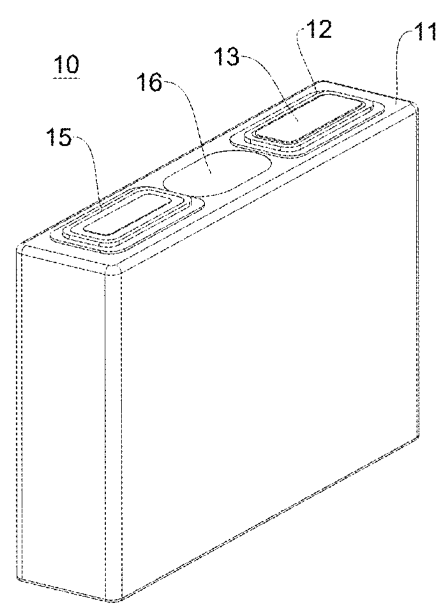
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FIG. 6

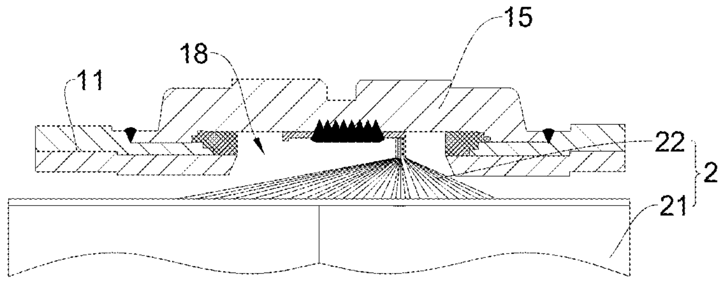
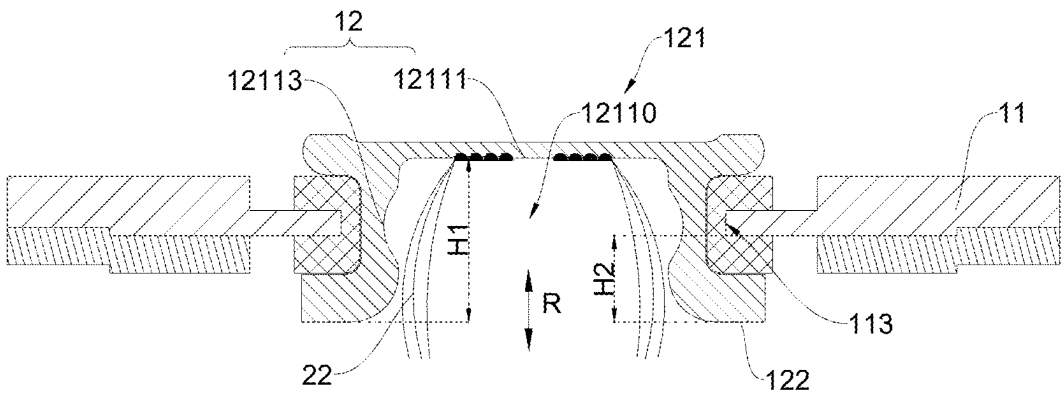


FIG. 7



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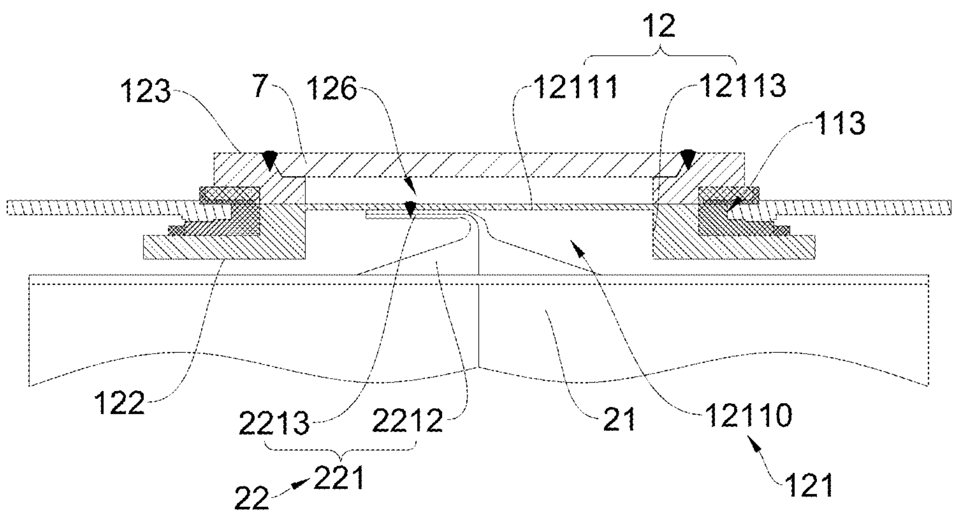

FIG. 9

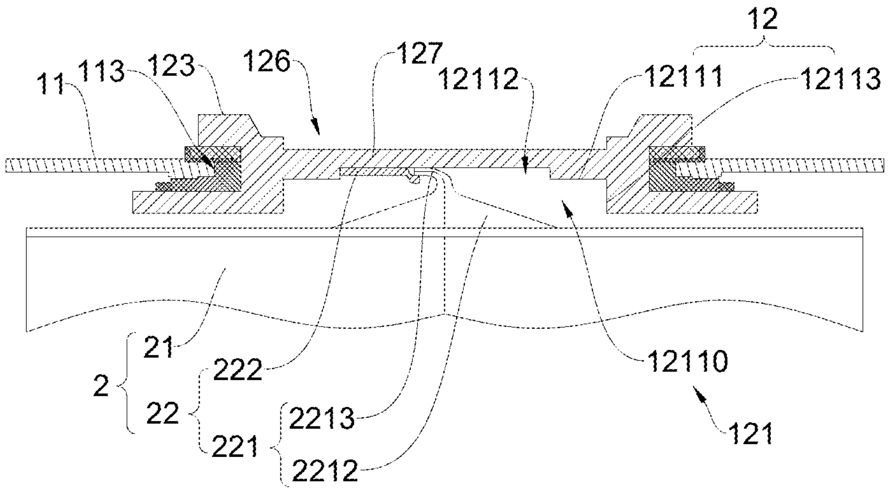

FIG. 10

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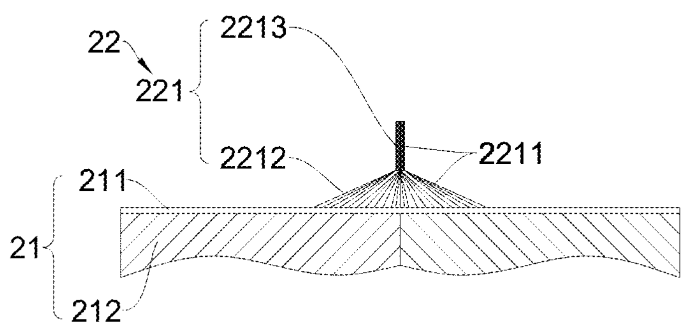

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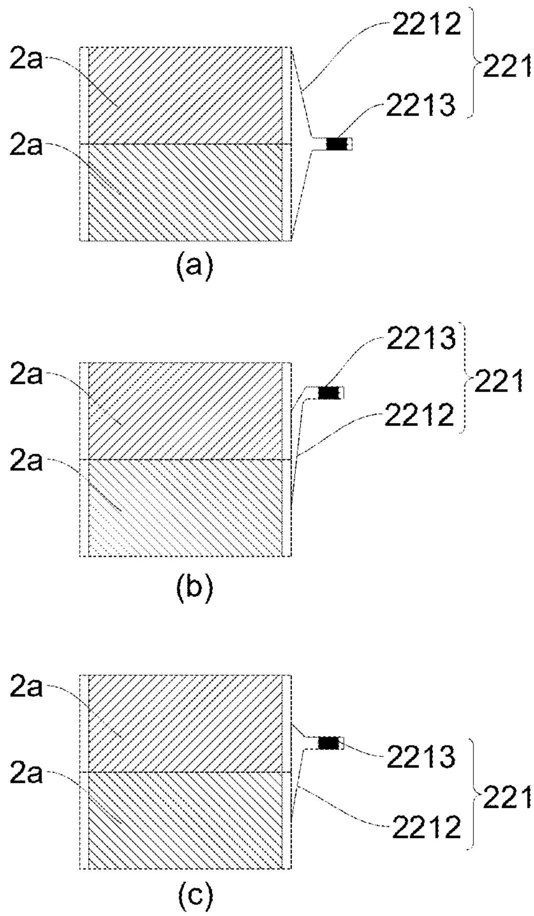
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FIG. 12

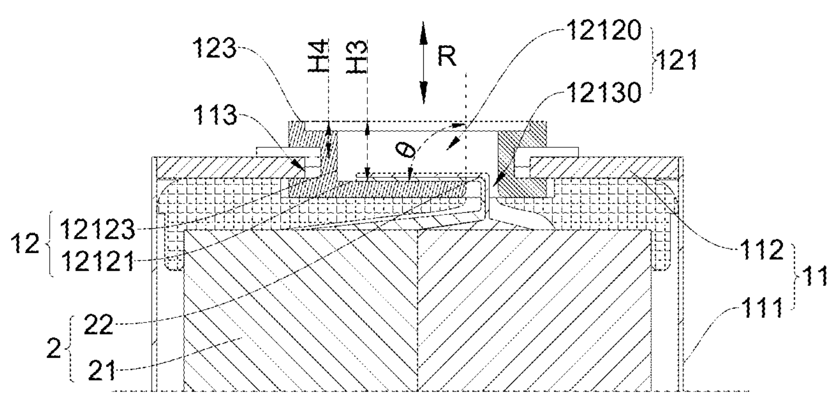
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FIG. 13

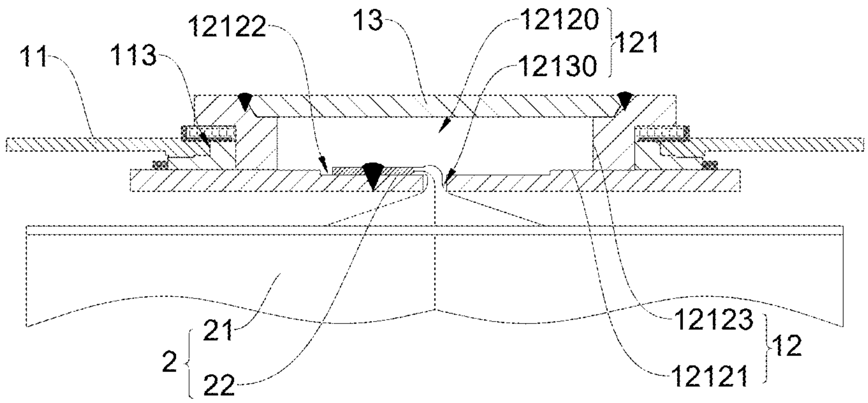

FIG. 14

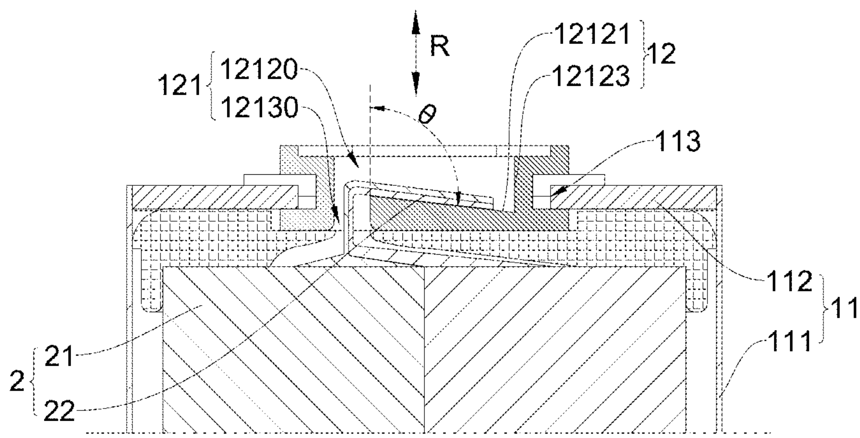

FIG. 15

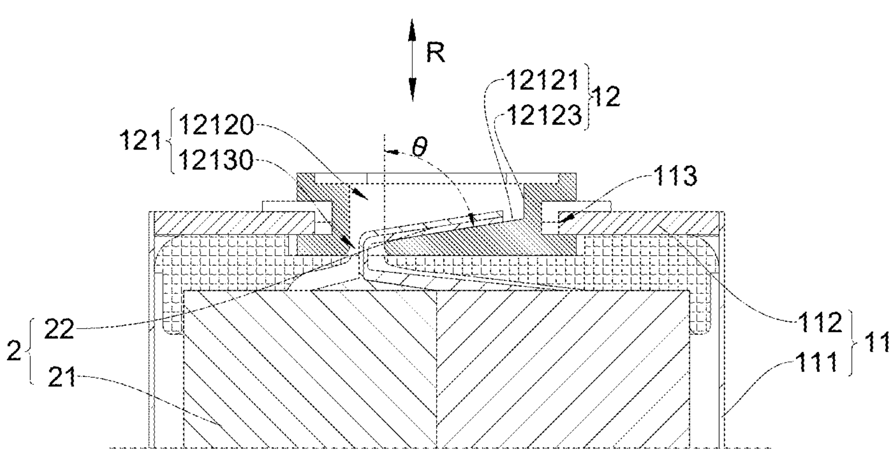

FIG. 16

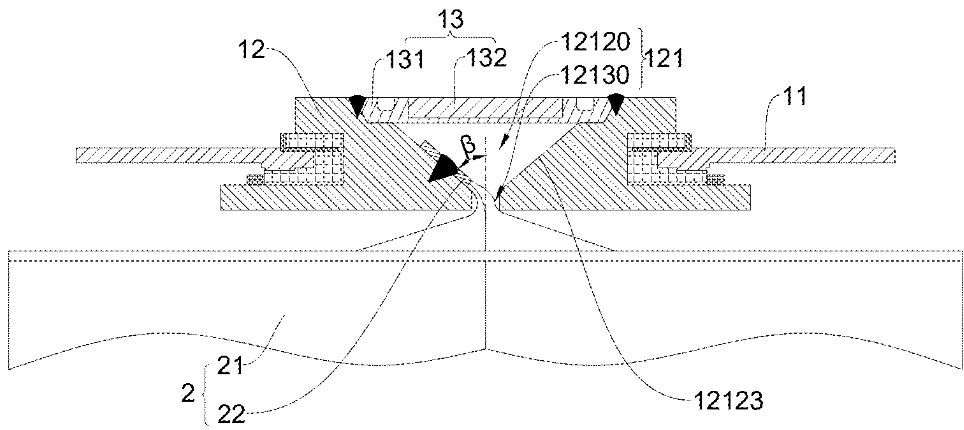
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FIG. 17

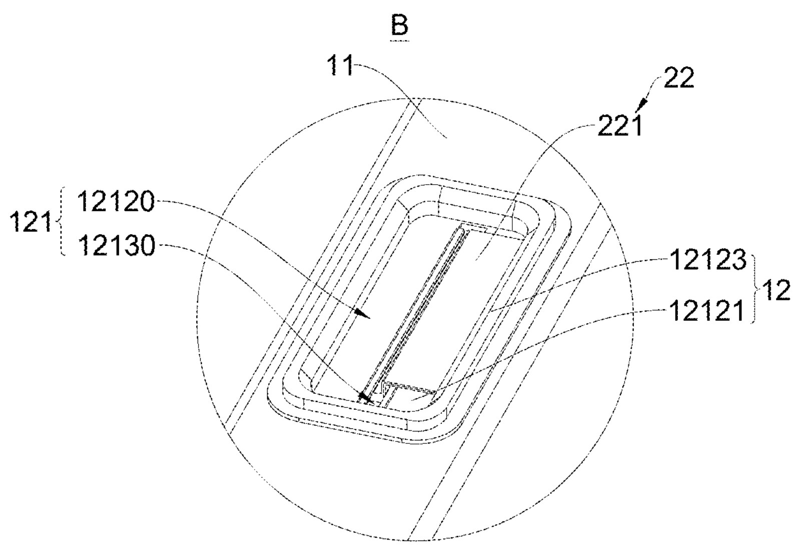
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FIG. 18

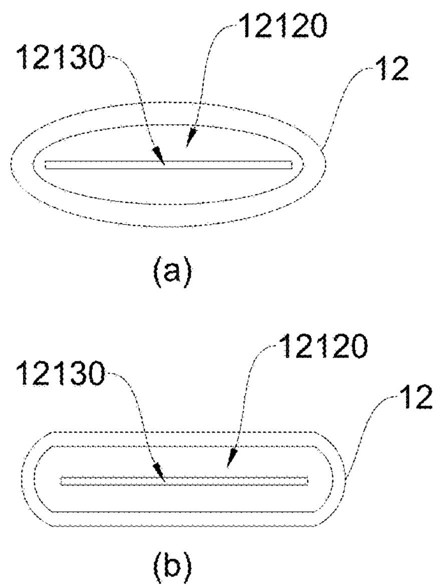


FIG. 19

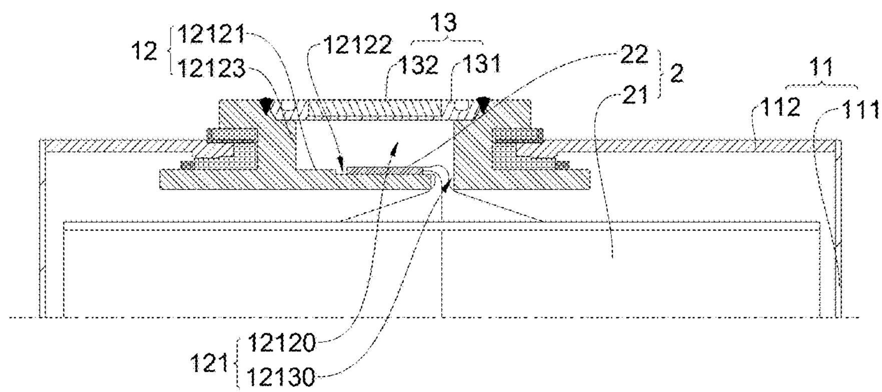
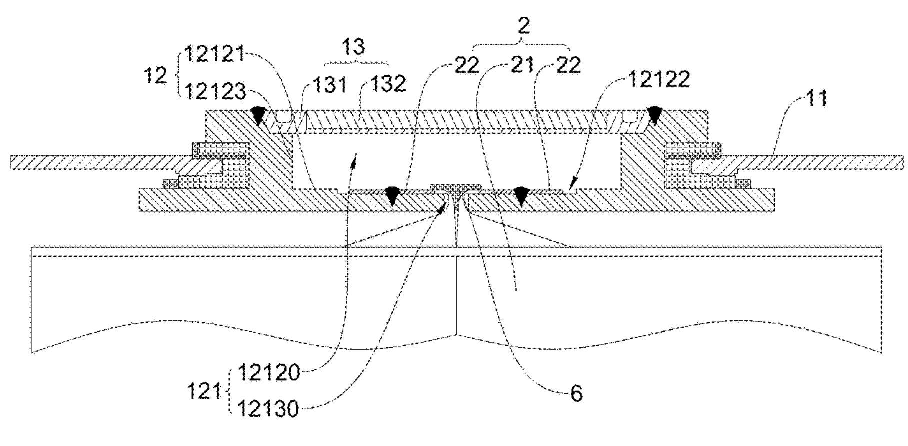


FIG. 20



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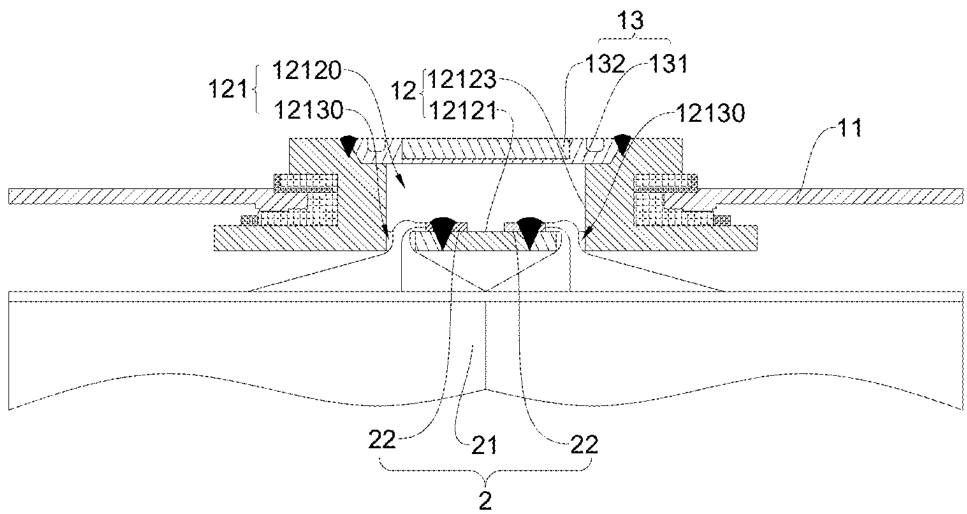
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FIG. 22

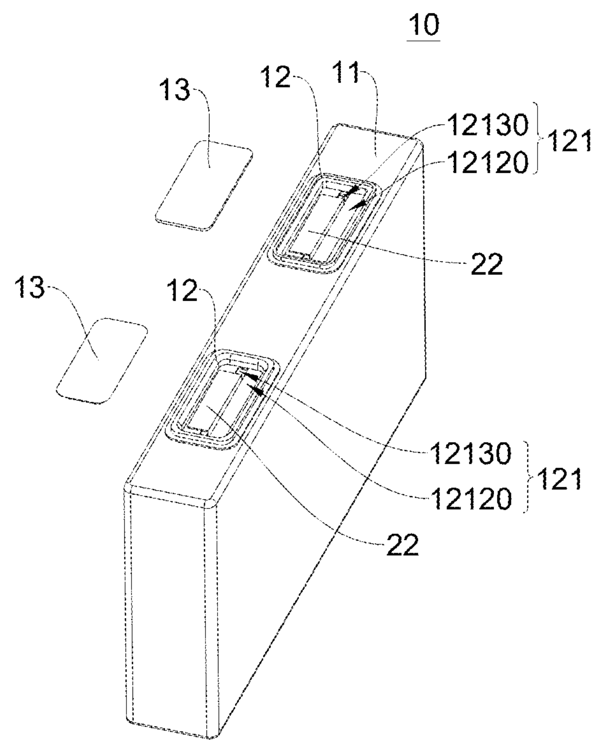
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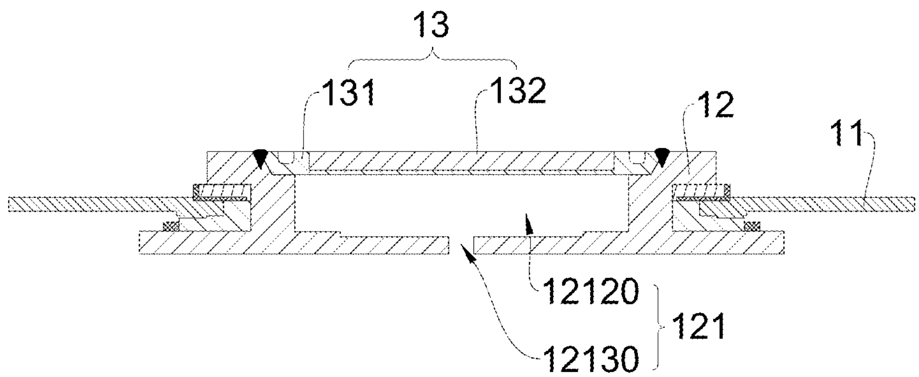
FIG. 23



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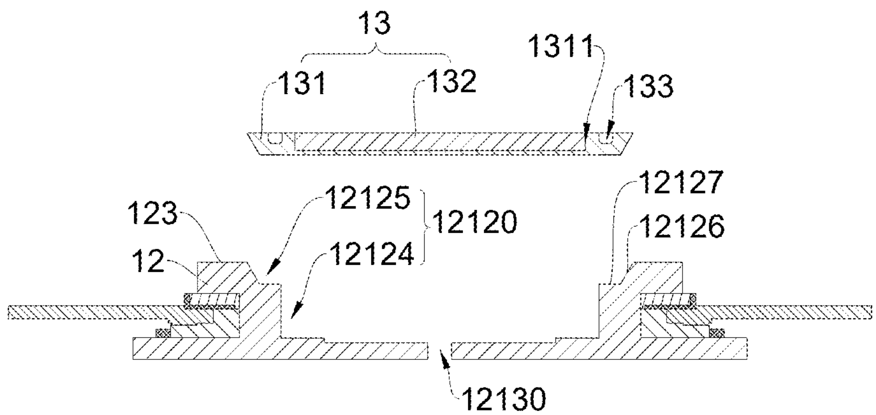

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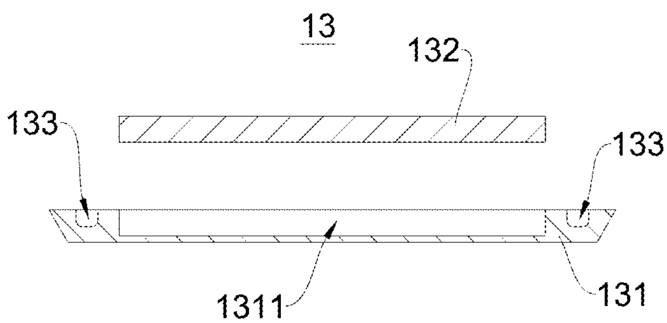

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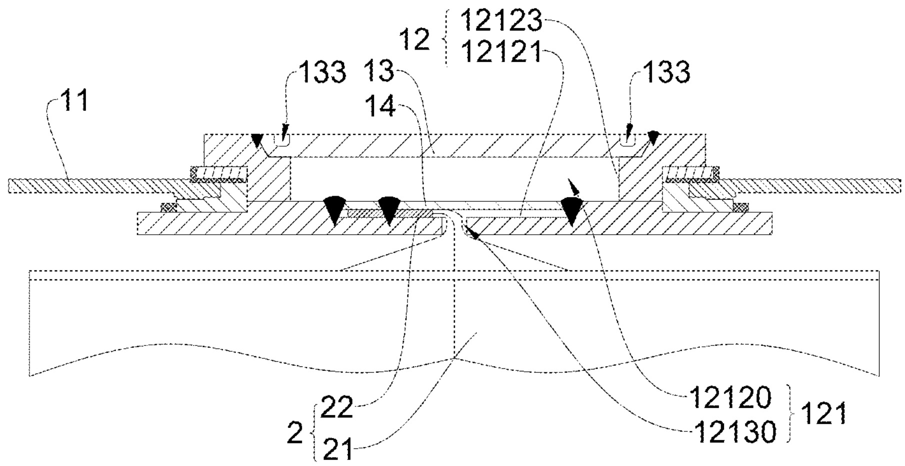

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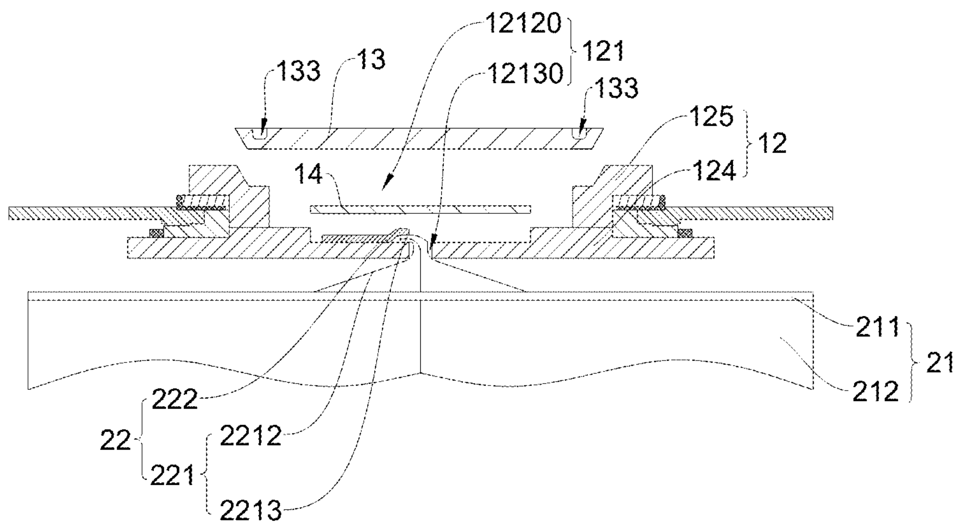

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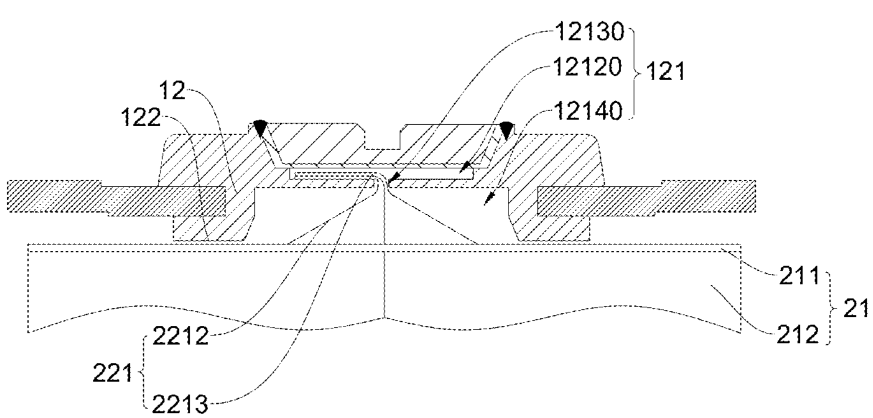

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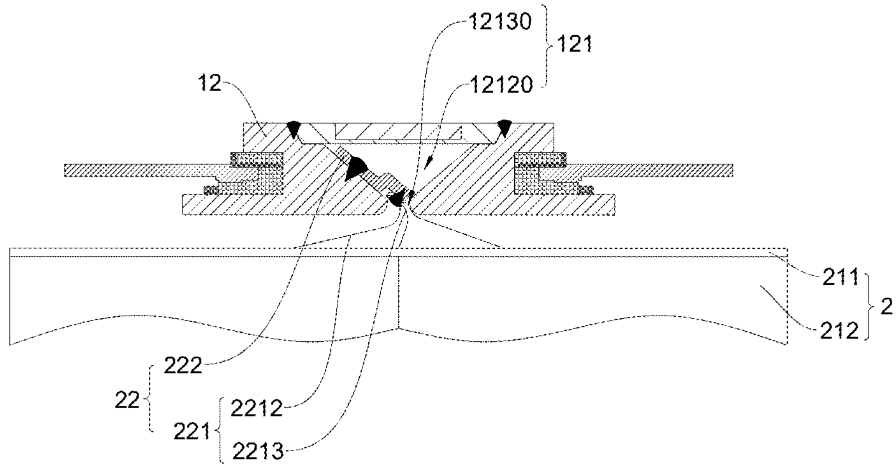

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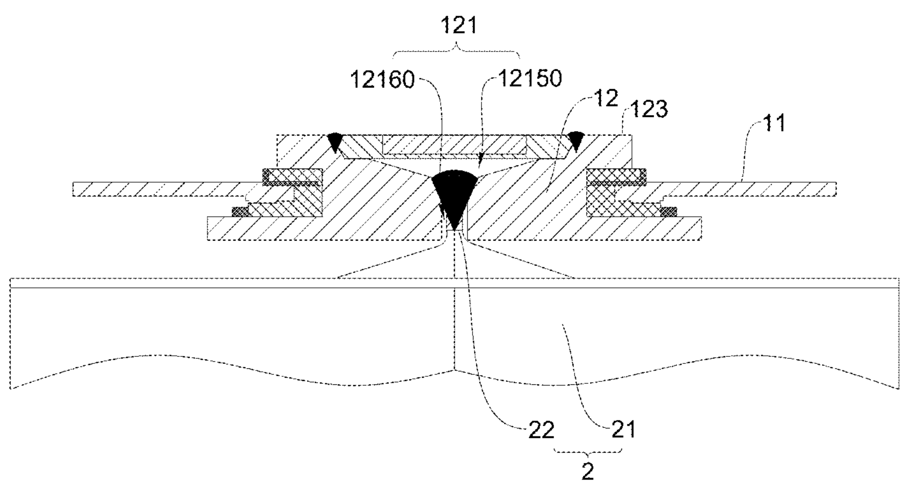


FIG. 31

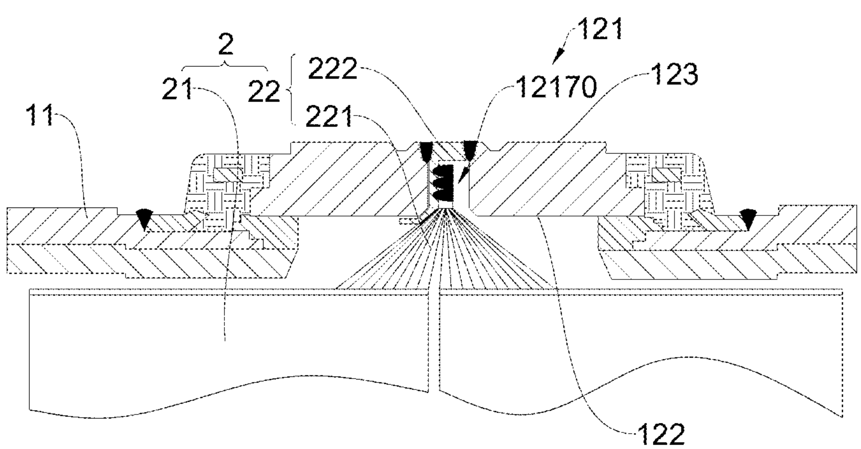


FIG. 32

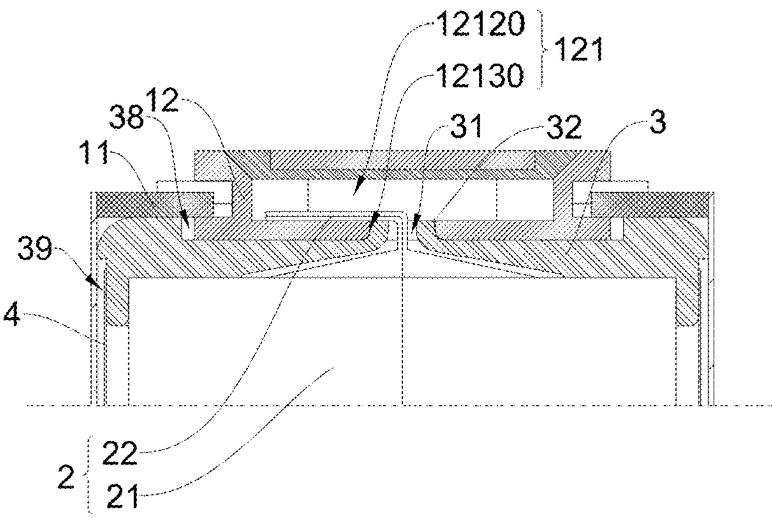


FIG. 33

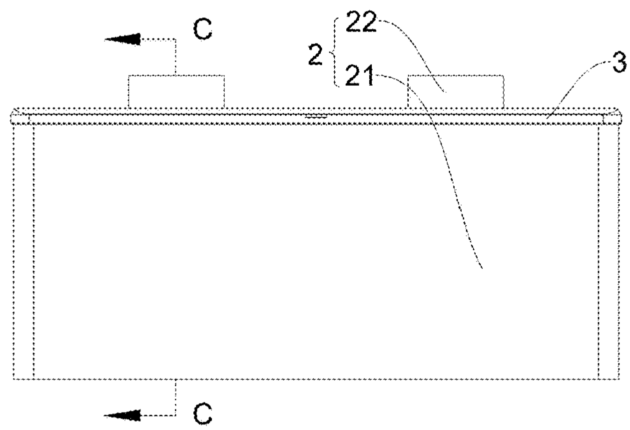


FIG. 34

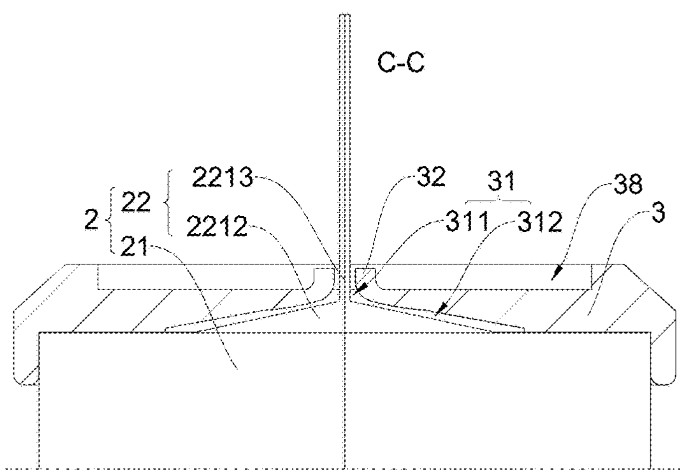
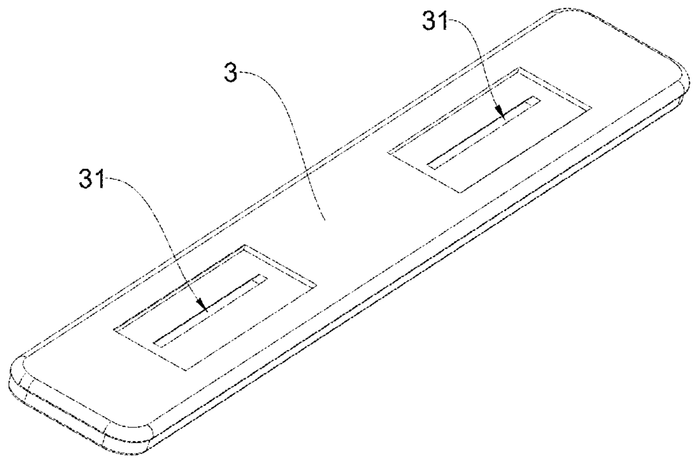


FIG. 35



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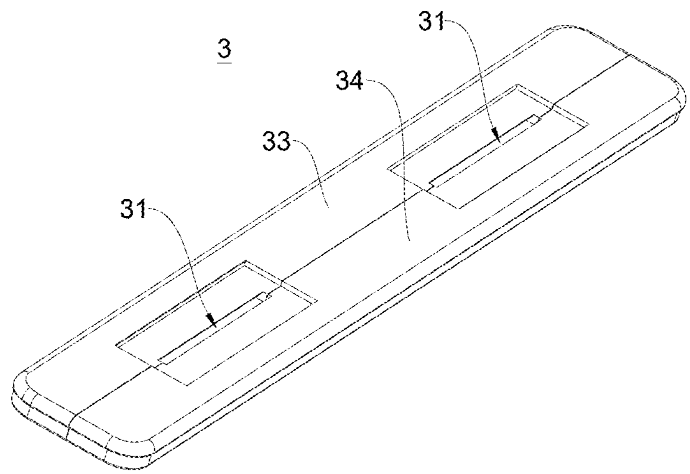

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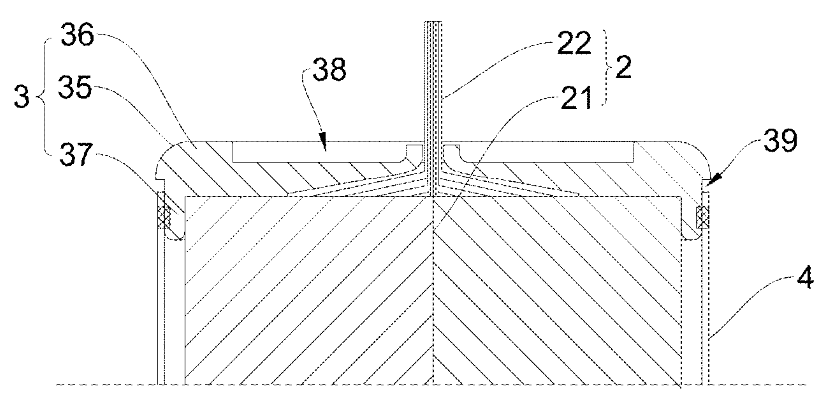

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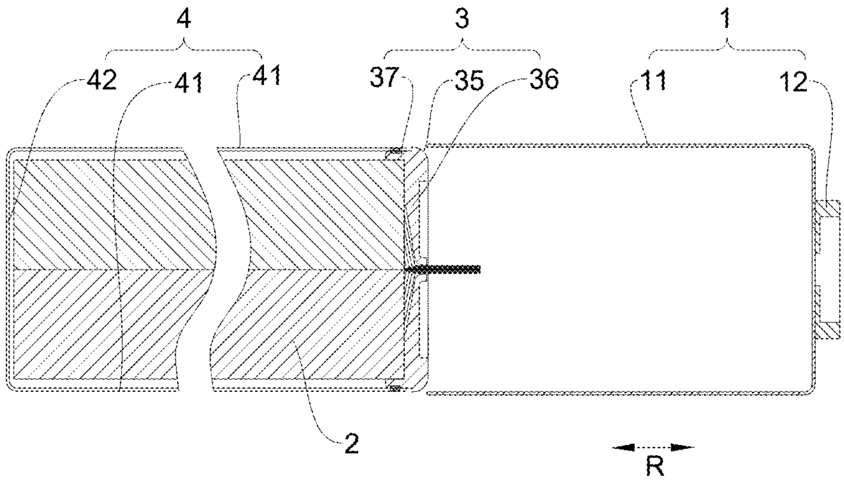

FIG. 39

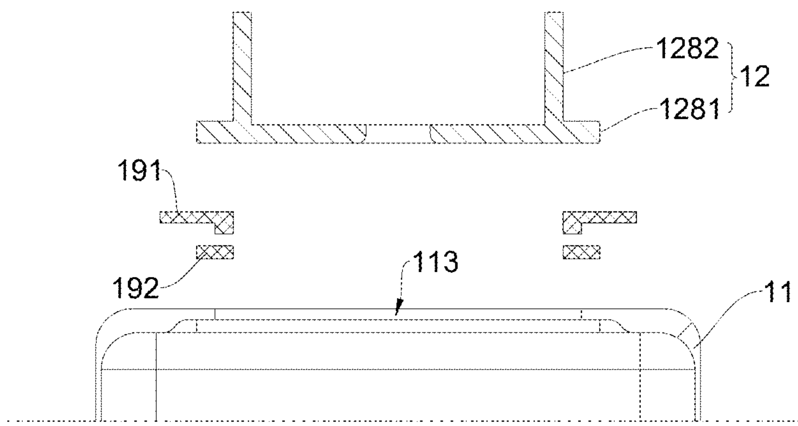

FIG. 40

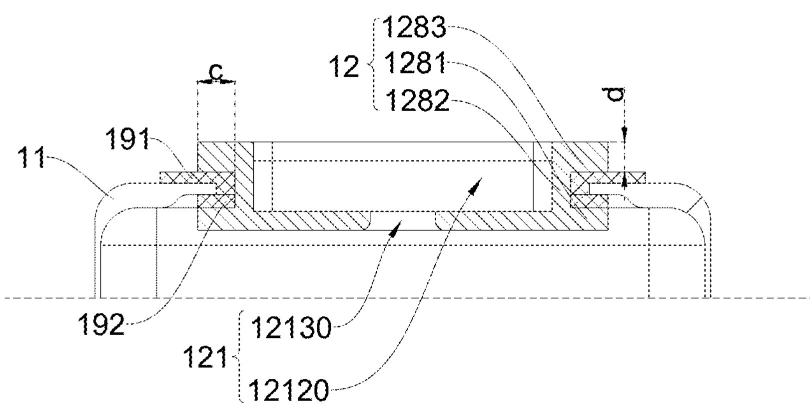

FIG. 41

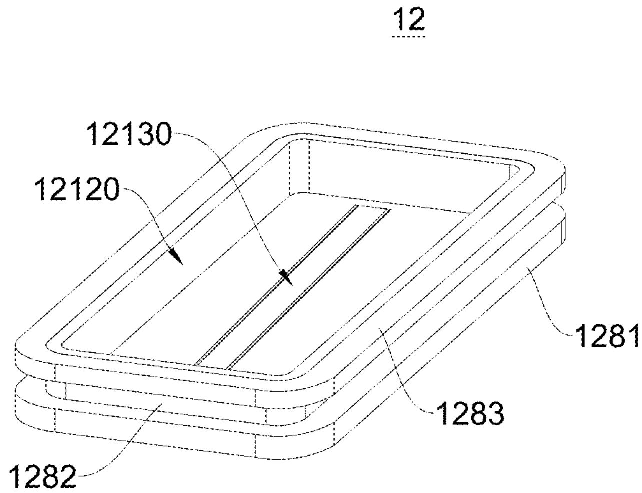

FIG. 42

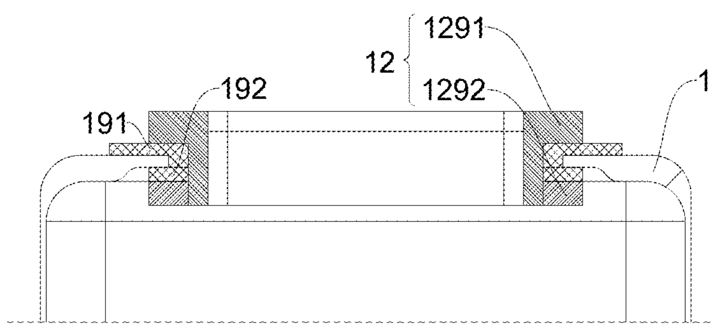

FIG. 43

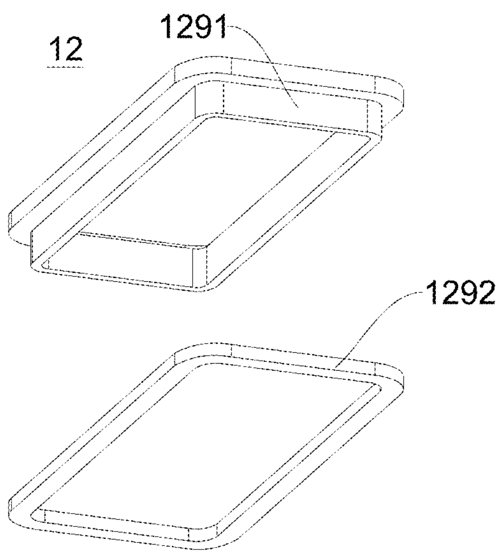

FIG. 44

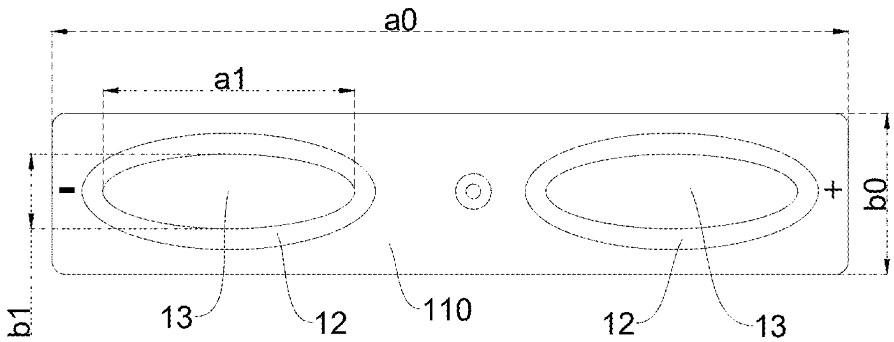

FIG. 45

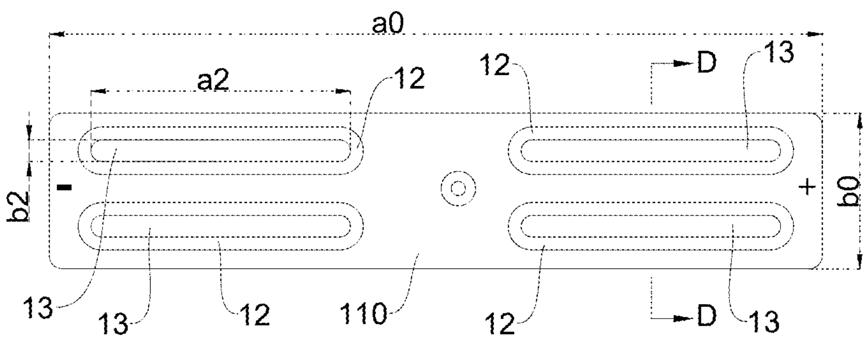

FIG. 46

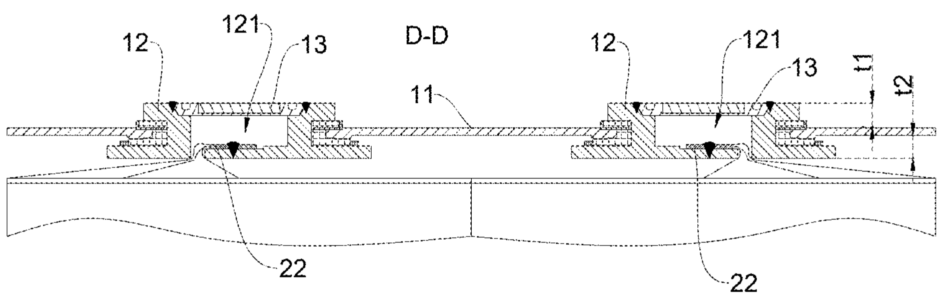

FIG. 47

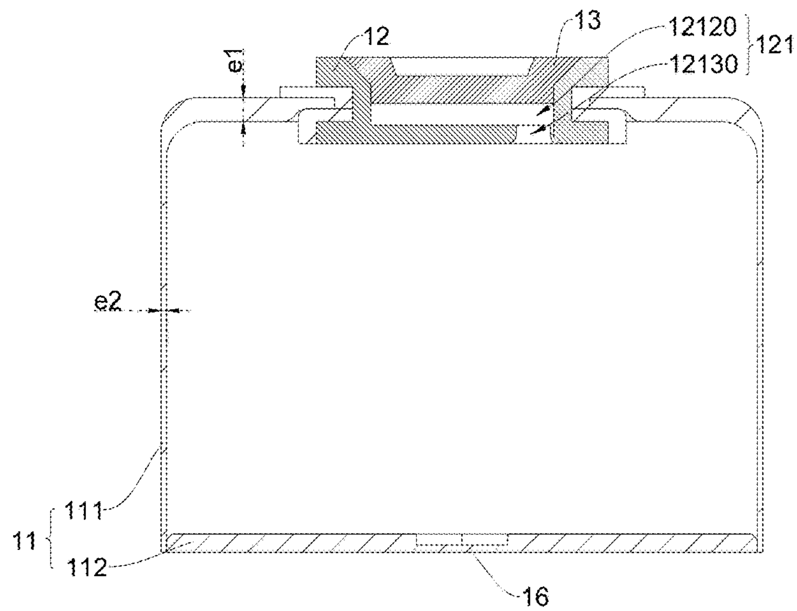

FIG. 48

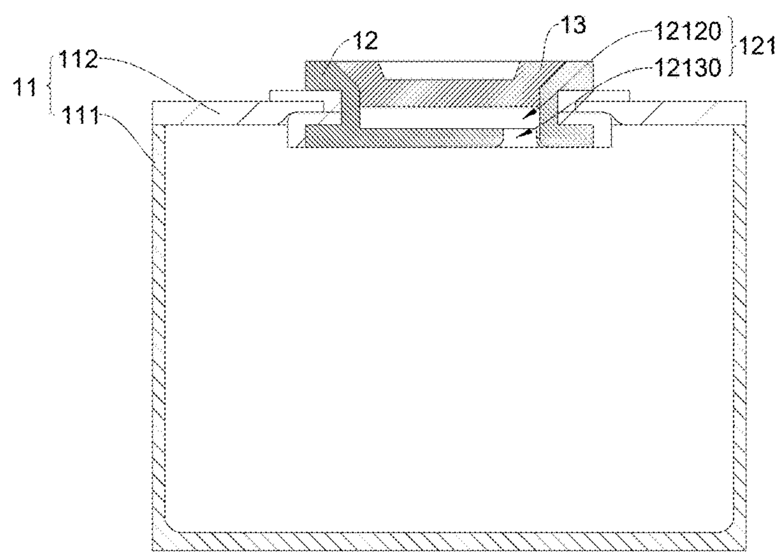

FIG. 49

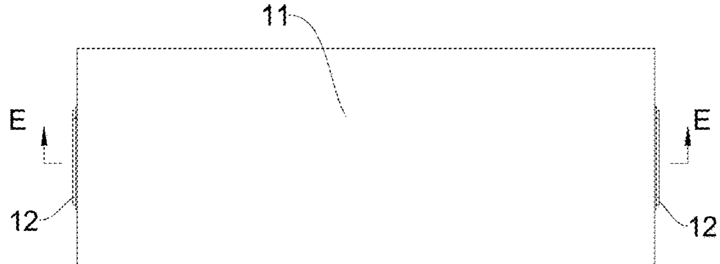

FIG. 50

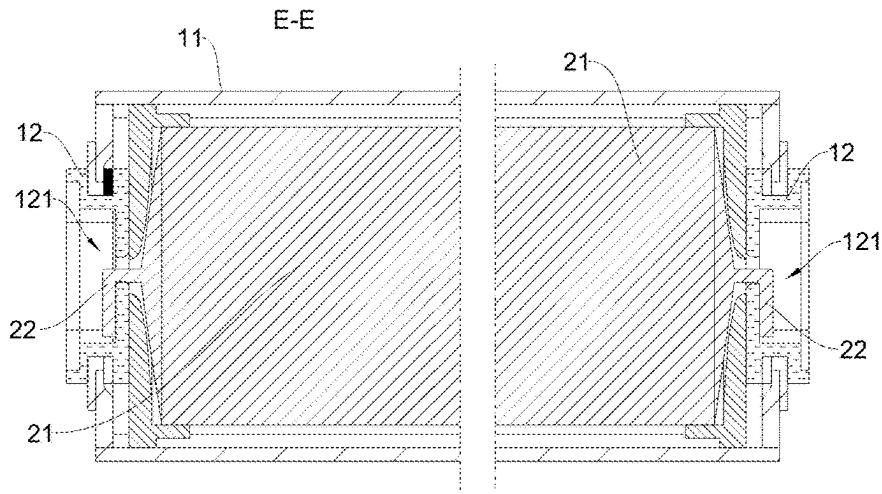

FIG. 51

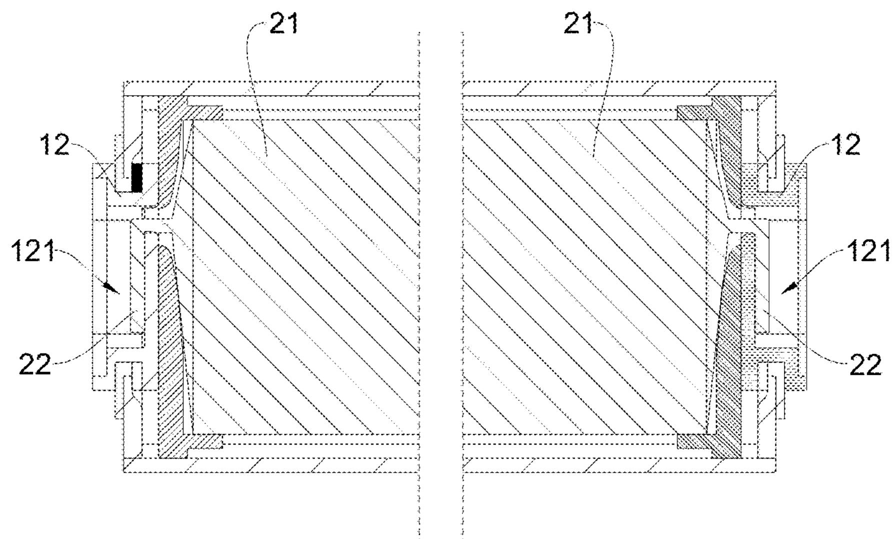

FIG. 52

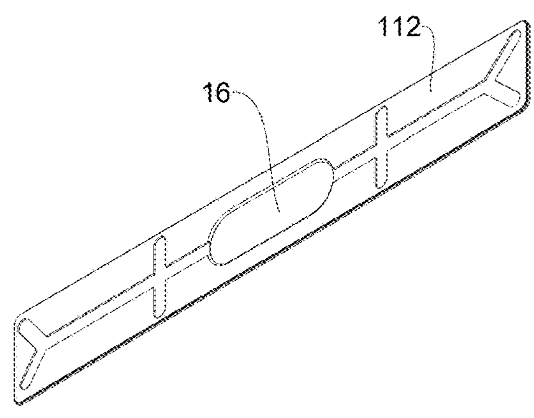

FIG. 53

BATTERY CELL, BATTERY AND ELECTRIC APPARATUS

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is a continuation of International Application No. PCT/CN2023/079679, filed on Mar. 3, 2023, the entire content of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] The present application relates to the technical field of batteries, and in particular, to a battery cell, a battery, and an electric device.

BACKGROUND

[0003] In recent years, new energy vehicles have developed by leaps and bounds. In the field of electric vehicles, batteries, as the power source of electric vehicles, play an irreplaceable and important role. Typically, a battery includes a plurality of battery cells. However, it is difficult to increase the energy density of the current battery cells, resulting in difficulty in increasing the endurance mileage of vehicles.

SUMMARY

[0004] Embodiments of the present application provide a battery cell, a battery, and an electric device, which are conducive to improving the energy density of battery cells.

[0005] In a first aspect, the embodiments of the present application provide a battery cell. The battery cell includes a housing assembly and a battery cell assembly. The housing assembly includes a housing and a first post terminal disposed on the housing. The battery cell assembly includes an active substance-coated part and a conductive part. The active substance-coated part is accommodated in the housing, and the conductive part is configured to electrically connect the active substance-coated part and the first post terminal, where the first post terminal is provided with an accommodating part, and at least a part of the conductive part is accommodated in the accommodating part.

[0006] In the above technical solution, in one aspect, the arrangement of the accommodating part in the first post terminal can reduce the weight of the first post terminal to a certain extent, so as to improve the gravimetric energy density of the battery cell and the battery. In another aspect, by accommodating at least a part of the conductive part in the accommodating part to occupy the space in the first post terminal, the space occupied by the conductive part in the housing can be reduced, and in the case of given dimensions of the housing, some space can be saved in the housing to accommodate a larger-sized active substance-coated part, thereby improving the volumetric energy density of the battery cell. Meanwhile, the accommodation of at least a part of the conductive part in the accommodating part can reduce the space occupied by the battery cell, so as to accommodate a greater number of battery cells in a battery of a given volume and improve the volumetric energy density of the battery. Moreover, the accommodation of at least a part of the conductive part in the accommodating part can reduce the redundancy of the conductive part in the housing to a certain extent, reduce the probability of short circuit between the conductive part and the active substance-coated part, and reduce the probability of short circuit of the electrode assembly, thus further improving the working reliability and stability of the battery cell and the battery.

[0007] In some embodiments, the accommodating part is provided with a first accommodating groove, a surface of the first post terminal on a side facing the active substance-coated part is a post terminal inner end surface, an opening of the first accommodating groove is formed on the post terminal inner end surface, and at least a part of the conductive part is accommodated in the first accommodating groove.

[0008] In the above technical solution, in one aspect, the formation of the first accommodating groove on the first post terminal can reduce the weight of the first post terminal to a certain extent, so as to improve the gravimetric energy density of the battery cell and the battery. In another aspect, since the groove opening of the first accommodating groove is formed on the post terminal inner end surface and the post terminal inner end surface is the surface of the first post terminal on a side proximal to the active substance-coated part, the first accommodating groove can be open in a direction toward the active substance-coated part, thereby facilitating the conductive part to extend into the first accommodating groove and improving the assembly efficiency. Meanwhile, as the first accommodating groove faces toward the active substance-coated part, the first accommodating groove can also serve as a buffer and temporary storage structure for the electrolyte, such that a larger amount of electrolyte can be accommodated in the housing. Since the electrolyte is consumed during the charging and discharging processes of the battery cell, the service life of the battery cell can be prolonged when there is a larger amount of electrolyte. Also, since the first accommodating groove faces toward the active substance-coated part, the first accommodating groove can serve as an accommodation and buffer structure for gases generated inside the electrode assembly, so as to reduce the expansion of the battery cell and improve the reliability and stability of the battery cell. Moreover, since the first accommodating groove is located on the inner side of the post terminal, external foreign matters and impurities can hardly enter the first accommodating groove, thereby reducing the impact of external foreign matters and impurities on the electrode assembly, ensuring the working stability and reliability of the electrode assembly, and further improving the stability and reliability of the battery cell and the battery.

[0009] In some embodiments, the housing is provided with a mounting hole, and the first post terminal is mounted in the mounting hole; in an axial direction of the first post terminal, a depth H1 of the first accommodating groove is greater than or equal to a minimum distance H2 from the post terminal inner end surface to the mounting hole.

[0010] In the above technical solution, since in the axial direction of the first post terminal, the depth of the first accommodating groove is greater than or equal to the minimum distance from the post terminal inner end surface to the mounting hole, the volume of the first post terminal can be fully utilized, such that the first accommodating groove has a relatively great depth, which is conducive to accommodating more conductive parts, thereby reducing the space occupied by the conductive parts in the housing to a greater extent, further improving the energy density of the battery cell, and further reducing the redundancy of the conductive parts in the housing. Meanwhile, since the first accommodating groove has a relatively great depth, the gas generated in the electrode assembly can be accommodated, so as to ensure the reliability and stability of the battery cell, and a larger amount of electrolyte can also be accommodated, so as to ensure the service life of the battery cell.

[0011] In some embodiments, the first post terminal includes a first end wall and a first side wall, the first end wall is located on a side of the first side wall distal to the active substance-coated part, the first end wall and the first side wall define, in an enclosing manner, the first accommodating groove, and the electrical connection position between the conductive part and the first post terminal is located on the first end wall and/or the first side wall.

[0012] In the above technical solution, setting the electrical connection position between the conductive part and the first post terminal on at least one of the first end wall and the first side wall will provide the first accommodating groove with the functionality of accommodating at least a part of the conductive part as well as the functionality of achieving the electrical connection to the conductive part, thereby simplifying the structure of the first post terminal, facilitating the processing of the first post terminal, simplifying the structure of the conductive part, reducing the redundancy of the conductive part, and reducing the cost of the conductive part. Moreover, utilizing the groove wall of the first accommodating groove to achieve the electrical connection to the conductive part may allow a relatively greater area of the electric connection between the conductive part and the first post terminal, which not only reduces the difficulty of electrical connection but also improves the reliability and stability of the electrical connection, thereby improving the performance of the battery cell.

[0013] In some embodiments, the first end wall is provided with a first recess, and at least a part of the electrical connection position between the conductive part and the first end wall is located in the first recess.

[0014] In the above technical solution, in one aspect, the arrangement of the first recess on the first end wall can achieve the pre-positioning of the conductive part via the first recess, which is conducive to accurately finding the position to achieve the electrical connection and improve manufacturing efficiency. In another aspect, by providing the first recess on the first end wall, the wall thickness of the part of the first end wall can be locally reduced, which is not only conducive to electrical connection by welding, but also conducive to reducing the weight of the first post terminal and improving the gravimetric energy density of the battery cell.

[0015] In some embodiments, the first post terminal is provided with a first groove, a surface of the first post terminal on a side distal to the active substance-coated part is a post terminal outer end surface, and an opening of the first groove is formed on the post terminal outer end surface.

[0016] In the above technical solution, in one aspect, the arrangement of the first groove on the first post terminal can further reduce the weight of the first post terminal, such that the gravimetric energy density of the battery cell and the battery can be improved. In another aspect, the first groove is located on the outer side of the first post terminal, which allows the accommodation and mounting of structural components electrically connecting various battery cells of the battery in the first groove, so as to make full use of the space in the first post terminal, thereby improving the space utilization and volumetric energy density of the battery. In addition, since the first post terminal is provided with both the first accommodating groove and the first groove, the first groove is located on a side of the first accommodating groove distal to the active substance-coated part, and the first groove is open in a direction away from the first accommodating groove, it is conducive to electrically connect the conductive part to the groove wall of the first accommodating groove through the first groove from the outside of the first post terminal. For example, it is conducive to externally welding the first post terminal to the conductive part through the first groove, which facilitates processing and manufacture of the battery cell and can save processing and manufacturing costs.

[0017] In some embodiments, the housing assembly further includes a groove cover, and the groove cover is arranged on the post terminal and closes the opening of the first groove.

[0018] In the above technical solution, the arrangement of the groove cover can facilitate the electrical connection between adjacent battery cells in the battery, and since the electrical connection position of the battery cells is separated from the electrical connection position of the conductive part and the first post terminal by the first groove, there is less interference between the two, which can further improve the stability and reliability of the battery cell. Meanwhile, the groove cover can also prevent foreign matters from entering the first groove, thereby reducing the interference of foreign matters with the battery cell assembly and further improving the reliability and stability of the battery cell.

[0019] In some embodiments, the active substance-coated part includes a current collector and an active substance layer disposed on the current collector, and the conductive part includes a tab part electrically connected to the current collector. The tab part includes a plurality of tab plates. Parts of the plurality of tab plates proximal to the current collector converge to form a first gathering part, and parts of the plurality of tab plates distal to the current collector converge and are connected to form a second gathering part. The first gathering part connects the second gathering part and the active substance-coated part, and at least a part of the second gathering part is accommodated in the first accommodating groove.

[0020] In the above technical solution, since the tab part includes the second gathering part formed by converging and connecting parts of the plurality of tab plates, at least a part of the second gathering part is accommodated in the first accommodating groove, which facilitates the connection between the conductive part and the first post terminal and can make full use of the space of the first post terminal, thereby improving the volumetric energy density of the battery cell.

[0021] In some embodiments, at least a part of the first gathering part is accommodated in the first accommodating groove.

[0022] In the above technical solution, at least a part of the first gathering part and at least a part of the second gathering part of the tab part are both accommodated in the first accommodating groove, such that the space in the first post terminal can be more fully utilized, the space occupied by the tab part in the housing can be further reduced, and the volumetric energy density of the battery cell can be improved.

[0023] In some embodiments, the conductive part further includes an adapting piece. The adapting piece is connected to the second gathering part, the conductive part is electrically connected to the first post terminal through the adapting piece, and at least a part of the adapting piece is accommodated in the first accommodating groove.

[0024] In the above technical solution, in one aspect, by accommodating at least a part of the second gathering part and at least a part of the adapting piece in the first accommodating groove, the space in the first post terminal can be more fully utilized, and the space occupied by the conductive part in the housing can be further reduced, so as to further improve the volumetric energy density of the battery cell. In another aspect, by using the adapting piece to achieve an indirect electrical connection between the second gathering part and the first post terminal, the adapting piece can be welded to the first post terminal at a part avoiding the second gathering part, such that the welding between the adapting piece and the first post terminal is firm, welding cracks are unlikely to occur, and the reliability and stability of the battery cell can be further improved. Meanwhile, by electrically connecting the first post terminal and the tab part through the adapting piece, the configuration of the tab part can also be simplified.

[0025] In some embodiments, the active substance-coated part includes a current collector and an active substance layer disposed on the current collector, and the conductive part includes a tab part and an adapting piece. The tab part includes a plurality of tab plates electrically connected to the current collector. Parts of the plurality of tab plates proximal to the current collector converge to form a first gathering part, and parts of the plurality of tab plates distal to the current collector converge and are connected to form a second gathering part. The adapting piece is electrically connected to the second gathering part, and at least a part of the adapting piece is accommodated in the first accommodating groove and electrically connected to the first post terminal.

[0026] In the above technical solution, in one aspect, by using the adapting piece to achieve an indirect electrical connection between the second gathering part and the first post terminal, the adapting piece can be welded to the first post terminal at a part avoiding the second gathering part, such that the welding between the adapting piece and the first post terminal is firm, welding cracks are unlikely to occur, and the reliability and stability of the battery cell can be further improved. Meanwhile, by electrically connecting the first post terminal and the tab part through the adapting piece, the configuration of the tab part can also be simplified. In another aspect, since at least a part of the adapting piece is accommodated in the first accommodating groove, the adapting piece can occupy the space in the first post terminal, such that the space occupied by the adapting piece in the housing can be reduced, so as to accommodate a larger-sized active substance-coated part, thereby improving the volumetric energy density of the battery cell. Moreover, the probability of short circuit between the adapting piece and the active substance-coated part can be reduced, and the risk of short circuit of the electrode assembly can be reduced, thereby improving the stability and reliability of the battery cell.

[0027] In some embodiments, the accommodating part is provided with a second accommodating groove, a surface of the first post terminal on a side distal to the active substance-coated part is a post terminal outer end surface, an opening of the second accommodating groove is formed on the post terminal outer end surface, the second accommodating groove is in communication with the interior of the housing through a first perforation, and the conductive part is provided in the first perforation in a penetrating manner and is at least partially accommodated in the second accommodating groove.

[0028] In the above technical solution, in one aspect, the arrangement of the second accommodating groove on the first post terminal can reduce the weight of the first post terminal to a certain extent, so as to improve the gravimetric energy density of the battery cell and the battery. In another aspect, since the opening of the second accommodating groove is formed on the post terminal outer end surface, and the post terminal outer end surface is the surface of the first post terminal on a side distal to the active substance-coated part, the second accommodating groove can be open in a direction away from the active substance-coated part. In this way, when at least a part of the conductive part is accommodated in the second accommodating groove, the accommodation and arrangement of the conductive part or operations on the electrical connection between the conductive part and the first post terminal can be easily achieved through the opening of the second accommodating groove, thereby reducing difficulties in producing the battery cell and improving the manufacturing efficiency of the battery cell. In addition, since the second accommodating groove is in communication with the housing through the first perforation, the second accommodating groove can also serve as a buffering and temporary storage structure for the electrolyte, such that more electrolyte can be accommodated in the housing. As the electrolyte is consumed during the charging and discharging of the battery cell, a greater amount of electrolyte can prolong the service life of the battery cell. It is also due to the fact that the second accommodating groove is in communication with the housing through the first perforation, that the second accommodating groove can also serve as an accommodating and buffering structure for the gas generated in the electrode assembly, reducing the expansion of the battery cell and improving the reliability and stability of the battery cell.

[0029] In some embodiments, the electrical connection position between the conductive part and the first post terminal is located on the wall of the first perforation formed on the first post terminal.

[0030] In the above technical solution, when the electrical connection position between the conductive part and the first post terminal is arranged on the wall of the first perforation, the electrical connection operation between the conductive part and the first post terminal may be performed through the second accommodating groove. Also, the electrical connection position between the conductive part and the first post terminal can achieve the sealing of the first perforation to save sealing costs and reduce electrolyte leakage.

[0031] In some embodiments, the first post terminal includes a second end wall and a second side wall, the second end wall is located on a side of the second side wall proximal to the active substance-coated part, the second end wall and the second side wall define, in an enclosing manner, the second accommodating groove, the first perforation is formed on the second end wall, and the electrical connection position between the conductive part and the first post terminal is located on the second end wall and/or the second side wall.

[0032] In the above technical solution, setting the electrical connection position between the conductive part and the first post terminal on at least one of the second end wall and the second side wall will provide the second accommodating groove with the functionality of accommodating at least a part of the conductive part and provide the wall of the second accommodating groove with the functionality of achieving the electrical connection to the conductive part, thereby simplifying the structure of the first post terminal and facilitating the processing of the first post terminal. In addition, forming the first perforation in the second end wall will facilitate the extension of the conductive part into the second accommodating groove through the first perforation, which may simplify the structure of the conductive part, reduce the redundancy of the conductive part, and reduce the cost of the conductive part. Furthermore, the opening direction of the second accommodating groove makes it easy to perform the electrical connection operation between the conductive part and the wall of the second accommodating groove through the opening of the second accommodating groove, thereby reducing the difficulties in the electrical connection. Moreover, utilizing the wall of the second accommodating groove to achieve the electrical connection to the conductive part may allow a relatively greater area of the electrical connection between the conductive part and the first post terminal, thereby improving the reliability and stability of the electrical connection and improving the performance of the battery cell.

[0033] In some embodiments, the second end wall is provided with a second recess, and at least a part of the electrical connection position between the conductive part and the second end wall is located in the second recess.

[0034] In the above technical solution, arranging the second recess on the second end wall may achieve the prepositioning of the conductive part via the second recess, which is beneficial to accurately finding the position to achieve the electrical connection and improving manufacture efficiency.

[0035] In some embodiments, the housing is provided with a mounting hole, and the first post terminal is mounted in the mounting hole; in an axial direction of the first post terminal, a depth H3 of the second accommodating groove is greater than or equal to a minimum distance H4 from the post terminal outer end surface to the mounting hole.

[0036] In the above technical solution, since in the axial direction of the first post terminal, the depth of the second accommodating groove is greater than or equal to the minimum distance from the post terminal outer end surface to the mounting hole, the volume of the first post terminal can be fully utilized, such that the second accommodating groove has a relatively great depth, which is conducive to accommodating more conductive parts, thereby reducing the space occupied by the conductive parts in the housing to a greater extent, further improving the energy density of the battery cell, and further reducing the redundancy of the conductive parts in the housing. Meanwhile, since the second accommodating groove has a relatively great depth, the gas generated in the electrode assembly can be accommodated, so as to improve the reliability and stability of the battery cell, and a larger amount of electrolyte can also be accommodated, so as to prolong the service life of the battery cell.

[0037] In some embodiments, the housing assembly further includes a first cover plate, the first cover plate fits the first post terminal and closes the opening of the second accommodating groove, and the first cover plate is electrically connected to the first post terminal.

[0038] In the above technical solution, arranging the first cover plate to close the opening of the second accommodating groove can prevent the electrolyte in the housing from leaking out of the opening of the second accommodating groove. Moreover, since the first cover plate closes the opening of the second accommodating groove and is electrically connected to the first post terminal, an indirect electrical connection between the first post terminal and a busbar component of the battery can be easily achieved by using the first cover plate, and the connection area of the electrical connection can also be increased, thereby reducing the resistance of the electrical connection.

[0039] In some embodiments, the first cover plate includes a first conductive member and a second conductive member made of different materials, the first conductive member fits and is electrically connected to the first post terminal, and the second conductive member fits and is electrically connected to the first conductive member.

[0040] In the above technical solution, the composite configuration of the first cover plate and the same material of the first conductive member and the first post terminal facilitate the electrical connection between the first conductive member and the first post terminal. Also, since the second conductive member is made of a different material from the first conductive member, the second conductive member may be used to electrically connect to the busbar component of the battery with a material different from that of the first post terminal.

[0041] In some embodiments, the first conductive member is provided with a second groove, the second conductive member is embedded in the second groove, and an opening of the second groove is formed on the surface of the first conductive member distal to the second accommodating groove, such that the second conductive member is exposed at the opening of the second groove.

[0042] In the above technical solution, embedding the second conductive member in the first conductive member can reduce the difficulties in assembling the first conductive member and the second conductive member, improve the stability and convenience of the fit of the first conductive member and the second conductive member, reduce the thickness of the first cover plate, and reduce the space occupied by the first cover plate, so as to improve the space utilization of the battery cell. Also, since the second conductive member can be exposed from the surface of the first conductive member on the side distal to the second accommodating groove through the opening of the second groove, it is conductive to achieving the electrical connection between the second conductive member and the busbar component of the battery outside the first post terminal. In addition, since the opening of the second groove is formed on the surface of the first conductive member on the side distal to the second accommodating groove, the first conductive member is arranged between the second accommodating groove and the second conductive member for separation, so as to prevent the electrolyte entering the second groove from contacting the second conductive member, thereby reducing the leakage of the electrolyte.

[0043] In some embodiments, the first cover plate is embedded in the opening of the second accommodating groove.

[0044] In the above technical solution, embedding the first cover plate in the second accommodating groove can reduce the difficulties in assembling the first cover plate and the first post terminal, improve the stability of the assembly of the first cover plate and the first post terminal and the reliability and convenience of the connection, and reduce the space occupied by the first cover plate outside the first post terminal. Moreover, since the first cover plate is embedded in the opening of the second accommodating groove, there is sufficient space in the second accommodating groove to accommodate the conductive part.

[0045] In some embodiments, the wall surface at the opening of the second accommodating groove formed on the first post terminal is an inclined guiding surface, and the inclined guiding surface is configured to guide the first cover plate to fit the opening of the second accommodating groove.

[0046] In the above technical solution, processing the wall surface at the opening of the second accommodating groove into an inclined surface with guidance functionality can reduce the difficulties in assembling the first cover plate and the second accommodating groove, and improve the assembly efficiency of the first cover plate and the second accommodating groove.

[0047] In some embodiments, the second accommodating groove includes a first groove segment and a second groove segment located on the side of the first groove segment proximal to the post terminal outer end surface, the cross-sectional area of the second groove segment is greater than the cross-sectional area of the first groove segment, so as to form a step surface between the first groove segment and the second groove segment, and the first cover plate is embedded in the second groove segment and supported by the step surface.

[0048] In the above technical solution, setting the second accommodating groove as a stepped groove form can stably fit the first cover plate to the opening of the second accommodating groove, thereby improving the connection stability between the first cover plate and the first post terminal. Also, limiting the groove depth of the first groove segment may provide the second accommodating groove with sufficient space to accommodate the conductive part.

[0049] In some embodiments, the first cover plate is provided with a stress relief groove, and the stress relief groove is located in an outer peripheral region of the first cover plate.

[0050] In the above technical solution, by disposing a stress relief groove on the first cover plate, the stress generated during the machining process of the first cover plate or during the electrical connection process between the first cover plate and the first post terminal can be relieved, so as to mitigate the problem of deformation or damage of the first cover plate caused by stress.

[0051] In some embodiments, the housing assembly further includes a second cover plate, and the second cover plate lids the first perforation and is located outside the conductive part in the second accommodating groove.

[0052] In the above technical solution, when the electrolyte enters the second accommodating groove through the first perforation, the problem of electrolyte overflow from the first post terminal can be mitigated through the second cover plate, thereby improving the reliability of the battery cell.

[0053] In some embodiments, the first post terminal includes a first post terminal part and a second post terminal part made of different materials and electrically connected, the second post terminal part is located on the side of the first post terminal part distal to the active substance-coated part, the accommodating part is arranged on the first post terminal part or on the first post terminal part and the second post terminal part, and the conductive part is electrically connected to the first post terminal part.

[0054] In the above technical solution, by configuring the first post terminal as a composite form composed of different materials, the first post terminal part located on the inner side is fitted in a receiving manner and electrically connected to the conductive part, and the second post terminal part located on the outer side is electrically connected to the busbar component of the battery and the like, which is conductive to realizing the assembly and electrical connection of the first post terminal to related components, reducing the mutual interference between the electrical connection position of the post terminal and the conductive part, and the electrical connection position of the post terminal and the busbar component of the battery, and improving the reliability and stability of the battery cell.

[0055] In some embodiments, the active substance-coated part includes a current collector and an active substance layer disposed on the current collector, and the conductive part includes a tab part electrically connected to the current collector. The tab part includes a plurality of tab plates. Parts of the plurality of tab plates proximal to the current collector converge to form a first gathering part, and parts of the plurality of tab plates distal to the current collector converge and are connected to form a second gathering part. The first gathering part connects the second gathering part and the active substance-coated part, and at least a part of the second gathering part is accommodated in the second accommodating groove.

[0056] In the above technical solution, since the tab part includes a second gathering part formed by converging and connecting a plurality of tab plates, at least a part of the second gathering part can be easily accommodated in the second accommodating groove, which facilitates the assembly of the conductive part and the first post terminal.

[0057] In some embodiments, the accommodating part is further provided with a third accommodating groove, the surface of the first post terminal on a side facing the active substance-coated part is a post terminal inner end surface, the third accommodating groove is located on a side of the second accommodating groove proximal to the active substance-coated part, the groove opening of the third accommodating groove is formed on the post terminal inner end surface, the third accommodating groove is in communication with the second accommodating groove through a first perforation, and at least a part of the first gathering part is accommodated in the third accommodating groove.

[0058] In the above technical solution, at least a part of the first gathering part of the tab part is accommodated in the third accommodating groove, and at least a part of the second gathering part of the tab part is accommodated in the second accommodating groove, such that the space in the first post terminal can be further fully utilized.

[0059] In some embodiments, the conductive part further includes an adapting piece. The adapting piece is connected to the second gathering part, the conductive part is electrically connected to the first post terminal through the adapting piece, and at least a part of the adapting piece is accommodated in the second accommodating groove.

[0060] In the above technical solution, by accommodating at least a part of the second gathering part and at least a part of the adapting piece in the second accommodating groove, the space in the first post terminal can be more fully utilized, and the space occupied by the conductive part in the housing can be further reduced, so as to further improve the volumetric energy density of the battery cell. In another aspect, by using the adapting piece to achieve an indirect electrical connection between the second gathering part and the first post terminal, the adapting piece can be welded to the first post terminal at a part avoiding the second gathering part, such that the welding between the adapting piece and the first post terminal is firm, welding cracks are unlikely to occur, and the reliability and stability of the battery cell can be further improved. Meanwhile, by using the adapting piece to achieve an electrical connection between the tab part and the first post terminal, the configuration of the tab part can also be simplified.

[0061] In some embodiments, the active substance-coated part includes a current collector and an active substance layer disposed on the current collector, and the conductive part includes a tab part and an adapting piece. The tab part includes a plurality of tab plates electrically connected to the current collector. Parts of the plurality of tab plates proximal to the current collector converge to form a first gathering part, and parts of the plurality of tab plates distal to the current collector converge and are connected to form a second gathering part. The adapting piece is electrically connected to the second gathering part, and at least a part of the adapting piece is accommodated in the second accommodating groove and electrically connected to the first post terminal.

[0062] In the above technical solution, the conductive part includes an adapting piece that enables an electrical connection between the tab part and the first post terminal, thereby simplifying the configuration of the tab part. In addition, by accommodating at least a part of the adapting piece in the second accommodating groove, the adapting piece can occupy the space in the first post terminal, such that the space occupied by the adapting piece in the housing can be reduced, so as to accommodate a larger-sized active substance-coated part, thereby improving the energy density of the battery cell. Moreover, the probability of short circuit between the adapting piece and the active substance-coated part can be reduced.

[0063] In some embodiments, the accommodating part is provided with a fourth accommodating groove, the surface of the first post terminal on a side distal to the active substance-coated part is a post terminal outer end face, the groove opening of the fourth accommodating groove is formed on the post terminal outer end surface, the fourth accommodating groove is in communication with the interior of the housing through a second perforation, the conductive part is provided in the second perforation in a penetrating manner, and the electrical connection position between the conductive part and the first post terminal is located on the wall of the second perforation formed on the first post terminal.

[0064] In the above technical solution, the arrangement of the fourth accommodating groove can easily achieve the electrical connection between the conductive part and the wall of the second perforation. Furthermore, in some cases, the sealing of the second perforation can be achieved by the electrical connection of the conductive part to the first post terminal.

[0065] In some embodiments, the battery cell further includes: a support, located in the housing and on a side of the active substance-coated part proximal to the first post terminal, where the support is provided with a clearance hole configured to avoid the conductive part, and the conductive part is suitable for extending to a side of the support distal to the active substance-coated part through the clearance hole.

[0066] In the above technical solution, by forming the clearance hole on the support, the conductive part can be guided and constrained to pass through the clearance hole to fit the first post terminal, which can not only simplify the arrangement of the conductive part, save the material of the conductive part, and reduce the cost, but also allows the support to support and guide the conductive part to fit the first post terminal. As a result, the risk of short-circuit connection between the conductive part and the active substance-coated part can be reduced, further improving the reliability of the battery cell.

[0067] In some embodiments, the support is provided with a guiding part, and the guiding part defines, in an enclosing manner, at least a part of the clearance hole and at least partially extends to the accommodating part.

[0068] In the above technical solution, since the support is provided with the guiding part that at least partially extends into the accommodating part, and at least a part of the clearance hole is defined by the guiding part in an enclosing manner, at least a part of the conductive part can be easily accommodated in the accommodating part, thereby improving the assembly efficiency of the conductive part. Meanwhile, through the arrangement of the guiding part, the fits between the support and the post terminal, and between the support and the conductive part become tighter and more reliable, such that the structure of the battery cell becomes more compact, which is more conducive to improving the energy density of the battery cell.

[0069] In some embodiments, the support is provided with a third groove, and at least a part of the first post terminal located in the housing is accommodated in the third groove.

[0070] In the above technical solution, by forming the third groove on the support, at least a part of the part of the first post terminal located in the housing is accommodated. In one aspect, it is conducive to reducing the space occupied by the support in the housing, and in another aspect, it is also conducive to improving the stability and reliability of the first post terminal, so as to ensure the reliability and stability of the electrical connection between the first post terminal and the battery cell assembly, thereby improving the reliability and stability of the charging and discharging operations of the battery cell.

[0071] In some embodiments, the clearance hole includes a first hole segment and a second hole segment, the second hole segment is located on a side of the first hole segment proximal to the active substance-coated part, and the cross-sectional area of the second hole segment gradually increases in a direction away from the first hole segment, the active substance-coated part includes a current collector and an active substance layer provided on the current collector, the conductive part includes a tab part electrically connected to the current collector, the tab part includes a plurality of tab plates, parts of the plurality of tab plates proximal to the current collector converge to form a first gathering part, parts of the plurality of tab plates distal to the current collector converge and are connected to form a second gathering part, the first gathering part connects the second gathering part and the active substance-coated part, at least a part of the first gathering part is accommodated in the second hole segment, and the second gathering part is provided in the first hole segment in a penetrating manner.

[0072] In the above technical solution, arranging the clearance hole to include the second hole segment that gradually expands in the direction toward the active substance-coated part facilitates the second hole segment to accommodate more first gathering parts, so as to improve the compactness of the fit between the support and the battery cell assembly, such that the overall volume of the battery cell is smaller, and the battery can accommodate a greater number of battery cells, thereby improving the volumetric energy density of the battery.

[0073] In some embodiments, the support is of an integrated structure; or the support is of a split-type structure and includes a first support and a second support that are separable, and the clearance hole is defined between the first support and the second support.

[0074] In the above technical solution, when the support is of the integrated structure, the support is convenient to process and has good reliability, and the support can be conveniently assembled with the housing assembly, thereby improving the assembly efficiency and fitting stability. When the support is of the split-type structure, the clearance hole is defined through the cooperation of the first support and the second support. When the support and the battery cell assembly are assembled, it is not necessary to pass the conductive part from one end of the clearance hole to the other end. Instead, the first support and the second support can be assembled at the position of the conductive part to clamp the conductive part, such that the clearance hole surrounds the conductive part, thereby facilitating the assembly of the support and the battery cell assembly and improving the assembly efficiency.

[0075] In some embodiments, the battery cell further includes: an inner insulating member, located in the housing, wrapped outside the active substance-coated part, and connected to the support.

[0076] In the above technical solution, in one aspect, by wrapping the active substance-coated part with the inner insulating member, the insulation reliability between the active substance-coated part and the housing can be improved, the corrosion of the housing caused by the contact between the active substance-coated part and the housing can be reduced or prevented, and the leakage of the electrolyte caused by the corrosion of the housing can be reduced, thereby improving the reliability of the battery cell. In another aspect, connecting the inner insulating member to the support can reduce the difficulty of fixing the inner insulating member and improve the reliability of the inner insulating member wrapped outside the active substance-coated part.

[0077] In some embodiments, the housing assembly includes a plurality of post terminals, and at least one of the plurality of post terminals is the first post terminal.

[0078] In the above technical solution, at least one of all the post terminals of the housing assembly is the first post terminal, such that a part of the post terminals in the overall battery cell can be the first post terminals with the accommodating parts, or all the post terminals can be the first post terminals with the accommodating parts, so as to allow flexible selection and fit according to actual requirements such as energy density and costs, to improve the applicability of the battery cell.

[0079] In some embodiments, the housing is provided with a pressure relief part, and the pressure relief part and the post terminal are located on a surface on the same side of the housing; or the pressure relief part and the post terminal are separately located on two surfaces on different sides of the housing.

[0080] In the above technical solution, when the post terminal and the pressure relief part are arranged on the same side, processing and assembly are facilitated, while when the post terminal and the pressure relief part are arranged on different sides, the space can be saved, the volume of the post terminal can be increased, and the adverse effects on the post terminal when the pressure relief part releases pressure can be reduced.

[0081] In some embodiments, the housing is provided with a pressure relief part, the housing includes a housing body and a housing cover, one end of the housing body is open, the housing cover is arranged at the open end of the housing body, and the pressure relief part is arranged on the housing cover.

[0082] In the above technical solution, the pressure relief part is convenient to process and has good pressure relief reliability.

[0083] In a second aspect, the embodiments of the present application further provide a battery. The battery includes the above battery cell.

[0084] In the above technical solution, since the battery is provided with the battery cell described above and the first post terminal of the battery cell is provided with the accommodating part, the weight of the first post terminal can be reduced to a certain extent, thereby improving the gravimetric energy density of the battery cell and the battery. In another aspect, by accommodating at least a part of the conductive part in the accommodating part to occupy the space in the first post terminal, it is therefore conducive to improving the volumetric energy density of the battery cell, or the space occupied by the battery cell can be reduced. This allows for the accommodation of a greater number of battery cells in the battery of a given volume, thereby improving the volumetric energy density of the battery. Moreover, the accommodation of at least a part of the conductive part in the accommodating part can reduce the redundancy of the conductive part in the housing to a certain extent, reduce the probability of short circuit between the conductive part and the active substance-coated part, thus further improving the working reliability and stability of the battery cell and the battery.

[0085] In a third aspect, the embodiments of the present application further provide an electric device. The electric device includes the above battery.

[0086] In the above technical solution, the electric device is provided with the battery described above and the energy density of the battery can be improved, which is therefore conducive to prolonging the service life of the electric device; and the working reliability and stability of the battery can be improved, thereby enhancing the working reliability and stability of the electric device.

BRIEF DESCRIPTION OF THE DRAWINGS

[0087] In order to more clearly illustrate the technical solutions in embodiments of the present disclosure, the drawings required for use in the embodiments will be briefly described below. It should be understood that the following drawings only illustrate some embodiments of the present disclosure and therefore shall not be considered as limiting the scope of the present disclosure, and other related drawings can be derived from these drawings by those of ordinary skill in the art without creative efforts.

[0088] FIG. **1** is a schematic structural diagram of a vehicle provided according to some embodiments of the present application;

[0089] FIG. **2** is an exploded view of the structure of a battery according to some embodiments of the present application;

[0090] FIG. **3** is a schematic structural diagram of a battery cell according to some embodiments of the present application;

[0091] FIG. **4** is an orthographic projection view of a battery cell according to some embodiments of the present application;

[0092] FIG. **5** is a cross-sectional view along the line A-A in FIG. **4**;

[0093] FIG. **6** is a schematic structural diagram of a battery cell according to some embodiments of the present application;

[0094] FIG. **7** is an assembly diagram of a second post terminal, a battery cell assembly, and a housing according to some embodiments of the present application;

[0095] FIG. **8** is a partial cross-sectional schematic diagram of a battery cell according to some embodiments of the present application;

[0096] FIG. **9** is a partial cross-sectional schematic diagram of a battery cell according to some embodiments of the present application;

[0097] FIG. **10** is a partial cross-sectional schematic diagram of a battery cell according to some embodiments of the present application;

[0098] FIG. **11** is a partial cross-sectional schematic diagram of a battery cell assembly according to some embodiments of the present application;

[0099] FIG. **12** is a diagram of various tab gathering solutions of a battery cell assembly according to some embodiments of the present application;

[0100] FIG. **13** is a partial cross-sectional schematic diagram of a battery cell according to some embodiments of the present application;

[0101] FIG. **14** is a partial cross-sectional schematic diagram of a battery cell according to some embodiments of the present application;

[0102] FIG. **15** is a partial cross-sectional schematic diagram of a battery cell according to some embodiments of the present application;

[0103] FIG. **16** is a partial cross-sectional schematic diagram of a battery cell according to some embodiments of the present application;

[0104] FIG. **17** is a partial cross-sectional schematic diagram of a battery cell according to some embodiments of the present application;

[0105] FIG. **18** is a partial enlarged view of the portion B in FIG. **3**;

[0106] FIG. **19** is an orthographic projection view of various first post terminals according to some embodiments of the present application;

[0107] FIG. **20** is a partial cross-sectional schematic diagram of a battery cell according to some embodiments of the present application;

[0108] FIG. **21** is a partial cross-sectional schematic diagram of a battery cell according to some embodiments of the present application;

[0109] FIG. **22** is a partial cross-sectional schematic diagram of a battery cell according to some embodiments of the present application;

[0110] FIG. **23** is an exploded view of the structure of a battery cell according to some embodiments of the present application;

[0111] FIG. **24** is a partial cross-sectional schematic diagram of a housing assembly according to some embodiments of the present application;

[0112] FIG. **25** is an exploded view of the structure of the housing assembly shown in FIG. **24**;

[0113] FIG. **26** is an exploded view of the structure of the first cover plate shown in FIG. **25**;

[0114] FIG. **27** is a partial cross-sectional schematic diagram of a battery cell according to some embodiments of the present application;

[0115] FIG. **28** is an exploded view of the structure of the battery cell shown in FIG. **27**;

[0116] FIG. **29** is a partial cross-sectional schematic diagram of a battery cell according to some embodiments of the present application;

[0117] FIG. **30** is a partial cross-sectional schematic diagram of a battery cell according to some embodiments of the present application;

[0118] FIG. **31** is a partial cross-sectional schematic diagram of a battery cell according to some embodiments of the present application;

[0119] FIG. **32** is a partial cross-sectional schematic diagram of a battery cell according to some embodiments of the present application;

[0120] FIG. **33** is a partial cross-sectional schematic diagram of a battery cell according to some embodiments of the present application;

[0121] FIG. **34** is a schematic diagram of the fit between a battery cell assembly and a support according to some embodiments of the present application;

[0122] FIG. **35** is a cross-sectional view along the line C-C in FIG. **34**;

[0123] FIG. **36** is a schematic structural diagram of an integrated support according to some embodiments of the present application;

[0124] FIG. **37** is a schematic structural diagram of a split-type support according to some embodiments of the present application;

[0125] FIG. **38** is a partial cross-sectional schematic diagram of a battery cell assembly and a support according to some embodiments of the present application;

[0126] FIG. **39** is an exploded view of the structure of a battery cell assembly, a support, and a housing assembly according to some embodiments of the present application;

[0127] FIG. **40** is an exploded view of the structure of a first post terminal, a housing, and a sealing gasket according to some embodiments of the present application;

[0128] FIG. **41** is an assembly diagram of the first post terminal, the housing, and the sealing gasket shown in FIG. **40**;

[0129] FIG. **42** is a schematic structural diagram of a first post terminal according to some embodiments of the present application;

[0130] FIG. **43** is an assembly diagram of a first post terminal, a housing, and a sealing gasket according to some embodiments of the present application;

[0131] FIG. **44** is an exploded view of the structure of the first post terminal shown in FIG. **43**;

[0132] FIG. **45** is an orthographic projection view of a battery cell according to some embodiments of the present application;

[0133] FIG. **46** is an orthographic projection view of a battery cell according to some embodiments of the present application;

[0134] FIG. **47** is a cross-sectional view along the line D-D in FIG. **46**;

[0135] FIG. **48** is a cross-sectional schematic diagram of a housing assembly according to some embodiments of the present application;

[0136] FIG. **49** is a cross-sectional schematic diagram of a housing assembly according to some embodiments of the present application;

[0137] FIG. **50** is an orthographic projection view of a battery cell according to some embodiments of the present application;

[0138] FIG. **51** is a cross-sectional view along the line E-E in FIG. **50**;

[0139] FIG. **52** is a partial cross-sectional schematic diagram of a battery cell according to some embodiments of the present application; and

[0140] FIG. **53** is a schematic structural diagram of a housing cover according to some embodiments of the present application.

REFERENCE NUMERALS

[0141] electric device **1000**; battery **100**; controller **200**; motor **300**;

[0142] first direction Z; second direction X; third direction Y;

[0143] axial direction R of the first post terminal;

[0144] battery cell **10**; case **20**; first case **201**; second case **202**;

[0145] housing assembly **1**;

[0146] housing **11**; first wall surface **110**; housing body **111**; housing cover **112**; mounting hole **113**;

[0147] first post terminal **12**;

[0148] accommodating part **121**;

[0149] first accommodating groove **12110**; first end wall **12111**; first recess **12112**; first side wall **12113**;

[0150] second accommodating groove **12120**; second end wall **12121**; second recess **12122**; second side wall **12123**;

[0151] first groove segment **12124**; second groove segment **12125**; inclined guiding surface **12126**; step surface **12127**;

[0152] first perforation **12130**; third accommodating groove **12140**; fourth accommodating groove **12150**;

[0153] second perforation **12160**; third perforation **12170**;

[0154] post terminal inner end surface **122**; post terminal outer end surface **123**;

[0155] first post terminal part **124**; second post terminal part **125**;

[0156] first groove **126**; spacer part **127**;

[0157] stopper part **1281**; penetration part **1282**; flange part **1283**;

[0158] first part **1291**; second part **1292**;

[0159] first cover plate **13**; first conductive member **131**; second groove **1311**; second conductive member **132**; stress relief groove **133**;

[0160] second cover plate **14**; second post terminal **15**; pressure relief part **16**; clearance groove **18**;

[0161] first sealing gasket **191**; second sealing gasket **192**;

[0162] battery cell assembly **2**; electrode assembly **2***a;*

[0163] active substance-coated part **21**; current collector **211**; active substance layer **212**; conductive part **22**;

[0164] tab part **221**; tab plate **2211**; first gathering part **2212**; second gathering part **2213**; adapting piece **222**;

[0165] support **3**; clearance hole **31**; first hole segment **311**; second hole segment **312**;

[0166] guiding part **32**; first support **33**; second support **34**; housing entry guiding surface **35**;

[0167] body part **36**; extending part **37**; third groove **38**; positioning groove **39**;

[0168] inner insulating member **4**; body part **41**; connecting part **42**;

[0169] sealing member **6**; groove cover **7**.

DETAILED DESCRIPTION

[0170] To make the objectives, technical solutions, and advantages of the embodiments of the present application clearer, the technical solutions in the embodiments of the present application will be clearly and completely described below with reference to the drawings in the embodiments of the present application. It is obvious that the described embodiments are some, but not all, embodiments of the present application. Based on the embodiments in the present application, all other embodiments obtained by those of ordinary skill in the art without creative work shall fall within the protection scope of the present application.

[0171] Unless otherwise defined, all technical and scientific terms used in the present application have the same meaning as commonly understood by those of ordinary skill in the art to which the present application belongs. The terms used in the specification of the present application are only used to describe specific embodiments and are not intended to limit the present application. The terms "include", "comprise", "have", and any variants thereof in the specification and claims of the present application and the descriptions of the above drawings are intended to cover a non-exclusive inclusion. The terms "first", "second", and the like in the specification and claims of the present application and the above drawings are defined to distinguish different objects and are not intended to describe a specific order or priority.

[0172] Reference in the present application to "embodiment" means that a particular feature, structure, or characteristic described in connection with the embodiment can be included in at least one embodiment of the present application. The references of the word in the context of the specification do not necessarily refer to the same embodiment, nor to separate or alternative embodiments exclusive of other embodiments.

[0173] In the present application, the term "and/or" is only an association relationship that describes the associated objects, and indicates that there may be three relationships. For example, A and/or B may indicate that: only A is present, both A and B are present, and only B is present. In addition, the character "/" in the present application generally indicates an "or" relationship between the associated objects before and after the "/". In this disclosure, unless otherwise specified, phrases like "at least one of A, B, and C" and "at least one of A, B, or C" both mean only A, only B, only C, or any combination of A, B, and C.

[0174] In the embodiments of the present application, the same reference numerals represent the same components, and for the sake of brevity, detailed descriptions of the same components are omitted in different embodiments. It should be understood that the thickness, length, width, and other dimensions of various components in the embodiments of the present application shown in the drawings, as well as the overall thickness, length, width, and other dimensions of the integrated device are only exemplary and should not impose any limitation on the present application.

[0175] The term "plurality" used in the present application refers to two or more (including two).

[0176] In the present application, battery cells may include lithium-ion secondary batteries, lithium-ion primary batteries, lithium-sulfur batteries, sodium-lithium-ion batteries, sodium-ion batteries, magnesium-ion batteries, etc. This is not limited in the embodiments of the present application. The battery cell may be cylindrical, flat, rectangular parallelepiped-shaped, or in other shapes, which is also not limited in the embodiments of the present application. According to the packaging method, battery cells are typically divided into three types: cylindrical battery cells, square battery cells, and soft-pack battery cells, which are not limited in the embodiments of the present application.

[0177] The battery mentioned in the embodiments of the present application refers to a single physical module including one or more battery cells to provide higher voltage and capacity. For example, the battery mentioned in the present application may be a battery module, a battery pack, or the like. A battery module generally includes a plurality of battery cells. A battery pack generally includes a case used for encapsulating one or more battery cells or one or more battery modules. The case can prevent liquid or other foreign substances from affecting the charging or discharging of the battery cells.

[0178] Illustratively, a battery cell may generally include a housing, a battery cell assembly, and an electrolyte. The housing is configured to accommodate the battery cell assembly and the electrolyte, and at least one positive electrode post terminal and at least one negative electrode post terminal are provided on the housing. The battery cell assembly includes one or more electrode assemblies, and the electrode assembly is formed by stacking or winding a positive electrode plate, a negative electrode plate, and a separation film.

[0179] The positive electrode plate may generally include a positive electrode current collector and a positive electrode active substance layer. The positive electrode current collector is directly or indirectly coated with the positive electrode active substance layer, the positive electrode current collector not coated with the positive electrode active substance layer protrudes from the positive electrode current collector coated with the positive electrode active substance layer, the positive electrode current collector not coated with the positive electrode active substance layer serves as a positive electrode tab plate, and a plurality of positive electrode tab plates are stacked together and electrically connected to the positive electrode post terminal. Illustratively, the plurality of positive electrode tab plates stacked together may be directly welded to the positive electrode post terminal to form an electrical connection. Alternatively, the battery cell assembly may further include a positive electrode adapting piece, the plurality of positive electrode tab plates stacked together are welded to one end of the positive electrode adapting piece, and the other end of the positive electrode adapting piece is welded to the positive electrode post terminal to form an electrical connection between the positive electrode tab plates and the positive electrode post terminal.

[0180] The negative electrode plate may generally include a negative electrode current collector and a negative electrode active substance layer. The negative electrode current collector is directly or indirectly coated with the negative electrode active substance layer, the negative electrode current collector not coated with the negative electrode active substance layer protrudes from the negative electrode current collector coated with the negative electrode active substance layer, the negative electrode current collector not coated with the negative electrode active substance layer serves as a negative electrode tab plate, and a plurality of negative electrode tab plates are stacked together and electrically connected to the negative electrode post terminal. Illustratively, the plurality of negative electrode tab plates stacked together may be directly welded to the negative electrode post terminal to form an electrical connection. Alternatively, the battery cell assembly may further include a negative electrode adapting piece, the plurality of negative electrode tab plates stacked together are welded to one end of the negative electrode adapting piece, and the other end of the negative electrode adapting piece is welded to the negative electrode post terminal to form an electrical connection between the negative electrode tab plates and the negative electrode post terminal. The material of the separation film is not limited, such as polypropylene or polyethylene.

[0181] Meanwhile, a battery cell primarily works by the movement of metal ions between the positive electrode plate and the negative electrode plate. Taking lithium-ion batteries as an example, the material of the positive electrode current collector may be aluminum, the material of the positive electrode active substance layer may be lithium cobaltate, lithium iron phosphate, ternary lithium, or lithium manganate, etc., the material of the negative electrode current collector may be copper, and the material of the negative electrode active substance layer may be carbon or silicon, etc. During the charging and discharging processes, Li^+ is intercalated or deintercalated back and forth between the two electrodes: during charging, Li^+ is deintercalated from the positive electrode, and intercalated into the negative electrode through the electrolyte, and the negative electrode is in a lithium-rich state; during discharging, the process is reversed.

[0182] In recent years, new energy vehicles have developed by leaps and bounds. In the field of electric vehicles, batteries, as the power source of electric vehicles, play an irreplaceable and important role. Batteries, as core components of new energy vehicles, have high requirements in terms of both energy density and reliability.

[0183] In the related art, during the manufacturing of a battery cell, a current collector is coated with an active substance layer and then cut to obtain an electrode plate consisting of a current collector coated with an active substance layer (referred to as an active substance-coated part) and a current collector not coated with an active substance layer (referred to as a tab plate). The positive and negative electrode plates and a separation film are then stacked or wound in sequence to obtain an electrode assembly. In the electrode assembly, a plurality of tab plates are stacked to form a tab part. The housing of the battery cell is provided with a post terminal, and a surface of the post terminal on a side facing the active substance-coated part is a post terminal inner end surface. When the battery cell is manufactured, the tab part is typically welded to the post terminal inner end surface directly, or welded to the post terminal inner end surface indirectly through the adapting piece, so as to ensure normal charging and discharging operations.

[0184] However, the inventors found that when the battery cell adopts the structure described above, the tab part and the adapting piece accumulate between the active substance-coated part and the post terminal inner end surface, occupying a large space. As a result, in the case of given dimensions of the housing, the size of the active substance-coated part cannot be increased, making it difficult to improve the energy density of the battery cell. Moreover, due to design or manufacturing reasons, the length of the tab part is typically long. When the space between the active substance-coated part and the post terminal inner end surface is small, the problem of redundancy of the tab part arises after disposing the electrode assembly in the housing, which can easily cause short circuit between the tab part or the adapting piece and the active substance-coated part, thereby affecting the reliability and stability of the battery cell.

[0185] Based on the above considerations, in order to improve the energy density, reliability, and stability of the battery cell, the inventors have designed a battery cell after in-depth research, in which at least one of the post terminals of the battery cell is provided with an accommodating part. At least a part of the tab part or the adapting piece that does not have an active substance and plays a conductive role is accommodated in the accommodating part to reduce the space occupied by this part in the housing, such that more space can be saved in the housing to accommodate the active substance-coated part and the volume of the active substance-coated part can be increased, which is conducive to improving the energy density of the battery cell. In addition, the redundancy of the tab part or adapting piece in the housing can be reduced to at least a certain extent and the probability of short circuit between the tab part or adapting piece and the active substance-coated part can be reduced, thereby reducing the probability of some reliability-related problems caused by short circuit and improving the working reliability and stability of the battery cell.

[0186] The embodiments of the present application provide an electric device using the battery cell disclosed herein as a power source. The electric device may be, but is not limited to, a mobile phone, a tablet, a laptop computer, an electric toy, an electric tool, an electric bike, an electric vehicle, a ship, a spacecraft, or the like. The electric toy may include a stationary or mobile electric toy, such as a game console, an electric car toy, an electric ship toy, and an electric airplane toy, and the spacecraft may include an airplane, a rocket, a space shuttle, a spaceship, and the like.

[0187] For ease of explanation in the following embodiments, the structures of the electric device **1000**, the battery **100**, and the battery cell **10** of the present application are described by taking the electric device as a vehicle as an example.

[0188] Referring to FIG. **1**, FIG. **1** is a schematic structural diagram of an electric device **1000** according to some embodiments of the present application as a vehicle. The vehicle may be a fuel vehicle, a gas vehicle, or a new energy vehicle. The new energy vehicle may be a pure electric vehicle, a hybrid vehicle, an extended-range vehicle, or the like. A battery **100** is arranged in the vehicle, and the battery **100** may be arranged at the bottom, head, or tail of the vehicle. The battery **100** may be used for powering the vehicle. For example, the battery **100** may serve as an operation power source for the vehicle. The vehicle may further include a controller **200** and a motor **300**. The controller **200** is configured to control the battery **100** to power the motor **300**, e.g., for operation power needed by the vehicle for start-up, navigation, and driving. In some embodiments of the present application, the battery **100** may not only serve as the operation power source for the vehicle, but also as a driving power source for the vehicle to, instead of or in part instead of fuel or natural gas, provide driving power for the vehicle.

[0189] Referring to FIG. **2**, FIG. **2** illustrates an exploded view of a battery cell **10** used in the battery **100** according to some embodiments of the present application. The battery **100** includes a case **20** and a plurality of battery cells **10**, and the battery cells **10** are accommodated in the case **20**. The case **20** is configured to provide an assembly space for the battery cell **10**, and the case **20** may be in various structures. In some embodiments, the case **20** may include a first case **201** and a second case **202**. The first case **201** and the second case **202** are lidded with each other. The first case **201** and the second case **202** jointly define an assembly space for accommodating the battery cell **10**. The second case **202** may be of a hollow structure with one end open, the first case **201** may be of a plate structure, and the first case **201** lids the open side of the second case **202**, such that the first case **201** and the second case **202** jointly define an assembly space; or both the first case **201** and the second case **202** may be of hollow structures with one side open (for example, as shown in FIG. **2**), and the open side of the first case **201** lids the open side of the second case **202**. Certainly, the case **20** formed by the first case **201** and the second case **202** may be in various shapes, such as a cylinder or a rectangular parallelepiped.

[0190] In the battery **100**, a plurality of battery cells **10** may be connected in series, in parallel, or in series-parallel. The series-parallel connection means that both series connection and parallel connection are present for the connection among the plurality of battery cells **10**. The plurality of battery cells **10** can be directly connected in series, in parallel, or in a mixed connection, and then the whole formed by the plurality of battery cells **10** is accommodated in the case **20**. Certainly, the battery **100** may also be in a form where a plurality of battery cells **10** are first connected in series, in parallel, or in a mixed connection to give a battery module, and then a plurality of battery modules are connected in series, in parallel, or in a mixed connection to form a whole and accommodated in the case **20**. The battery

100 may further include other structures. For example, the battery **100** may further include a busbar component for achieving electrical connection between the plurality of battery cells **10**.

[0191] Referring to FIG. **3**, FIG. **3** illustrates a schematic view of a battery cell **10** according to some embodiments of the present application. The battery cell **10** is in the shape of a rectangular parallelepiped, and the height direction of the battery cell **10** is the first direction Z, the length direction of the battery cell **10** is the second direction X, and the thickness direction of the battery cell **10** is the third direction Y The first direction Z, the second direction X, and the third direction Y are perpendicular to each other, However, they are not limited to this. In other embodiments of the present application, the battery cell **10** may also be cylindrical, flat, or in other shapes.

[0192] Referring to FIGS. **4** and **5**, FIG. **4** is an orthographic projection view of a battery cell **10** according to some embodiments of the present application; FIG. **5** is a cross-sectional view along the line A-A in FIG. **4**. In the embodiments of the present application, the battery cell **10** includes a housing assembly **1** and a battery cell assembly **2**, and the housing assembly **1** includes a housing **11** and a first post terminal **12** disposed on the housing **11**.

[0193] The shape of the housing **11** is adjusted according to the type of the battery cell **10**, and the type of the battery cell **10** in the embodiments of the present application is not limited. For example, when the battery cell **10** is a square battery, the housing **11** is square, and when the battery cell **10** is a cylindrical battery, the housing **11** is cylindrical. The embodiments of the present application are all described by taking the housing **11** as a square as an example. The housing **11** is provided with post terminals for electrical connection to the battery cell assembly **2** to ensure normal charging and discharging operations of the battery cell **10**. Generally, the number of post terminals is at least two, specifically including at least one positive electrode post terminal and at least one negative electrode post terminal. For example, when the number of post terminals is two, one is a positive electrode post terminal and the other is a negative electrode post terminal, and the two are electrically connected to positive and negative output positions of the battery cell assembly **2**, separately. For another example, when the number of post terminals is four, two may be positive electrode post terminals and two may be negative electrode post terminals. In this case, the two positive electrode post terminals are both electrically connected to a positive output position of the battery cell assembly **2**, and the two negative electrode post terminals are both electrically connected to a negative output position of the battery cell assembly **2**.

[0194] Meanwhile, FIG. **6** is a schematic structural diagram of a battery cell **10** according to some embodiments of the present application; and FIG. **7** is an assembly diagram of a second post terminal **15**, a battery cell assembly **2**, and a housing **11** according to some embodiments of the present application. Referring to FIGS. **1** to **7**, in the embodiments of the present application, at least one of the plurality of post terminals is a first post terminal **12**, and the first post terminal **12** can serve as either a positive electrode post terminal or a negative electrode post terminal. The first post terminal **12** is a post terminal formed with an accommodating part **121**. The accommodating part **121** is a hollow structure with an accommodating space, and may be a groove-shaped structure, a hole-shaped structure, or a composite structure of the groove-shaped structure and the hole-shaped structure. That is, all the post terminals on the housing **11** may be the first post terminals **12** having the accommodating parts **121** formed therein, or a part of the post terminals on the housing **11** may be the first post terminals **12** having the accommodating parts **121** formed therein. When part of the post terminals on the housing **11** are the first post terminals **12** having the accommodating parts **121** formed therein, the remaining post terminals on the housing **11** are second post terminals **15** without the accommodating parts **121** (referring to FIGS. **6** and **7**).

[0195] Regardless of whether the post terminal on the housing **11** is the first post terminal **12** or the second post terminal **15**, the first post terminal **12** and the second post terminal **15** can be both electrically connected to the battery cell assembly **2** to ensure the normal charging and discharging processes of the battery cell **10**. Certainly, in other embodiments of the present application, the housing assembly **1** may be provided with only one post terminal, which is the first post terminal **12**. The first post terminal **12** includes two parts, which are connected in an insulating manner and serve separately as a positive electrode post terminal and a negative electrode post terminal. In order to simplify the description, the following description is mainly made with all the post terminals on the housing **11** being the first post terminals **12** which have the accommodating parts **121** formed therein as an example.

[0196] Referring to FIGS. **3** to **7** again, in the embodiments of the present application, the battery cell assembly **2** includes an active substance-coated part **21** and a conductive part **22**. The active substance-coated part **21** is accommodated in the housing **11**. The active substance-coated part **21** is a part of the battery cell assembly **2** coated with an active substance, and can assist in the deintercalation of metal ions during the charging and discharging processes of the battery cell **10**. The conductive part **22** is a metal structure that electrically connects the active substance-coated part **21** and the post terminal, and is not coated with the active substance. Both the first post terminal **12** and the second post terminal **15** can be electrically connected to the active substance-coated part **21** through the conductive part **22** to enable the charging and discharging operations of the battery cell **10**.

[0197] It should be noted that, in the embodiments of the present application, the active substance-coated part **21** is divided into a positive electrode active substance-coated part and a negative electrode active substance-coated part. The positive electrode active substance-coated part includes a part in which the positive electrode current collector is coated with a positive electrode active substance layer, and the negative electrode active substance-coated part includes a part in which the negative electrode current collector is coated with a negative electrode active substance layer. The conductive part **22** is divided into a positive electrode conductive part and a negative electrode conductive part. The positive electrode conductive part electrically connects the positive electrode active substance-coated part and the positive electrode post terminal, and the negative electrode conductive part electrically connects the negative electrode active substance-coated part and the negative electrode post terminal.

[0198] In the embodiments of the present application, referring to FIGS. **4** and **5** again, at least a part of the conductive part **22** is accommodated in the corresponding accommodating part **121**. Here, "at least a part" means that the conductive part **22** may be completely accommodated in the accommodating part **121**, or only a part of the conductive part **22** may be accommodated in the accommodating part **121**. Since the first post terminal **12** is provided with the accommodating part **121**, the hollow structure of the accommodating part **121** can reduce the weight of the post terminal **12** to a certain extent, so as to improve the gravimetric energy density of the battery cell **10** and the battery **100**.

[0199] Furthermore, the conductive part **22** is partially or completely accommodated in the accommodating part **121**, such that a part of the conductive part **22** located in the accommodating part **121** can occupy the space in the first post terminal **12**, which can thereby reduce the space occupied by the conductive part **22** in the housing **11**. When the dimension of the housing **11** is fixed, some space can be saved in the housing **11** to accommodate a larger active substance-coated part **21**, thereby improving the volumetric energy density of the battery cell **10**. For example, when the conductive part **22** is led out from a side of the active substance-coated part **21** proximal to the first post terminal **12**, the space occupied by the conductive part **22** between the active substance-coated part **21** and the first post terminal **12** can be saved, such that the dimension of the active substance-coated part **21** in the direction in which the conductive part **22** is led out can be increased, and the spacing between the active substance-coated part **21** and the first post terminal **12** can be reduced, thereby improving the energy density of the battery cell **10**.

[0200] In addition, by accommodating at least a part of the conductive part **22** in the accommodating part **121**, the space occupied by the battery cell **10** can be reduced, such that a battery **100** of the same volume can accommodate a larger number of battery cells **10**, thereby further improving the volumetric energy density of the battery **100**. In addition, by accommodating at least a part of the conductive part **22** in the accommodating part **121** to occupy a space in the first post terminal **12**, the redundancy of the conductive part **22** in the housing **11** can be reduced to at least a certain extent, the probability of short circuit between the conductive part **22** and the active substance-coated part **21** can be reduced, the probability of short circuit of the battery cell **10** can be reduced, and working reliability and stability of the battery cell **10** and the battery **100** can be improved. In addition, by accommodating at least a part of the conductive part **22** in the accommodating part **121**, it is also conducive to stabilizing and limiting the conductive part **22**, so as to improve the stability of the conductive part **22**, thereby facilitating welding of the conductive part **22** with the first post terminal **12** to improve the assembly efficiency.

[0201] In some optional embodiments of the present application, referring to FIGS. **3** to **5** again, each one of the plurality of post terminals on the housing assembly **1** is a first post terminal **12** with an accommodating part **121**. In this case, more conductive parts **22** can be accommodated in all the first post terminals **12** to better improve the volumetric energy density of the battery cell **10**.

[0202] In other optional embodiments of the present application, referring to FIGS. **6** and **7**, at least one of the plurality of post terminals on the housing assembly **1** is a first post terminal **12** with an accommodating part **121** and at least one is a second post terminal **15** without an accommodating part **121**, such that the first post terminal **12** and the second post terminal **15** can be flexibly selected and matched according to actual requirements such as energy density and costs, so as to improve the applicability of the battery cell **10**.

[0203] It should be noted that, referring to FIGS. **6** and **7**, when the housing **11** is provided with a second post terminal **15**, a clearance groove **18** may be provided between the second post terminal **15** and the housing **11** as required, such that at least a part of the conductive part **22** is received in the clearance groove **18**. As such, the space occupied by the conductive part **22** in the housing **11** can also be reduced to a certain extent, which is conducive to improving the energy density and mitigating the problems such as short circuit caused by the redundancy of the conductive part **22**.

[0204] In the embodiments of the present application, the accommodating part **121** may be located on the side of the first post terminal **12** proximal to the active substance-coated part **21**, or may be located on the side of the first post terminal **12** distal to the active substance-coated part **21**. Illustratively, FIG. **8** is a partial cross-sectional schematic diagram of the battery cell **10** according to some embodiments of the present application; FIG. **9** is a partial cross-sectional schematic diagram of the battery cell **10** according to some embodiments of the present application. Referring to FIGS. **8** and **9**, when the accommodating part **121** is located on a side of the first post terminal **12** facing the active substance-coated part **21**, the accommodating part **121** includes a first accommodating groove **12110**, and the surface of the first post terminal **12** on a side facing the active substance-coated part **21** is the post terminal inner end surface **122**. The opening of the first accommodating groove **12110** is formed on the post terminal inner end surface **122**, and at least a part of the conductive part **22** is accommodated in the first accommodating groove **12110**.

[0205] Illustratively, the first accommodating groove **12110** is a groove body, and the groove body is a groove-shaped structure having a certain depth. For example, when the first post terminal **12** is provided on the upper end wall of the housing **11** and the post terminal inner end surface **122** is the lower surface of the first post terminal **12**, the first accommodating groove **12110** is formed as an accommodating groove with an opening open downward and a groove wall recessed upward. For another example, when the first post terminal **12** is provided on the lower end wall of the housing **11** and the post terminal inner end surface **122** is the upper surface of the first post terminal **12**, the first accommodating groove **12110** is formed as an accommodating groove with an opening open upward and a groove wall recessed downward.

[0206] In the above technical solution, in one aspect, the provision of the first accommodating groove **12110** on the first post terminal **12** can reduce the weight of the first post terminal **12** to a certain extent, so as to improve the gravimetric energy density of the battery cell **10** and the battery **100**. In another aspect, as the opening of the first accommodating groove **12110** is formed on the post terminal inner end surface **122** and the post terminal inner end surface **122** is the surface of the first post terminal **12** on a side proximal to the active substance-coated part **21**, the first accommodating groove **12110** can be open toward the active substance-coated part **21**, thereby facilitating the conductive part **22** to extend into the first accommodating groove **12110** and improving the assembly efficiency. Moreover, the first accommodating groove **12110** in this type facilitates processing and thus improves production efficiency.

[0207] Furthermore, it is easy to process the first accommodating groove **12110** to have a relatively large volume, such that more conductive parts **22** can be accommodated. Meanwhile, since the first accommodating groove **12110** is open toward the active substance-coated part **21**, the first accommodating groove **12110** can also serve as a buffering and temporary storage structure for the electrolyte, such that more electrolyte can be accommodated in the housing **11**. As the electrolyte will be consumed during the charging and discharging processes of the battery cell **10**, a greater amount of electrolyte can prolong the service life of the battery cell **10**. Also, since the first accommodating groove **12110** is open toward the active substance-coated part **21**, the first accommodating groove **12110** can also serve as an accommodating and buffering structure for gases produced in the battery cell assembly **2**, reducing the expansion of the battery cell **10** and improving the reliability and stability of the battery cell **10**.

[0208] In addition, as the first accommodating groove **12110** is located on the inner side of the first post terminal **12**, external foreign matters and impurities can hardly enter the first accommodating groove **12110**, thereby reducing the impact of external foreign matters and impurities on the battery cell assembly **2**, improving the stability and reliability of the battery cell assembly **2**, and further improving the stability and reliability of the battery cell **10** and the battery **100**.

[0209] Referring to FIG. **8** again, in the embodiments of the present application, the method for connecting the first post terminal **12** to the housing **11** is not limited. For example, the method may be welding or riveting. For example, when the first post terminal and the housing are fitted by riveting, the housing **11** is provided with a mounting hole **113**, and the first post terminal **12** is riveted to and mounted at the mounting hole **113**. It will be appreciated that when the two are fitted by welding or other methods, the housing **11** may also be provided with a mounting hole **113** to allow the first post terminal **12** to be mounted on the housing **11** through the mounting hole **113**, which is not limited herein. Meanwhile, the first accommodating groove **12110** can be provided corresponding to the position of the mounting hole **113**, or in other words, on a projection plane perpendicular to the axial direction R of the first post terminal **12**, the orthographic projection of the first accommodating groove **12110** is located within the range of orthographic projection of the mounting hole **113**, such that the first accommodating groove **12110** can have a large depth to accommodate more conductive parts **22**, thereby reducing the space occupied by the conductive part **22** in the housing **11** to a greater extent.

[0210] Specifically, when the housing **11** is provided with the mounting hole **113** and the first post terminal **12** is mounted in the mounting hole **113**, in the axial direction R of the first post terminal **12**, the depth H1 of the first accommodating groove **12110** is greater than or equal to the minimum distance H2 from the post terminal inner end surface **122** to the mounting hole **113**.

[0211] It should be noted that the specific shape of the first accommodating groove **12110** is not limited, and may be a regular shape or an irregular shape, for example, a columnar groove of constant cross-section with a rectangular, elliptical, or track-shaped cross-section (a structure defined by two arcs and two straight lines in an enclosing manner), a trapezoidal groove with a rectangular cross-section and a gradually changing cross-sectional dimension, a hemispherical groove with a circular cross-section and a gradually changing cross-sectional dimension, a semi-ellipsoidal groove with an elliptical cross-section and a gradually changing cross-sectional dimension, or the like. Therefore, the depth H1 of the first accommodating groove **12110** refers to: the maximum depth of the first accommodating groove **12110** in the axial direction R of the first post terminal **12**.

[0212] Since in the axial direction R of the first post terminal **12**, the depth H1 of the first accommodating groove **12110** is greater than or equal to the minimum distance H2 from the post terminal inner end surface **122** to the mounting hole **113**, a volume of the first post terminal **12** can be fully utilized, such that the first accommodating groove **12110** has a larger depth, which is conducive to accommodating more conductive parts **22**, and then can reduce the space occupied by the conductive parts **22** in the housing **11** to a greater extent, further improve the energy density of the battery cell **10**, and further reduce the redundancy of the conductive part **22** in the housing **11**. Meanwhile, since the first accommodating groove **12110** has a larger depth, gases produced by the battery cell assembly **2** can also be accommodated, which ensures the reliability and stability of the battery cell **10**, and can also accommodate a larger amount of electrolyte to ensure the service life of the battery cell **10**.

[0213] It should be further noted that the volume of the first accommodating groove **12110** is not limited. For example, in some specific examples, the volume of the first accommodating groove **12110** that can be used for accommodating the conductive part **22** (referred to as a first volume V1) may be greater than or equal to 298 mm^3, such that the first accommodating groove **12110** may have sufficient space to accommodate the conductive part **22** and facilitate welding of the conductive part **22** to the first post terminal **12**. However, when the first volume V1 of the first accommodating groove **12110** is less than 298 mm^3, the accommodating capacity of the first accommodating groove **12110** for the conductive part **22** is relatively weakened, and the welding difficulty of the conductive part **22** with the first post terminal **12** is increased.

[0214] In addition, it is worth noting that the first volume V1 of the first accommodating groove **12110** is: the difference between the total volume V2 of the first accommodating groove **12110** and the volume of other components required to be accommodated by the first accommodating groove **12110** except the conductive part **22** (referred to as a second volume V3), that is, $V1=V2-V3$. It can be understood that when the first accommodating groove **12110** does not need to accommodate other components except the conductive part **22**, the second volume V3 may be 0 mm^3. For example, the first volume V1 of the first accommodating groove **12110** may be 300 mm^3-1500 mm^3, for example, 300 mm^3, 400 mm^3, 500 mm^3, 600 mm^3, 700 mm^3, 800 mm^3, 1000 mm^3, 1200 mm^3, 1400 mm^3, or 1500 mm^3.

[0215] Referring to FIGS. **8** and **9** again, in order to ensure the stability and reliability of the electrical connection between the active substance-coated part **21** and the first post terminal **12**, in the embodiments of the present application, the electrical connection position of the conductive part **22** and the first post terminal **12** may be located on the groove wall of the first accommodating groove **12110**. Illustratively, the conductive part **22** and the first post terminal **12** may be electrically connected by welding, and the electrical connection position is the welding position of the conductive part **22** and the first post terminal **12**. Meanwhile, the welding method for the conductive part **22** and the first post terminal **12** is not limited, which may be laser welding, for example. Moreover, depending on factors such as the position, angle, or structure of the welding part, vertical welding or inclined welding, and overlap welding or edge sealing welding can be selected. In other embodiments of the present application, the conductive part **22** and the first post terminal **12** may be electrically connected by other means instead of welding, such as by providing conductive adhesive or conductive nails. In order to simplify the description, the following description will take an example that the conductive part **22** is welded to the first post terminal **12** to form an electrical connection, and that the welding position is the electrical connection position of the conductive part **22** and the first post terminal **12**.

[0216] Specifically, the first post terminal **12** includes a first end wall **12111** and a first side wall **12113**, the first end wall **12111** is located on a side of the first side wall **12113** distal to the active substance-coated part **21**, the first end wall **12111** and the first side wall **12113** defines, in an enclosing manner, the first accommodating groove **12110**, and the electrical connection position of the conductive part **22** and the first post terminal **12** is located on the first end wall **12111** and/or the first side wall **12113**. That is, the conductive part **22** may be welded to at least one of the first end wall **12111** and the first side wall **12113**.

[0217] In the above technical solution, by arranging the electrical connection position of the conductive part **22** and the first post terminal **12** on at least one of the first end wall **12111** and the first side wall **12113**, not only does the first accommodating groove **12110** have a function of accommodating at least a part of the conductive part **22**, but also the wall of the first accommodating groove **12110** has a function of achieving an electrical connection to the conductive part **22**, which can simplify the structure of the first post terminal **12**, facilitate processing of the first post terminal **12**, simplify the structure of the conductive part **22**, reduce the redundancy of the conductive part **22**, and reduce the costs of the conductive part **22**. Moreover, utilizing the groove wall of the first accommodating groove **12110** to achieve the electrical connection to the conductive part **22** may allow a relatively large region of the electrical connection between the conductive part **22** and the first post terminal **12**, which not only reduces the difficulty of electrical connection, but also improves the reliability and stability of the electrical connection, thereby improving the performance of the battery cell **10**.

[0218] In addition, since the electrical connection position of the conductive part **22** and the first post terminal **12** is located in the first accommodating groove **12110**, not only can the electrical connection position be prevented from protruding out of the first post terminal **12** and occupying the space outside the first post terminal **12**, but also the electrical connection position can be protected by the first post terminal **12**, thereby improving the reliability and stability of the electrical connection between the conductive part **22** and the first post terminal **12**.

[0219] In addition, in the embodiments of the present application, the first end wall **12111** is configured as a closed structure without any perforations, such that the first accommodating groove **12110** is isolated from the space outside the housing **11**, thereby avoiding the problem of the electrolyte in the housing **11** leaking from the first accommodating groove **12110**.

[0220] Referring to FIGS. **8** and **9** again, in the embodiments of the present application, a local shape of the conductive part **22** matches a local shape of the first end wall **12111**, and the conductive part and the first end wall are arranged in a fitting manner and electrically connected, such that the electrical connection position of the conductive part **22** and the first end wall **12111** extends in the length or width direction of the first end wall **12111**. For example, when the first end wall **12111** is flat, a part of the conductive part **22** may also be flat and fitted to the first end wall **12111**, and the fitted position is electrically connected, such as welded. As such, the area of the electrical connection can be increased, and the reliability and stability of the electrical connection can be improved.

[0221] In addition, when the electrical connection between the conductive part **22** and the first end wall **12111** is welding, since the first end wall **12111** is located on a side of the first accommodating groove **12110** distal to the active substance-coated part **21**, a welding operation is convenient. For example, welding may be performed from a side of the first post terminal **12** distal to the active substance-coated part **21**.

[0222] It is worth noting that the shape of the first end wall **12111** is not limited. For example, the first end wall may be in a flat-plate shape, an arc-plate shape or the like. When the first end wall **12111** is of a flat plate structure, the first end wall **12111** is provided at an included angle to an axial direction R of the first post terminal **12**. For example, the first end wall may be of a flat plate structure perpendicular to the axial direction R of the first post terminal **12**, or an inclined plate structure that is not perpendicular to the axial direction R of the first post terminal **12**, but the inclination direction is not limited.

[0223] Certainly, in other embodiments of the present application, the electrical connection position of the conductive part **22** and the first end wall **12111** may not extend in the length or width direction of the first end wall **12111**, and may be, for example, a plurality of discretely arranged spots. For example, the conductive part **22** is provided with a plurality of parts that are spaced apart from each other and separately welded to the first end wall **12111**, which will not be described in detail herein.

[0224] FIG. **10** is a partial cross-sectional schematic diagram of a battery cell **10** according to some embodiments of the present application. Referring to FIG. **10**, in the embodiments of the present application, when the conductive part **22** is electrically connected to the first end wall **12111**, a first recess **12112** may be formed on the first end wall **12111**, and the sinking direction of the first recess **12112** is the direction away from the active substance-coated part **21**. At least a part of the electrical connection position between the conductive part **22** and the first end wall **12111** is located in the first recess **12112**. Illustratively, at least a part of the conductive part **22** may be arranged within the first recess **12112** and connected to a portion of the first end wall **12111** configured to define the first recess **12112**.

[0225] In the above technical solution, in one aspect, the first recess **12112** can be used to realize pre-positioning and limiting of the electrical connection position of the conductive part **22**, which is not only conducive to accurately finding the position to achieve the electrical connection and improve the production efficiency, but also conducive to improving the stability and reliability of the conductive part **22** and ensuring the stability and reliability of the battery cell **10** during the charging and discharging processes. In another aspect, by arranging the first recess **12112** on the first end wall **12111**, the local wall thickness of the first end wall **12111** can be locally reduced, which is not only conducive to welding, but also conducive to reducing the weight of the first post terminal **12** and improving the gravimetric energy density of the battery cell **10**.

[0226] In some optional embodiments, a part of the conductive part **22** matches the first side wall **12113** in shape and is provided in a fitting manner. For example, when the first side wall **12113** is a curved surface, a part of the conductive part **22** may also be a curved surface and fitted to the first side wall **12113**, and the fitted position may be electrically connected (e.g., welded), such that the electrical connection position of the conductive part **22** and the first side wall **12113** extends along the first side wall **12113**. As such, the area of the electrical connection can be increased, thereby improving the reliability and stability of the electrical connection.

[0227] Certainly, in other embodiments of the present application, the electrical connection position of the conductive part **22** and the first side wall **12113** may not extend along the first side wall **12113**, and may be, for example, a plurality of discretely arranged spots. For example, the conductive part **22** is provided with a plurality of parts that are spaced apart from each other and separately welded to the first side wall **12113**, which will not be described in detail herein.

[0228] It should be noted that the number of first side walls **12113** is not limited and may be determined according to the shape of the first accommodating groove **12110**, as long as one end of each first side wall **12113** distal to the opening of the first accommodating groove **12110** is connected to the first end wall **12111**. Illustratively, when the cross-sectional shape of the first accommodating groove **12110** is circular or elliptical, the first end wall **12111** is circular or elliptical. The number of the first side wall **12113** is one, and the first side wall is annular and provided around the circumferential edge of the first end wall **12111**. Further illustratively, when the cross-sectional shape of the first accommodating groove **12110** is rectangular or track-shaped, the first end wall **12111** is rectangular or track-shaped. The number of the first side walls **12113** is four, and the first side walls are separately connected to four sides of the first end wall **12111**.

[0229] It should be further noted that the first accommodating groove **12110** is not limited to the form defined by the first end wall **12111** and the first side wall **12113**. For example, in some embodiments, the first end wall **12111** may not exist. In this case, one end of each first side wall **12113** distal to the opening of the first accommodating groove **12110** converges together, such that the first accommodating groove **12110** is defined only by a plurality of first side walls **12113**. In this case, the conductive part **22** may be electrically connected to the first side wall **12113** to ensure normal charging and discharging processes of the battery cell **10**.

[0230] In addition, it is worth noting that in other embodiments of the present application, the electrical connection position of the conductive part **22** and the first post terminal **12** may not be located in the first accommodating groove **12110**. For example, the electrical connection position of the conductive part **22** and the first post terminal **12** may also be located on the post terminal inner end surface **122**. In this case, a part of the conductive part **22** is accommodated in the first accommodating groove **12110**, which can also save space to a certain extent to improve the energy density of the battery cell **10**.

[0231] Referring to FIGS. **9** and **10** again, in the embodiments of the present application, the first post terminal **12** may also be provided with a first groove **126** as required. The first groove **126** is located on a side of the first post terminal **12** distal to the active substance-coated part **21**, that is, the surface of the side of the first post terminal **12** distal to the active substance-coated part **21** is the post terminal outer end surface **123**, and the groove opening of the first groove **126** is formed on the post terminal outer end surface **123**.

[0232] It can be understood that the first groove **126** is a groove body, and the groove body is a groove-shaped structure having a certain depth. Furthermore, when the first post terminal **12** is provided on an upper end wall of the housing **11** and the post terminal outer end surface **123** is an upper surface of the first post terminal **12**, the first groove **126** is formed as a groove with an opening open upward and a wall recessed downward (that is, recessed toward a square proximal to the battery cell assembly **2**). For another example, when the first post terminal **12** is provided on a lower end wall of the housing **11** and the post terminal outer end surface **123** is a lower surface of the first post terminal **12**, the first groove **126** is formed as a groove with an opening downward and a wall recessed upward (that is, recessed toward a square proximal to the battery cell assembly **2**).

[0233] In the above technical solution, in one aspect, since the first post terminal **12** is provided with the first groove **126**, the weight of the first post terminal **12** can be further reduced, such that the gravimetric energy density of the battery cell **10** and the battery **100** can be improved. In another aspect, the first groove **126** is located on the outer side of the first post terminal **12**, that is, the first groove is open to a side of the first post terminal **12** distal to the interior of the housing **11**, and the first groove **126** can be used for accommodating or mounting structural components of the battery **100** that electrically connect various battery cells **10**, so as to make full use of the space in the first post terminal **12**, thereby improving the space utilization and volumetric energy density of the battery **100**.

[0234] In addition, since the first post terminal **12** is provided with both the first accommodating groove **12110** and the first groove **126**, the first groove **126** is located on a side of the first accommodating groove **12110** distal to the active substance-coated part **21**, and the first groove **126** is open in a direction away from the first accommodating groove **12110**, the conductive part **22** can be laser-welded to the first end wall **12111** through the first groove **126** from the outside of the first post terminal **12**, that is, a side of the first post terminal **12** distal to the active substance-coated part **21**, so as to facilitate the electrical connection between the conductive part **22** and the first post terminal **12** through external welding. That is, by the above structural arrangement, external welding of the first post terminal **12** to the conductive part **22** through the first groove **126** can be facilitated, which facilitates processing and manufacture of the battery cell **10** and can save processing and manufacturing costs.

[0235] Further, in order to conveniently and effectively weld the conductive part **22** to the groove wall of the first accommodating groove **12110** through the first groove **126** and improve the welding reliability of the conductive part **22** and the groove wall of the first accommodating groove **12110**, in the embodiments of the present application, the part between the first groove **126** and the first accommodating groove **12110** can be laser-welded to the conductive part **22**, that is, a spacer part **127** shown in FIG. **10** can be laser-welded to the conductive part **22** to achieve an electrical connection between the battery cell assembly **2** and the first post terminal **12**. The spacer part **127** of the first post terminal **12** located between the first groove **126** and the first accommodating groove **12110** is thin. The spacer part **127** isolates the first groove **126** from the first accommodating groove **12110**. A wall surface of the spacer part **127** on a side proximal to the active substance-coated part **21** can be used as the first end wall **12111**. When the conductive part **22** needs to be welded to the first end wall **12111**, since the spacer part **127** is thin, the welding of the conductive part **22** to the first end wall **12111** can be achieved through the first groove **126**, thereby improving the convenience and reliability of welding.

[0236] In some embodiments, the first accommodating groove **12110** may be configured into a shape in which the length of the cross-section is greater than the width, such as a rectangle, an ellipse, or a track shape. The weld mark formed by welding the conductive part **22** to the first post terminal **12** may be a long strip-shaped weld mark parallel to the length direction of the first accommodating groove **12110**, so as to improve the welding reliability and increase the current passage performance. Illustratively, when the conductive part **22** is welded to the first end wall **12111** to form a weld mark as a long strip-shaped weld mark, the width of the weld mark may be greater than or equal to 6 mm, and the distance between the weld mark and the first side wall **12113** may be greater than or equal to 1 mm, so as to ensure the current passage capacity of the battery cell **10** while ensuring the convenience and reliability of welding.

[0237] Referring to FIG. **9** again, further, the housing assembly **1** may further include a groove cover **7**. The groove cover **7** is disposed on the first post terminal **12** and closes the groove opening of the first groove **126**.

[0238] In the above technical solution, by providing the groove cover **7** that closes the first groove **126**, the first post terminal **12** can be indirectly electrically connected to the busbar component through the groove cover **7**. The electrical connection between the groove cover **7** and the busbar component can be more convenient and the electrical connection area can be larger through the arrangement of the position and structure of the groove cover **7**. As such, arranging the groove cover **7** can facilitate the electrical connection between adjacent battery cells **10** in the battery **100**, and since the electrical connection positions of the battery cells **10** are located at the groove cover **7**, which can be separated from the electrical connection position of the conductive part **22** and the first post terminal **12** by the first groove **126**, there is less interference between the two, which can further improve the stability and reliability of the battery cell **10**.

[0239] It should be noted that, based on the solution that the accommodating part **121** includes the first accommodating groove **12110**, the specific structure of the battery cell assembly **2** in the embodiments of the present application is not limited. For example, the battery cell assembly may include but is not limited to the following two embodiments.

[0240] FIG. **11** is a partial cross-sectional schematic diagram of the battery cell assembly **2** according to some embodiments of the present application. Referring to FIGS. **9-11**, in a first embodiment, the active substance-coated part **21** includes a current collector **211** and an active substance layer **212** arranged on the current collector **211**. The conductive part **22** includes a tab part **221** electrically connected to the current collector **211**. The tab part **221** includes a plurality of tab plates **2211**. The tab plates **2211** are structures that are electrically connected to the current collector **211** but not coated with active substances, and may be formed by direct die cutting of the current collector **211**. Parts of the plurality of tab plates **2211** proximal to the current collector **211** converge (that is, get together in a direction towards each other) to form a first gathering part **2212**. Parts of the plurality of tab plates **2211** distal to the current collector **211** converge and are connected to form a second gathering part **2213**. The first gathering part **2212** connects the second gathering part **2213** and the active substance-coated part **21**. When the accommodating part **121** is provided with the first accommodating groove **12110**, at least a part of the second gathering part **2213** may be accommodated in the first accommodating groove **12110**.

[0241] In the above technical solution, the plurality of tab plates **2211** only converge (that is, get together in a direction towards each other) and gather but are not connected when forming the first gathering part **2212**, while the plurality of tab plates **2211** not only converge and gather but are also connected into an integrated structure when forming the second gathering part **2213**. For example, the plurality of tab plates **2211** may be connected into an integrated plate structure by welding (such as ultrasonic welding) to form the second gathering part **2213**. For example, the plurality of tab plates **2211** may also converge and be connected to form the second gathering part **2213** by conductive adhesive bonding, etc., which will not be described in detail herein.

[0242] It should be noted that in the embodiments of the present application, the tab plates **2211** are divided into positive electrode tab plates **2211** and negative electrode tab plates **2211**. The positive electrode tab plates **2211** that need to gather together are stacked together and ultrasonically pre-welded to form a second gathering part **2213** of the positive electrode, which can reduce the gap between layers and enable the plurality of fluffy positive electrode tab plates **2211** to form a plate structure of a certain rigidity. Similarly, the negative electrode tab plates **2211** that need to gather together are stacked together and ultrasonically pre-welded to form a second gathering part **2213** of the negative electrode, which can reduce the gap between layers and enable the plurality of fluffy negative electrode tab plates **2211** to form a plate structure of a certain rigidity.

[0243] In the above technical solution, "parts of the plurality of tab plates **2211** proximal to the current collector **211** converge to form a first gathering part **2212**, and parts of the plurality of tab plates **2211** distal to the current collector **211** converge and are connected to form a second gathering part **2213**" is intended to illustrate that: in the extension direction of the tab plates **2211**, the first gathering part **2212** and the second gathering part **2213** are sequentially provided in a direction away from the current collector **211**, and the specific positions of the first gathering part **2212** and the second gathering part **2213** are not limited, that is, it is not required how close the first gathering part **2212** is to the current collector **211**, or how far the second gathering part **2213** is from the current collector **211**. In some optional examples, the current collector **211** and the tab plates **2211** may be an integrated member, such as an integrally formed aluminum foil for the positive electrode plates, and an integrally formed copper foil for the negative electrode plates.

[0244] In the above technical solution, since the tab part **221** includes the second gathering part **2213** formed by converging and connecting the plurality of tab plates **2211**, at least a part of the second gathering part **2213** is accommodated in the first accommodating groove **12110**, which facilitates the connection between the conductive part **22** and the first post terminal **12**, and can make full use of the space of the first post terminal **12**, thereby improving the volumetric energy density of the battery cell **10**.

[0245] Referring to FIGS. **10-11**, in the first embodiment, at least a part of the first gathering part **2212** is accommodated in the first accommodating groove **12110**. In the above technical solution, at least a part of the first gathering part **2212** and at least a part of the second gathering part **2213** of the tab part **221** are both accommodated in the first accommodating groove **12110**, such that the space in the first post terminal **12** can be more fully utilized, and the space occupied by the tab part **221** in the housing **11** can be further reduced, so as to accommodate a larger active substance-coated part **21**, thereby improving the volumetric energy density of the battery cell **10**. Moreover, the redundancy of the tab part **221** in the housing **11** can be better reduced, and the probability of short circuit between the tab part **221** and the active substance-coated part **21** can be further reduced.

[0246] In this embodiment, the second gathering part **2213** is electrically connected to the first post terminal **12** directly or indirectly. For example, referring to FIG. **9**, when the second gathering part **2213** is electrically connected to the first post terminal **12** directly, such as when the second gathering part **2213** is welded (such as laser-welded) to the first post terminal **12**, the structure of the battery cell assembly **2** can be simplified, thereby reducing the number of components, simplifying the assembly process, and improving the assembly efficiency. The method and position of direct electrical connection between the second gathering part **2213** and the first post terminal **12** are not limited. For example, the electrical connection position of the second gathering part **2213** and the first post terminal **12** may be located at the first end wall **12111** and/or the first side wall **12113**. Further, the electrical connection position of the second gathering part **2213** and the first end wall **12111** may extend in the length direction or the width direction of the first end wall **12111**. Yet further, the first end wall **12111** is provided with the first recess **12112**, and the electrical connection position of the second gathering part **2213** and the first end wall **12111** may be located in the first recess **12112**, and the like. The corresponding technical effects can refer to the introduction of the above embodiments and will not be described in detail herein.

[0247] As an optional solution, the conductive part **22** may also be provided with an adapting piece **222** as required. In this case, the second gathering part **2213** is electrically connected to the first post terminal **12** indirectly. Specifically, referring to FIG. **10**, when the conductive part **22** includes the adapting piece **222**, the adapting piece **222** is connected to the second gathering part **2213**, and the conductive part **22** is electrically connected to the first post terminal **12** via the adapting piece **222**. In this case, at least a part of the adapting piece **222** is accommodated in the first accommodating groove **12110**. In this example, at least a part of the second gathering part **2213** is also accommodated in the first accommodating groove **12110**, but the first gathering part **2212** may be accommodated in the first accommodating groove **12110** or may not be accommodated in the first accommodating groove **12110**.

[0248] In the above technical solution, the active substance-coated part **21** can be electrically connected to the first post terminal **12** through the first gathering part **2212**, the second gathering part **2213**, and the adapting piece **222** in sequence. The electrical connection position of the conductive part **22** and the first post terminal **12** is located on the adapting piece **222**. For example, the electrical connection can be achieved by welding (for example, laser welding) the adapting piece **222** to the first post terminal **12**. In addition, the adapting piece **222** and the tab plate **2211** are two separate components and are connected by welding (such as ultrasonic welding) or the like.

[0249] In the above technical solution, in one aspect, by accommodating at least a part of the second gathering part **2213** and at least a part of the adapting piece **222** in the first accommodating groove **12110**, the space in the first post terminal **12** can be more fully utilized, and the space occupied by the conductive part **22** in the housing **11** can be further reduced, so as to improve the volumetric energy density of the battery cell **10**. When at least a part of the first gathering part **2212**, at least a part of the second gathering part **2213**, and at least a part of the adapting piece **222** are all accommodated in the first accommodating groove **12110**, the space in the first post terminal **12** can be further fully utilized, and the space occupied by the conductive part **22** in the housing **11** can be more effectively reduced, so as to further improve the volumetric energy density of the battery cell **10**.

[0250] In another aspect, by using the adapting piece **222** to achieve an indirect electrical connection between the second gathering part **2213** and the first post terminal **12**, the adapting piece **222** can be welded to the first post terminal **12** at a part avoiding the second gathering part **2213**, such that the welding between the adapting piece **222** and the first post terminal **12** is more secure, the risk of welding cracking is low, and the reliability and stability of the battery cell **10** can be further improved. Meanwhile, by electrically connecting the first post terminal **12** and the tab plates **2211** via the adapting piece **222**, the configuration of the tab plates **2211** can also be simplified.

[0251] The method and position of direct electrical connection between the adapting piece **222** and the first post terminal **12** are not limited. For example, the adapting piece **222** is electrically connected to the first post terminal **12** by welding. For example, the electrical connection position of the adapting piece **222** and the first post terminal **12** may be located at the first end wall **12111** and/or the first side wall **12113**. Further, the electrical connection position of the adapting piece **222** and the first end wall **12111** may extend in the length direction or the width direction of the first end wall **12111**. Yet further, the first end wall **12111** is provided with the first recess **12112**, and the electrical connection position of the adapting piece **222** and the first end wall **12111** may be located in the first recess **12112**, and the like. The corresponding technical effects can refer to the introduction of the above embodiments and will not be described in detail herein. When the electrical connection position of the adapting piece **222** and the first post terminal **12** is located at the first end wall **12111** and/or the first side wall **12113**, since at least a part of the adapting piece **222** is accommodated in the first accommodating groove **12110**, the structure of the adapting piece **222** can be simplified, thereby reducing the redundancy and lowering the costs.

[0252] Referring to FIGS. **10-11**, in a second embodiment, the active substance-coated part **21** includes a current collector **211** and an active substance layer **212** arranged on the current collector **211**. The conductive part **22** includes a tab part **221** and an adapting piece **222**. The tab part **221** includes a plurality of tab plates **2211** electrically connected to the current collector **211**. Parts of the plurality of tab plates **2211** proximal to the current collector **211** converge to form a first gathering part **2212**. Parts of the plurality of tab plates **2211** distal to the current collector **211** converge and are connected to form a second gathering part **2213**. The adapting piece **222** is electrically connected to the second gathering part **2213**. When the accommodating part **121** is provided with the first accommodating groove **12110**, at least a part of the adapting piece **222** may be accommodated in the first accommodating groove **12110** and electrically connected to the first post terminal **12**.

[0253] In the above technical solution, compared with the solution including the adapting piece **222** in the first embodiment, in the second embodiment, at least a part of the adapting piece **222** is accommodated in the first accommodating groove **12110**, but the relative positions of the tab part **221** and the first accommodating groove **12110** are not limited, that is, at least a part of the tab part **221** may be accommodated in the first accommodating groove **12110**, and the tab part **221** may also be completely located outside the first accommodating groove **12110**, thereby meeting different structural design requirements.

[0254] In the above technical solution, by accommodating at least a part of the adapting piece **222** in the first accommodating groove **12110**, the adapting piece **222** can occupy the space in the first post terminal **12**, such that the space occupied by the adapting piece **222** in the housing **11** can be reduced, so as to accommodate a larger active substance-coated part **21**, thereby improving the volumetric energy density of the battery cell **10**. Moreover, the probability of short circuit between the adapting piece **222** and the active substance-coated part **21** can be reduced, and the risk of short circuit of the battery cell assembly **2** can be reduced, thereby improving the stability and reliability of the battery cell **10**.

[0255] In addition, by using the adapting piece **222** to achieve an indirect electrical connection between the second gathering part **2213** and the first post terminal **12**, the adapting piece **222** can be welded to the first post terminal **12** at a part avoiding the second gathering part **2213**, such that the welding between the adapting piece **222** and the first post terminal **12** is secure, the risk of welding cracking is low, and the reliability and stability of the battery cell **10** can be further improved. Meanwhile, by electrically connecting the first post terminal **12** and the tab plates **2211** through the adapting piece **222**, the configuration of the tab plates **2211** can also be simplified.

[0256] Illustratively, in some optional embodiments, for example, in the first embodiment above or a third embodiment below, when the second gathering part **2213** is electrically connected to the first post terminal **12** directly, the conductive part **22** may be composed only of the positive electrode tab and the negative electrode tab in each electrode assembly **2***a*. For example, in other embodiments, for example, in the first or second embodiment above or a third or fourth embodiment below, when the second gathering part **2213** is electrically connected to the first post terminal **12** indirectly through the adapting piece **222**, the conductive part **22** may be composed simultaneously of the positive electrode tab and the negative electrode tab in each electrode assembly **2***a* and each adapting piece **222**.

[0257] FIG. **12** is a diagram of various tab gathering solutions of the battery cell assembly **2** according to some embodiments of the present application. Referring to FIGS. **11** and **12**, in some embodiments, when the battery cell assembly **2** includes two electrode assemblies **2***a*, the tab plates **2211** of the two electrode assemblies **2***a* may be gathered together, and the gathering position is located at the center position between the two electrode assemblies **2***a* to form a symmetrical gathering form (for example, as shown in FIGS. **11** and **12**(*a*)). Alternatively, in other embodiments, when the tab plates **2211** of the two electrode assemblies **2***a* are gathered together, the gathering position may also be closer to one of the electrode assemblies **2***a* to form an asymmetrical gathering form (for example, as shown in FIGS. **12**(*b*) and **12**(*c*)). In addition, the electrode assembly **2***a* may be in a full-tab form (for example, as shown in FIG. **12**(*a*)) or a half-tab form (for example, as shown in FIGS. **11**, **12**(*b*), and **12**(*c*)).

[0258] Certainly, the tab plates **2211** of the same polarity of the two electrode assemblies **2***a* may not be gathered together. For example, the tab plates **2211** of each electrode assembly **2***a* are gathered individually according to the positive and negative electrodes, that is, the positive electrode tabs of one electrode assembly **2***a* are gathered individually, and the positive electrode tabs of another electrode assembly **2***a* are also gathered individually, which will not be described in detail herein.

[0259] It should be noted that the accommodating part **121** in the embodiments of the present application is not limited to having the form of the first accommodating groove **12110**. For example, other optional embodiments will be given later.

[0260] Illustratively, FIG. **13** is a partial cross-sectional schematic diagram of the battery cell **10** according to some embodiments of the present application. Referring to FIG. **13**, in the embodiments of the present application, the accommodating part **121** may also be arranged to include a second accommodating groove **12120**. A surface of the first post terminal **12** on a side distal to the active substance-coated part **21** is the post terminal outer end surface **123**. An opening of the second accommodating groove **12120** is formed on the post terminal outer end surface **123**. The second accommodating groove **12120** is in communication with the interior of the housing **11** through a first perforation **12130**. The conductive part **22** is provided in the first perforation **12130** in a penetrating manner and is at least partially accommodated in the second accommodating groove **12120**.

[0261] It can be understood that the second accommodating groove **12120** is a groove body, and the groove body is a groove-shaped structure of a certain depth. For example, when the first post terminal **12** is provided on the upper end wall of the housing **11** and the post terminal outer end surface **123** is the upper surface of the first post terminal **12**, the second accommodating groove **12120** is formed as an accommodating groove with an opening open upward and a groove wall recessed downward. For another example, when the first post terminal **12** is provided on the lower end wall of the housing **11** and the post terminal outer end surface **123** is the lower surface of the first post terminal **12**, the second accommodating groove **12120** is formed as an accommodating groove with an opening open downward and a groove wall recessed upward.

[0262] In the above technical solution, referring to FIG. **13**, in one aspect, the first post terminal **12** is provided with the second accommodating groove **12120**, which can reduce the weight of the first post terminal **12** to a certain extent, so as to improve the gravimetric energy density of the battery cell **10** and the battery **100**. In another aspect, since the opening of the second accommodating groove **12120** is formed on the post terminal outer end surface **123** and the post terminal outer end surface **123** is a surface of the first post terminal **12** on a side distal to the active substance-coated part **21**, the second accommodating groove **12120** can be open in a direction away from the active substance-coated part **21**. In this way, when at least a part of the conductive part **22** is accommodated in the second accommodating groove **12120**, the accommodation and arrangement of the conductive part **22** can be easily achieved through the opening of the second accommodating groove **12120**, and the electrical connection operation between the conductive part **22** and the first post terminal **12** can be easily achieved through the opening of the second accommodating groove **12120**, etc., thereby reducing the difficulty in producing the battery cell **10** and improving the production efficiency of the battery cell **10**.

[0263] Meanwhile, since the second accommodating groove **12120** can be in communication with the interior of the housing **11** through the first perforation **12130**, the second accommodating groove **12120** can also serve as a buffering and temporary storage structure for the electrolyte, such that a larger number of electrolyte can be accommodated in the housing **11**. As the electrolyte will be consumed during the charging and discharging processes of the battery cell **10**, a greater amount of electrolyte can prolong the service life of the battery cell **10**. It is also due to the fact that the second accommodating groove **12120** can be in communication with the interior of the housing **11** through the first perforation **12130**, that the second accommodating groove **12120** can also serve as an accommodating and buffering structure for gas generated inside the battery cell assembly **2**, reducing the expansion of the battery cell **10** and improving the reliability and stability of the battery cell **10**.

[0264] It is worth noting that when the accommodating part **121** is provided with the second accommodating groove **12120** and the conductive part **22** is provided in the first perforation **12130** in a penetrating manner and is at least partially accommodated in the second accommodating groove **12120**, and the electrical connection position of the conductive part **22** and the first post terminal **12** is not limited.

[0265] Illustratively, when the conductive part **22** is provided in the first perforation **12130** in a penetrating manner and is at least partially accommodated in the second accommodating groove **12120**, in the embodiments of the present application, the electrical connection position between the conductive part **22** and the first post terminal **12** is located on the wall of the first perforation **12130** formed on the first post terminal **12**.

[0266] In the above technical solution, the arrangement of the electrical connection position of the conductive part **22** and the first post terminal **12** on the wall of the first perforation **12130** facilitates the electrical connection operation between the conductive part **22** and the first post terminal **12** through the second accommodating groove **12120**. In addition, when the electrical connection area between the conductive part **22** and the first post terminal **12** is large, the sealing of the first perforation **12130** can be achieved by utilizing the electrical connection between the conductive part **22** and the first post terminal **12**, so as to save sealing costs, reduce electrolyte leakage, and save sealing parts.

[0267] Specifically, the conductive part **22** may be welded to the wall of the first perforation **12130** at a position of the first perforation **12130** connected to the second accommodating groove **12120**, so as to facilitate the operation. In addition, by controlling a weld mark, the first perforation **12130** can be sealed by utilizing the weld mark and the conductive part **22**, so as to mitigate the problem of the electrolyte in the housing **11** leaking from the first perforation **12130**.

[0268] Further illustratively, when the conductive part **22** is provided in the first perforation **12130** in a penetrating manner and is at least partially accommodated in the second accommodating groove **12120**, in other embodiments of the present application, the electrical connection position of the conductive part **22** and the first post terminal **12** may also be located on the groove wall of the second accommodating groove **12120** formed on the first post terminal **12**. As such, the electrical connection operation is facilitated. For example, when the conductive part **22** is welded to the groove wall of the second accommodating groove **12120** formed on the first post terminal **12**, the conductive particles generated by welding can be prevented from entering the housing **11**, which may otherwise cause short circuit and other problems.

[0269] Specifically, FIG. **14** is a partial cross-sectional schematic diagram of the battery cell **10** according to some embodiments of the present application. Referring to FIGS. **13** and **14**, the first post terminal **12** includes a second end wall **12121** and a second side wall **12123**. The second end wall **12121** is located on a side of the second side wall **12123** proximal to the active substance-coated part **21**. The second end wall **12121** and the second side wall **12123** define, in an enclosing manner, a second accommodating groove **12120**. The first perforation **12130** is formed on the second end wall **12121**. The electrical connection position of the conductive part **22** and the first post terminal **12** is located on the second end wall **12121** and/or on the second side wall **12123**.

[0270] More specifically, the conductive part **22** and the first post terminal **12** may be electrically connected by welding. Therefore, the welding position is the electrical connection position of the conductive part **22** and the first post terminal **12**. In other embodiments of the present application, the conductive part **22** and the first post terminal **12** may be electrically connected by other means instead of welding, such as by providing conductive adhesive or conductive nails, which will not be described in detail herein.

[0271] In order to simplify the description, the following description will take an example that the conductive part **22**

is welded to the first post terminal **12** to form an electrical connection, and that the welding position is the electrical connection position of the conductive part **22** and the first post terminal **12**. For example, in some embodiments, the electrical connection position of the conductive part **22** and the first post terminal **12** is located on the second end wall **12121** and/or the second side wall **12123**, and the conductive part **22** may be welded to at least one of the second end wall **12121** and the second side wall **12123**.

[0272] In the above technical solution, by arranging the electrical connection position of the conductive part **22** and the first post terminal **12** on at least one of the second end wall **12121** and the second side wall **12123**, not only does the second accommodating groove **12120** have a function of accommodating at least a part of the conductive part **22**, but also the groove wall of the second accommodating groove **12120** has a function of achieving an electrical connection to the conductive part **22**, which can simplify the structure of the first post terminal **12** and facilitate processing of the first post terminal **12**. In addition, since the first perforation **12130** is formed on the second end wall **12121**, the conductive part **22** can easily extend into the second accommodating groove **12120** through the first perforation **12130**, thereby simplifying the structure of the conductive part **22**, reducing the redundancy of the conductive part **22**, and lowering the cost of the conductive part **22**. Furthermore, the opening direction of the opening of the second accommodating groove **12120** enables the electrical connection operation on the conductive part **22** and the groove wall of the second accommodating groove **12120** to be easily performed through the opening of the second accommodating groove **12120**, thereby reducing the difficulty of electrical connection. Furthermore, by utilizing the groove wall of the second accommodating groove **12120** to achieve electrical connection to the conductive part **22**, the electrical connection region of the conductive part **22** and the first post terminal **12** may be relatively large, thereby improving the reliability and stability of the electrical connection, and further improving the performance of the battery cell **10**.

[0273] In addition, since the electrical connection position of the conductive part **22** and the first post terminal **12** is located in the second accommodating groove **12120**, not only can the electrical connection position be prevented from protruding out of the first post terminal **12** and occupying the space outside the first post terminal **12**, but also the electrical connection position can be protected by the first post terminal **12**, thereby improving the reliability and stability of the electrical connection between the conductive part **22** and the first post terminal **12**.

[0274] Referring to FIGS. **13** and **14** again, in some embodiments, a local shape of the conductive part **22** matches a local shape of the second end wall **12121**, and the conductive part and the second end wall are arranged in a fitting manner and electrically connected, such that the electrical connection position of the conductive part **22** and the second end wall **12121** extends in the length or width direction of the second end wall **12121**. For example, when the second end wall **12121** is flat, a part of the conductive part **22** may also be flat and fitted to the second end wall **12121**, and the fitted position is electrically connected, such as welded. As such, the area of the electrical connection can be increased, and the reliability and stability of the electrical connection can be improved.

[0275] It is worth noting that the shape of the second end wall **12121** is not limited. For example, the second end wall may be of a flat plate structure or an arc plate structure. When the second end wall **12121** is of a flat plate structure, the second end wall **12121** is provided at an included angle to an axial direction R of the first post terminal **12**. For example, the second end wall may be of a flat plate structure perpendicular to the axial direction R of the first post terminal **12**, or an inclined flat plate structure that is not perpendicular to the axial direction R of the first post terminal **12**, but the inclination direction is not limited.

[0276] For example, referring to FIGS. **13** and **14**, when the second end wall **12121** is of a flat plate structure, an included angle θ between the second end wall **12121** and the axial direction R of the first post terminal **12** is equal to 90°, that is, in the direction from the first perforation **12130** to the second side wall **12123**, the second end wall **12121** is equally spaced from the active substance-coated part **21**. As such, the welding of the conductive part **22** to the second end wall **12121** can be facilitated.

[0277] For another example, FIG. **15** is a partial cross-sectional schematic diagram of the battery cell **10** according to some embodiments of the present application. Referring to FIG. **15**, the included angle θ between the second end wall **12121** and the axial direction R of the first post terminal **12** is greater than 90°, that is, in the direction from the first perforation **12130** to the second side wall **12123**, the second end wall **12121** extends obliquely in a direction close to the active substance-coated part **21**. As such, the extension distance of the conductive part **22** along the second end wall **12121** can be increased, so as to improve the reliability of the electrical connection. Illustratively, the included angle θ between the second end wall **12121** and the axial direction R of the first post terminal **12** may be 90°-145°, for example, 100°, 110°, 120°, 130°, 140°, such that in one aspect, the second end wall **12121** can be easy to process and convenient to be electrically connected to the conductive part **22**, and in another aspect, the space in the first post terminal **12** can be more fully utilized to accommodate the conductive part **22**.

[0278] For yet another example, FIG. **16** is a partial cross-sectional schematic diagram of the battery cell **10** according to some embodiments of the present application. Referring to FIG. **16**, the included angle θ between the second end wall **12121** and the axial direction R of the first post terminal **12** is less than 90°, that is, in the direction from the first perforation **12130** to the second side wall **12123**, the second end wall **12121** extends obliquely in a direction away from the active substance-coated part **21**. As such, the extension distance of the conductive part **22** along the second end wall **12121** can be increased, so as to improve the reliability of the electrical connection. Illustratively, the included angle θ between the second end wall **12121** and the axial direction R of the first post terminal **12** may be 45°-90°, for example, 50°, 60°, 70°, 80°, such that in one aspect, the second end wall **12121** can be easy to process and convenient to be electrically connected to the conductive part **22**, and in another aspect, the space in the first post terminal **12** can be more fully utilized to accommodate the conductive part **22**.

[0279] Certainly, in other embodiments of the present application, the electrical connection position of the conductive part **22** and the second end wall **12121** may not extend in the length or width direction of the second end wall **12121**, and may be, for example, a plurality of discretely arranged spots. For example, the conductive part **22** is provided with a plurality of parts that are spaced apart from each other and separately welded to the second end wall **12121**, which will not be described in detail herein.

[0280] Referring to FIG. **14** again, regardless of the specific value of the included angle θ between the second end wall **12121** and the first post terminal **12** in the axial direction R, in the embodiments of the present application, when the conductive part **22** is electrically connected to the second end wall **12121**, a second recess **12122** may be formed on the second end wall **12121** as required. The second recess **12122** is a groove formed by sinking a part of the second end wall **12121** toward an end proximal to the active substance-coated part. At least a part of the electrical connection position between the conductive part **22** and the second end wall **12121** is located in the second recess **12122**.

[0281] In the above technical solution, the part of the conductive part **22** located in the second recess **12122** is arranged to match the second recess **12122** in shape, and is arranged in a fitting manner to achieve the electrical connection, such that the second recess **12122** can be configured to pre-position and limit the electrical connection position of the conductive part **22**, which is conducive to accurately finding the position to achieve the electrical connection, thereby improving the production efficiency and improving the stability and reliability of the electrical connection position, so as to ensure the reliability and stability of charging and discharging operations of the battery cell **10**.

[0282] It should be noted that, in the embodiments of the present application, a part of the conductive part **22** may also be configured to match the second side wall **12123** in shape and be fitted thereto. For example, when the second side wall **12123** is a curved surface, the part of the conductive part **22** may also be a curved surface and fitted to the second side wall **12123**, and the fitted position may be electrically connected (for example, welded), such that the electrical connection position of the conductive part **22** and the second side wall **12123** extends along the second side wall **12123**. As such, the area of the electrical connection can be increased, thereby improving the reliability and stability of the electrical connection.

[0283] It should be further noted that, in other embodiments of the present application, the electrical connection position of the conductive part **22** and the second side wall **12123** may not extend along the second side wall **12123**, and may be, for example, a plurality of discretely arranged spots. For example, the conductive part **22** is provided with a plurality of parts that are spaced apart from each other and separately welded to the second side wall **12123**, which will not be described in detail herein.

[0284] It can be understood that the number of second side walls **12123** is not limited and may be determined according to the shape of the second accommodating groove **12120**, as long as an end of each second side wall **12123** distal to the opening of the second accommodating groove **12120** is connected to the second end wall **12121**. Illustratively, when the cross-sectional shape of the second accommodating groove **12120** is circular or elliptical, the second end wall **12121** is circular or elliptical. The number of the second side wall **12123** is one, and the second side wall is annular and provided around the circumferential edge of the second end wall **12121**. Further illustratively, when the cross-sectional shape of the second accommodating groove **12120** is rectangular or track-shaped, the second end wall **12121** is rectangular or track-shaped. The number of the second side walls **12123** is four, and the second side walls are separately connected to four sides of the second end wall **12121**.

[0285] It should be further noted that the second accommodating groove **12120** is not limited to the form defined by the second end wall **12121** and the second side wall **12123**. For example, in some embodiments, FIG. **17** is a partial cross-sectional schematic diagram of the battery cell **10** according to some embodiments of the present application. Referring to FIG. **17**, an end of each second side wall **12123** distal to the opening of the second accommodating groove **12120** extends to the first perforation **12130**, such that the second accommodating groove **12120** is defined only by a plurality of second side walls **12123**. In this case, the conductive part **22** may be electrically connected to the second side wall **12123**.

[0286] In some embodiments, referring to FIG. **29**, the accommodating part **121** may be provided with both a third accommodating groove **12140** and a second accommodating groove **12120**. The second accommodating groove **12120** is located on a side of the third accommodating groove **12140** distal to the active substance-coated part **21**. The third accommodating groove **12140** is a groove body, and the groove body is a groove-shaped structure of a certain depth. The opening of the third accommodating groove **12140** is formed on the post terminal inner end surface **122** of the first post terminal **12** on a side proximal to the active substance-coated part **21**, and the opening of the second accommodating groove **12120** is formed on the post terminal outer end surface **123** of the first post terminal **12** on a side distal to the active substance-coated part **21**. The third accommodating groove **12140** is in communication with the second accommodating groove **12120** through the first perforation **12130**. In this case, a part of the conductive part **22** is located in the third accommodating groove **12140**, and the conductive part **22** is also provided in the first perforation **12130** in a penetrating manner, and the rest of the conductive part **22** is located in the second accommodating groove **12120**, such that the space in the first post terminal **12** can be fully utilized and the space occupied by the conductive part **22** in the housing **11** can be reduced.

[0287] It is worth noting that when the conductive part **22** is connected to the second end wall **12121** or the second side wall **12123** by laser welding, referring to FIG. **17** again, the included angle R between a part of the conductive part **22** used for welding and the axis of the first perforation **12130** can be set to be greater than 5°, so as to mitigate the problem of laser entering the housing **11** through the first through hole **12130**, and facilitate the welding operation. In addition, when the included angle β between the part of the conductive part **22** used for welding and the axis of the first perforation **12130** is close to 5°, edge sealing welding may be used, and overlap welding may be used for the rest.

[0288] Referring to FIG. **13** again, in the embodiments of the present application, the method for connecting the first post terminal **12** to the housing **11** is not limited. For example, the method may be welding or riveting. For example, when the first post terminal **12** and the housing **11** are fitted by riveting, the housing is provided with a mounting hole **113** and the first post terminal **12** is mounted by riveting in the mounting hole **113**. It will be appreciated that when the two are fitted by welding or other methods, the housing **11** may be further provided with a mounting hole

113 to enable the first post terminal **12** to be mounted on the housing **11** through the mounting hole **113**.

[0289] Optionally, referring to FIG. **13**, the second accommodating groove **12120** can be provided corresponding to the position of the mounting hole **113**, or in other words, on a projection plane perpendicular to the axial direction R of the first post terminal **12**, the orthographic projection of the second accommodating groove **12120** is located within the range of orthographic projection of the mounting hole **113**, such that the second accommodating groove **12120** can have a larger depth to accommodate more conductive parts **22**, thereby reducing the space occupied by the conductive parts **22** in the housing **11** to a greater extent.

[0290] In some embodiments, referring to FIG. **13**, when the housing **11** is provided with a mounting hole **113** and the first post terminal **12** is mounted in the mounting hole **113**, in the axial direction R of the first post terminal **12**, the depth H3 of the second accommodating groove **12120** is greater than or equal to the minimum distance H4 from the post terminal outer end surface **123** to the mounting hole **113**.

[0291] It should be noted that the specific shape of the second accommodating groove **12120** is not limited, and may be a regular shape or an irregular shape, for example, a columnar groove of constant cross-section with a rectangular, elliptical, or track-shaped cross-section, a trapezoidal groove with a rectangular cross-section and a gradually changing cross-sectional dimension, a hemispherical groove with a circular cross-section and a gradually changing cross-sectional dimension, a semi-ellipsoidal groove with an elliptical cross-section and a gradually changing cross-sectional dimension, or the like. It is worth noting that the track shape described herein refers to a shape in which the two short sides of a rectangle are replaced by convex curves, for example, referring to the shape shown in FIG. **19**(*b*).

[0292] Therefore, the depth H3 of the second accommodating groove **12120** refers to: the maximum depth of the second accommodating groove **12120** in the axial direction R of the first post terminal **12**. Since in the axial direction R of the first post terminal **12**, the depth H3 of the second accommodating groove **12120** is greater than or equal to the minimum distance H4 from the post terminal outer end surface **123** to the mounting hole **113**, a volume of the first post terminal **12** can be fully utilized, such that the second accommodating groove **12120** has a larger depth, which is conducive to accommodating more conductive parts **22**, and then can reduce the space occupied by the conductive parts **22** in the housing **11** to a greater extent, further improve the energy density of the battery cell **10**, and further reduce the redundancy of the conductive part **22** in the housing **11**. Meanwhile, since the second accommodating groove **12120** has a larger depth, gases produced by the battery cell assembly **2** can also be accommodated, which ensures the reliability and stability of the battery cell **10**, and can also accommodate a larger amount of electrolyte to ensure the service life of the battery cell **10**.

[0293] It should be noted that the volume of the second accommodating groove **12120** is not limited. For example, in some specific examples, the volume of the second accommodating groove **12120** that can be used for accommodating the conductive part **22** (referred to as a third volume V4) may be greater than or equal to 298 mm^3, such that the second accommodating groove **12120** may have sufficient space to accommodate the conductive part **22** and facilitate welding of the conductive part **22** to the first post terminal **12**. However, when the third volume V4 of the second accommodating groove **12120** is less than 298 mm^3, the accommodating capacity of the second accommodating groove **12120** for the conductive part **22** is relatively weakened, and the welding difficulty of the conductive part **22** with the first post terminal **12** is increased. For example, the third volume V4 of the second accommodating groove **12120** may be 300 mm^3, 400 mm^3, 500 mm^3, 600 mm^3, 700 mm^3, 800 mm^3, 1000 mm^3, or the like.

[0294] It is worth noting that the third volume V4 of the second accommodating groove **12120** is: the difference between a total volume V5 of the second accommodating groove **12120** and a volume of other components (such as the first cover plate **13** and the second cover plate **14** described herein) required to be accommodated by the second accommodating groove **12120** except the conductive part **22** (referred to as a fourth volume V6), that is, V4=V5−V6. For example, in some specific examples, the total volume V5 of the second accommodating groove **12120** may be 1400 mm^3-1500 mm^3, such that the second accommodating groove **12120** may have more sufficient space to accommodate the conductive part **22** and other components. For example, the total volume V5 of the second accommodating groove **12120** may be 1420 mm^3, 1440 mm^3, 1460 mm^3, 1480 mm^3, 1490 mm^3, or the like.

[0295] In the embodiments of the present application, the shape of the first perforation **12130**, the number of the first perforation **12130**, and the relative positional relationship between the first perforation **12130** and the second accommodating groove **12120** are not limited.

[0296] Illustratively, FIG. **18** is a partially enlarged view of the part B in FIG. **3**; FIG. **19** is an orthographic projection view of various first post terminals **12** according to some embodiments of the present application. Referring to FIGS. **18** and **19**, for the shape of the first perforation **12130**, in the embodiments of the present application, the shape of the first perforation **12130** may be a long strip, so as to match the shape of the sheet-shaped part of the conductive part **22**, thereby facilitating the sheet-shaped part of the conductive part **22** to pass through. Meanwhile, when the first perforation **12130** is long strip-shaped, the second accommodating groove **12120** may also be configured as a shape in which the length of the cross-section is greater than the width, such as a rectangle, an ellipse, and a track shape. In this case, the length direction of the first perforation **12130** may be set to be consistent with the length direction of the cross-section of the second accommodating groove **12120**, such that the space can be fully utilized. In addition, the weld mark formed by welding the conductive part **22** to the first post terminal **12** may be a long strip-shaped weld mark parallel to the length direction of the first perforation **12130**, so as to improve the welding reliability and increase the current passage performance. Illustratively, when the conductive part **22** is welded to the second end wall **12121** to form a weld mark as a long strip-shaped weld mark, the width of the weld mark may be greater than or equal to 6 mm, and the distance between the weld mark and the second side wall **12123** may be greater than or equal to 1 mm, so as to ensure the current passage capacity of the battery cell **10** while ensuring the convenience and reliability of welding.

[0297] With regard to the dimension and number of the first perforations **12130**, in the embodiments of the present application, the formation dimension and specific position of the first perforations **12130** on the second accommodating groove **12120** are not limited and may be designed according to the number of the formed first perforations **12130**. For example, the width of the first perforation **12130** may be greater than or equal to 2 mm, which facilitates the conductive part **22** to pass through. For example, when only one first perforation **12130** is formed on the second accommodating groove **12120**, in some examples, referring to FIGS. **18** and **19**, the first perforation **12130** may be centered relative to the second accommodating groove **12120**. In other examples, FIG. **20** is a partial cross-sectional schematic diagram of the battery cell **10** according to some embodiments of the present application, and referring to FIG. **20**, the first perforation **12130** may also be offset relative to the center of the second accommodating groove **12120**. For example, in some embodiments, referring to FIG. **20**, the first perforation **12130** may be formed at the edge of the second end wall **12121** to be provided proximal to the second side wall **12123**, such that the available area of the second end wall **12121** can be increased, thereby increasing the welding area between the conductive part **22** and the second end wall **12121**.

[0298] It can be understood that after passing through the first perforation **12130**, the conductive part **22** will be folded to be fitted to the second end wall **12121**, but the folding direction is not limited. For example, when the first perforation **12130** is centered relative to the second accommodating groove **12120**, after passing through the first perforation **12130**, the conductive part **22** may be folded toward any side of the first perforation **12130** (referring to FIG. **18**), such that the dimension of the second accommodating groove **12120** can be appropriately reduced, thereby enhancing the compactness and strength of the structure. Alternatively, FIG. **21** is a partial cross-sectional schematic diagram of the battery cell **10** according to some embodiments of the present application, and referring to FIG. **21**, after passing through the first perforation **12130**, the conductive part **22** may also be folded toward opposite sides at the same time, so as to reduce the thickness at the welding position and reduce the welding heat input, thereby reducing problems such as particle splashing.

[0299] For example, when a plurality of first perforations **12130** are formed on the second accommodating groove **12120**, the length directions of the plurality of first perforations **12130** are provided in parallel or substantially in parallel to make full use of the space. In this case, the folding direction of the conductive part **22** after passing through the first perforation **12130** may be set according to the relative positional relationship of the plurality of first perforations **12130**. For example, FIG. **22** is a partial cross-sectional schematic diagram of a battery cell **10** according to some embodiments of the present application. Referring to FIG. **22**, when two first perforations **12130** are formed on the second accommodating groove **12120** and are distal to each other, the two conductive parts **22** passing through the two first perforations **12130** may be folded in a direction towards each other; and when two first perforations **12130** are formed on the second accommodating groove **12120** and are proximal to each other, the two conductive parts **22** passing through the two first perforations **12130** may be folded in a direction away from each other.

[0300] It can be understood that when a plurality of first perforations **12130** are formed on the second accommodating groove **12120**, the number of first post terminals **12** may be appropriately reduced, so as to reduce costs and processes.

[0301] In addition, in some embodiments, referring to FIGS. **20** and **21**, the first perforation **12130** may be centered relative to the active substance-coated part **21**, but the position of the first perforation **12130** relative to the second accommodating groove **12120** is not limited, and may be centered or offset. Since the first perforation **12130** is centered relative to the active substance-coated part **21**, the conductive part **22** may be gathered corresponding to the midline position of the active substance-coated part **21**.

[0302] In some embodiments, referring to FIG. **21**, a sealing member **6** may be provided at the first perforation **12130**, so as to mitigate the problem of the electrolyte in the housing **11** leaking from the first perforation **12130**. The material and shape of the sealing member **6**, as well as the method for connecting the sealing member to the first perforation **12130** are not limited. Illustratively, the sealing member **6** may be a metal member made of the same material as the first post terminal **12** or the conductive part **22**, which can fit the wall surface at the first perforation **12130** of the first post terminal **12** by welding to seal the first perforation **12130**; further illustratively, the sealing member **6** may be rubber, plastic, or the like, and is inserted into the first perforation **12130**, such that an interference fit is formed to seal the first perforation **12130**. In the embodiments of the present application, all design can be made based on actual requirements and is not limited.

[0303] FIG. **23** is an exploded view of the structure of a battery cell **10** according to some embodiments of the present application; FIG. **24** is a partial cross-sectional schematic diagram of the housing assembly **1** according to some embodiments of the present application; and FIG. **25** is an exploded view of the structure of the housing assembly **1** shown in FIG. **24**. Referring to FIGS. **23**, **24** and **25**, in the embodiments of the present application, when the accommodating part **121** is provided with the second accommodating groove **12120** according to any one of the embodiments described above, optionally, the housing assembly **1** may further include a first cover plate **13**, the first cover plate **13** fits the first post terminal **12** and closes the groove opening of the second accommodating groove **12120**, and the first cover plate **13** is electrically connected to the first post terminal **12**.

[0304] In the above technical solution, by providing the first cover plate **13** to close the opening of the second accommodating groove **12120**, the electrolyte in the housing **11** can be prevented from leaking from the opening of the second accommodating groove **12120**. Moreover, since the first cover plate **13** closes the opening of the second accommodating groove **12120** and is electrically connected to the first post terminal **12**, an indirect electrical connection between the first post terminal **12** and a busbar component can be easily achieved by using the first cover plate **13**, and the connection area of the electrical connection can also be increased, thereby helping reduce the resistance of the electrical connection.

[0305] It is worth noting that the fitting manner and fitting position of the first cover plate **13** and the first post terminal **12** are not limited, as long as the first cover plate **13** can close the opening of the second accommodating groove **12120**. For example, in some embodiments, referring to FIG. **22**, the first cover plate **13** may be welded to the first post terminal

12. During processing, the conductive part **22** may be first passed through the first perforation **12130** and welded to the groove wall of the second accommodating groove **12120**, and then the first cover plate **13** is welded to the first post terminal **12** to close the opening of the second accommodating groove **12120**.

[0306] It should be further noted that the specific structure of the first cover plate **13** is not limited. For example, in some optional embodiments, FIG. **26** is an exploded view of the structure of the first cover plate **13** shown in FIG. **25**. Referring to FIGS. **24** to **26**, the first cover plate **13** includes a first conductive member **131** and a second conductive member **132** made of different materials. The first conductive member **131** fits and is electrically connected to the first post terminal **12**, and the second conductive member **132** fits and is electrically connected to the first conductive member **131**.

[0307] In the above technical solution, the first cover plate **13** is provided in a composite form, and the first conductive member **131** is set to be made of the same material as that of the first post terminal **12**, so as to facilitate the electrical connection between the first conductive member **131** and the first post terminal **12**. For example, the first conductive member **131** can be easy to be reliably and stably connected to the first post terminal **12** through welding. Furthermore, since the second conductive member **132** is made of a different material from that of the first conductive member **131**, the electrical connection between the second conductive member **132** and a busbar component and the like made of a different material from that of the first post terminal **12** is facilitated. For example, the second conductive member **132** can be easy to be reliably and stably connected to a busbar component made of the same material as that of the second conductive member **132** through welding.

[0308] For example, when the first post terminal **12** is a negative electrode post terminal, the first post terminal **12** is a copper column, and when the busbar component is an aluminum sheet, the first conductive member **131** may be set to be made of copper, and the second conductive member **132** may be set to be made of aluminum. In this case, the first post terminal **12** and the first conductive member **131** are made of the same material and can be effectively welded, and the second conductive member **132** and the busbar component are made of the same material and can be effectively welded, thereby effectively achieving an indirect electrical connection between the first post terminal **12** and the busbar component through the first cover plate **13**. Moreover, the welding of the first post terminal **12** to the first conductive member **131** is welding between the copper materials, which has good fluidity and is not prone to cracks, thereby helping improve the sealing effect of the welding position.

[0309] Referring to FIGS. **24-26** again, in some optional examples, the first conductive member **131** is located between the second accommodating groove **12120** and the second conductive member **132**. In the above technical solution, since the first conductive member **131** is located between the second accommodating groove **12120** and the second conductive member **132**, the second accommodating groove **12120** and the second conductive member **132** can be separated. Therefore, when the electrolyte in the housing **11** enters the second accommodating groove **12120** from the first perforation **12130**, the first conductive member **131** can be used to prevent the part of the electrolyte from contacting the second conductive member **132**, thereby solving the problem of corrosion of the second conductive member **132** by the electrolyte.

[0310] It is worth noting that the fitting manner of the first conductive member **131** and the second conductive member **132** is not limited. For example, in some embodiments, referring to FIGS. **24-26**, the first conductive member **131** is provided with a second groove **1311**, the second conductive member **132** is embedded in the second groove **1311**, and an opening of the second groove **1311** is formed on a surface of the first conductive member **131** on a side distal to the second accommodating groove **12120**, such that the second conductive member **132** is exposed from the opening of the second groove **1311**. Alternatively, in other embodiments, the first conductive member **131** and the second conductive member **132** may be connected in a fastening connection, a snap connection, or the like.

[0311] It should be further noted that "exposed" mentioned in the second conductive member **132** being exposed from the opening of the second groove **1311** means that the first conductive member **131** does not block the second conductive member **132** at the opening position of the second groove **1311**, and the second conductive member **132** is not required to protrude from the opening of the second groove **1311**. For example, the second conductive member **132** may be flush with the surface of the first conductive member **131** on a side distal to the second accommodating groove **12120**, or the second conductive member **132** may protrude from the surface of the first conductive member **131** on a side distal to the second accommodating groove **12120**.

[0312] In the above technical solution, embedding the second conductive member **132** in the first conductive member **131** can reduce the difficulties in assembling the first conductive member **131** and the second conductive member **132**, improve the stability and convenience of the fit of the first conductive member **131** and the second conductive member **132**, reduce the thickness of the first cover plate **13**, and reduce the space occupied by the first cover plate **13**, so as to improve the space utilization of the battery cell **10**. In another aspect, since the second conductive member **132** can be exposed from the surface of the first conductive member **131** on a side distal to the second accommodating groove **12120** through the opening of the second groove **1311**, it is conducive to achieving the electrical connection between the second conductive member **132** and the busbar component outside the first post terminal **12**.

[0313] In addition, since the opening of the second groove **1311** is formed on the surface of the first conductive member **131** on a side distal to the second accommodating groove **12120**, it means that the second groove **1311** is open in a direction away from the active substance-coated part **21**, such that the part of the first conductive member **131** configured to define the groove wall of the second groove **1311** is located between the second accommodating groove **12120** and the second conductive member **132**, so as to separate the second accommodating groove **12120** from the second conductive member **132**, thereby preventing the electrolyte entering the second groove **1311** from contacting the second conductive member **132** and reducing leakage of the electrolyte.

[0314] Certainly, in other embodiments, the first cover plate **13** may not be in a composite form composed of a plurality of materials. For example, in other embodiments of the present application, FIG. **27** is a partial cross-sectional schematic diagram of a battery cell **10** according to some embodiments of the present application, and FIG. **28** is an exploded view of the structure of the battery cell **10** shown in FIG. **27**. Referring to FIGS. **27** and **28**, the first cover plate **13** as a whole may also be provided in a non-composite form made of the same material, for example, for matching the positive electrode post terminal, which will not be described in detail herein.

[0315] Referring to FIGS. **24-26** again, in some embodiments, the first cover plate **13** is also embedded in the opening of the second accommodating groove **12120**. In the above technical solution, by embedding the first cover plate **13** in the second accommodating groove **12120**, the difficulty in assembling the first cover plate **13** and the first post terminal **12** can be reduced, such that the stability of assembly of the first cover plate **13** and the first post terminal **12** and the reliability and convenience of the connection are improved, and the space occupied by the first cover plate **13** outside the first post terminal **12** is reduced. Moreover, since the first cover plate **13** is embedded in the opening of the second accommodating groove **12120**, there is sufficient space in the second accommodating groove **12120** to accommodate the conductive part **22**.

[0316] Certainly, in other embodiments of the present application, the fitting manner of the first cover plate **13** and the first post terminal **12** is not limited to being embedded in the second accommodating groove **12120**. The first cover plate **13** may also be directly provided outside the first post terminal **12** as a covering, that is, directly lids the opening of the second accommodating groove **12120**, as long as it is conducive to the fit to the busbar component of the battery **100**, which is not limited in the embodiments.

[0317] Referring to FIGS. **24** to **26** again, optionally, in the embodiments of the present application, at least a part of the wall surface at the groove opening of the second accommodating groove **12120** formed on the first post terminal **12** is an inclined guiding surface **12126**, and the inclined guiding surface **12126** is configured to guide the first cover plate **13** to fit the groove opening of the second accommodating groove **12120**. In the above technical solution, by processing the wall surface at the opening of the second accommodating groove **12120** into an inclined surface with guidance functionality, the difficulty in assembling the first cover plate **13** and the second accommodating groove **12120** can be reduced, thereby improving the assembly efficiency of the first cover plate **13** and the second accommodating groove **12120**. Moreover, when the first cover plate **13** is welded to the inclined guiding surface **12126**, the area at the welding position can be increased, thereby improving the reliability of the welding connection between the first cover plate **13** and the first post terminal **12** and mitigating the problem of weld pool collapse or laser penetration into the first post terminal **12** during welding.

[0318] Specifically, referring to FIGS. **24-26**, the second accommodating groove **12120** includes a first groove segment **12124** and a second groove segment **12125** located on a side of the first groove segment **12124** proximal to the post terminal outer end surface **123**. The cross-sectional area of the second groove segment **12125** is greater than the cross-sectional area of the first groove segment **12124**, such that the second accommodating groove **12120** is in the shape of a stepped groove, and a step surface **12127** is formed at the connection position of the first groove segment **12124** and the second groove segment **12125**, such that when the first cover plate **13** is embedded in the second accommodating groove **12120**, the first cover plate can be specifically embedded in the second groove segment **12125** and supported by the step surface **12127**.

[0319] In the above technical solution, by providing the second accommodating groove **12120** in a stepped groove form, the first cover plate **13** can be stably fitted at the opening position of the second accommodating groove **12120**, thereby improving the connection stability between the first cover plate **13** and the first post terminal **12**. In addition, by defining the groove depth of the first groove segment **12124**, the second accommodating groove **12120** is provided with sufficient space to accommodate the conductive part **22**.

[0320] Further, when the wall surface at the opening of the second accommodating groove **12120** formed by the first post terminal **12** is the inclined guiding surface **12126**, the cross-sectional area of the second groove segment **12125** may be set to gradually increase in a direction close to the post terminal outer end surface **123**, such that the side wall of the second groove segment **12125** is formed as the inclined guiding surface **12126**, which facilitates processing and can easily and effectively meet the guidance requirements.

[0321] Referring to FIGS. **24-26** again, in the embodiments of the present application, a stress relief groove **133** may be further formed on the first cover plate **13** as required. The stress relief groove **133** is located in an outer peripheral region of the first cover plate **13** to assist the first cover plate **13** in stress relief. In the above technical solution, by providing the stress relief groove **133** on the first cover plate **13**, the stress generated during the processing of the first cover plate **13** itself or the electrical connection between the first cover plate **13** and the first post terminal **12** can be relieved, so as to alleviate the related problems such as deformation or damage of the first cover plate **13** caused by stress.

[0322] Specifically, when the first cover plate **13** and the second accommodating groove **12120** are embedded and welded, the stress generated by welding can be relieved by utilizing the stress relief groove **133**, thereby improving the lateral heat conduction and reducing the probability of damage or deformation of the first cover plate **13**. Meanwhile, when the first cover plate **13** is in the above composite form including the first conductive member **131** and the second conductive member **132**, the stress relief groove **133** may be provided on the first conductive member **131** and located in the outer peripheral region of the second conductive member **132**. When the first conductive member **131** and the second accommodating groove **12120** are embedded and welded, the stress generated by welding can be relieved by utilizing the stress relief groove **133**, thereby improving the lateral heat conduction and reducing the probability of damage or deformation of the second conductive member **132**. Moreover, when the second conductive member **132** and the first conductive member **131** are embedded and welded, the stress generated by welding can be relieved by utilizing the stress relief groove **133**, thereby improving the lateral heat conduction and reducing the probability of causing the first conductive member **131** to deform that may cause the first conductive member **131** to be unable to be embedded with the second accommodating groove **12120**.

[0323] Referring to FIGS. **27-28**, in the embodiments of the present application, the housing assembly **1** may be further provided with a second cover plate **14** as required. The second cover plate **14** lids the first perforation **12130** and is also located outside the conductive part **22** in the second accommodating groove **12120**.

[0324] It is worth noting that when the housing assembly **1** includes the second cover plate **14**, the housing assembly **1** may also include the first cover plate **13**, or may not include the first cover plate **13**. Further, when the housing assembly **1** includes both the second cover plate **14** and the first cover plate **13**, the first cover plate **13** may be in the composite form made of a plurality of materials, or may be in the non-composite form made of the same material.

[0325] In the above technical solution, at least a part of the conductive part **22** is located in the second accommodating groove **12120**, and the second cover plate **14** lids the part of the conductive part **22**. The second cover plate **14** also lids the first perforation **12130**, such that when the electrolyte enters the second accommodating groove **12120** from the first perforation **12130**, the problem of the part of the electrolyte overflowing from the first post terminal **12** can be mitigated through the second cover plate **14**, thereby improving the reliability of the battery cell **10**.

[0326] For example, as shown in FIGS. **27-28**, when a part of the conductive part **22** is sandwiched between the second cover plate **14** and the second end wall **12121**, the part of the conductive part **22**, the second cover plate **14**, and the second end wall **12121** can be welded together by using laser welding, so as to improve the reliability of the connection between the first post terminal **12** and the conductive part **22**. Furthermore, since the second cover plate **14** can press the conductive part **22**, the second cover plate **14** may be used to improve the stability of the conductive part **22** accommodated in the second accommodating groove **12120**.

[0327] In the embodiments of the present application, the first post terminal **12** may be an integrally formed post terminal, or a separately formed composite post terminal. Referring to FIGS. **27** and **28** again, for example, the first post terminal **12** may include a first post terminal part **124** and a second post terminal part **125** made of different materials and electrically connected, the second post terminal part **125** is located on a side of the first post terminal part **124** distal to the active substance-coated part **21**, the accommodating part **121** is disposed on the first post terminal part **124** or the accommodating part **121** is disposed on the first post terminal part **124** and the second post terminal part **125**, and the conductive part **22** is electrically connected to the first post terminal part **124**.

[0328] In the above technical solution, by providing the first post terminal **12** in a composite form composed of different materials, the first post terminal part **124** located on the inner side is fitted in an accommodating manner and electrically connected to the conductive part **22**, and the second post terminal part **125** located on the outer side is electrically connected to the busbar component and the like, which is conducive to realizing the assembly and electrical connection of the first post terminal **12** to related components, reducing the mutual interference between the electrical connection position of the first post terminal **12** and the conductive part **22**, and the electrical connection position of the first post terminal **12** and the busbar component of the battery **100**, and improving the reliability and stability of the battery cell **10**.

[0329] For example, when the material of the conductive part **22** is different from that of the busbar component, the first post terminal part **124** may be set to be made of the same material as that of the conductive part **22**, and the second post terminal part **125** may be set to be made of the same material as that of the busbar component, which is conducive to realizing the welding of the second post terminal part **125** to the busbar component, and the welding of the first post terminal part **124** to the conductive part **22**, improving the reliability and stability of the electrical connection between the conductive part **22** and the first post terminal **12**, and the reliability and stability of the electrical connection between the first post terminal **12** and the busbar component.

[0330] Further, when the first post terminal **12** is in the composite form of the above embodiments and is provided with the second accommodating groove **12120** and the first perforation **12130** of any one of the above embodiments, in some embodiments, the housing assembly **1** may also include the second cover plate **14** of any one of the above embodiments. In this case, the second cover plate **14** may be set to be made of the same material as that of the first post terminal part **124**, and the first post terminal part **124** may be electrically connected to the second cover plate **14**, so as to improve the reliability and stability of the electrical connection between the first post terminal part **124** and the second cover plate **14**. For example, the first post terminal part **124** and the second cover plate **14** may be connected by welding.

[0331] For example, referring to FIGS. **27-28**, when the first post terminal **12** is a negative electrode post terminal, the first post terminal part **124** is made of copper, the second post terminal part **125** is made of aluminum, and the busbar component is an aluminum sheet, the second cover plate **14** may be set to be made of copper, and the first cover plate **13** may be set to be made of aluminum. In this case, the second cover plate **14** and the first post terminal part **124** are made of the same material and can be effectively welded, the second post terminal part **125** and the first cover plate **13** are made of the same material and can be effectively welded, and the first cover plate **13** and the busbar component are made of the same material and can be effectively welded.

[0332] In the embodiments of the present application, when the accommodating part **121** is provided with a second accommodating groove **12120**, depending on the structure of the battery cell assembly **2**, the fitting condition of the battery cell assembly **2** and the second accommodating groove **12120** is not limited, and for example, may include but is not limited to the following third and fourth embodiments.

[0333] FIG. **29** is a partial cross-sectional schematic diagram of the battery cell **10** according to some embodiments of the present application. Referring to FIG. **29**, in the third embodiment, the active substance-coated part **21** includes a current collector **211** and an active substance layer **212** arranged on the current collector **211**. The conductive part **22** includes a tab part **221** electrically connected to the current collector **211**. The tab part **221** includes a plurality of tab plates **2211**. Parts of the plurality of tab plates **2211** proximal to the current collector **211** converge to form a first gathering part **2212**, and parts of the plurality of tab plates **2211** distal to the current collector **211** converge and are connected to form a second gathering part **2213**. The first gathering part **2212** connects the second gathering part **2213** and the active substance-coated part **21**. When the accommodating part **121** is provided with the second accommodating groove **12120**, at least a part of the second gathering part **2213** is accommodated in the second accommodating groove **12120**.

[0334] It is worth noting that the specific structure of the battery cell assembly **2** in the third embodiment is basically the same as that of the battery cell assembly **2** in the first embodiment above, and may refer to the description of the first embodiment, and will not be described in detail herein. In the third embodiment, since the tab part **221** includes the second gathering part **2213** formed by the convergence and connection of a plurality of tab plates **2211**, at least a part of the second gathering part **2213** can be easily accommodated in the second accommodating groove **12120**, which facilitates the assembly of the conductive part **22** and the first post terminal **12**.

[0335] In some optional examples, referring to FIG. **29**, the position where the first gathering part **2212** is connected to the second gathering part **2213** can be provided corresponding to the first perforation **12130**, that is, on a projection plane perpendicular to the axial direction R of the first post terminal **12**, the orthographic projection of the position where the first gathering part **2212** is connected to the second gathering part **2213** is located within the range of orthographic projection of the first perforation **12130**, such that the second gathering part **2213** can extend into the first perforation **12130** at a relatively short distance and enter the second accommodating groove **12120**, thereby reducing the redundancy and reducing costs.

[0336] It can be understood that the gathering position of the tab plates **2211** may be designed according to the position of the first perforation **12130**, for example, by adopting the symmetrical gathering form or the asymmetrical gathering form described above, such that the position where the first gathering part **2212** is connected to the second gathering part **2213** is provided corresponding to the first perforation **12130**, which will not be described in detail herein. In addition, referring to the above description, when the second gathering part **2213** forms a plate structure by ultrasonic pre-welding, it is convenient for the second gathering part **2213** to pass through the first perforation **12130**.

[0337] Referring to FIG. **29**, in the third embodiment, in addition to the second accommodating groove **12120**, the accommodating part **121** may be further provided with a third accommodating groove **12140**. The third accommodating groove **12140** is located on a side of the second accommodating groove **12120** proximal to the active substance-coated part **21**. The surface of the first post terminal **12** on a side facing the active substance-coated part **21** is the post terminal inner end surface **122**, and the opening of the third accommodating groove **12140** is formed on the post terminal inner end surface **122**. The third accommodating groove **12140** is in communication with the second accommodating groove **12120** through the first perforation **12130**. In this case, at least a part of the first gathering part **2212** may be accommodated in the third accommodating groove **12140**.

[0338] In the above technical solution, at least a part of the first gathering part **2212** of the tab part **221** is accommodated in the third accommodating groove **12140**, and at least a part of the second gathering part **2213** is accommodated in the second accommodating groove **12120**, such that the space in the first post terminal **12** can be more fully utilized, and the space occupied by the tab part **221** in the housing **11** can be further reduced, so as to accommodate a larger active substance-coated part **21**, thereby improving the energy density of the battery cell **10**. Moreover, the redundancy of the tab part **221** in the housing **11** can be better reduced, the probability of short circuit between the tab part **221** and the active substance-coated part **21** can be further reduced, and thus the risk of the tab part **221** being inverted toward the active substance-coated part **21** can be further reduced.

[0339] It should be noted that the specific shape of the third accommodating groove **12140** is not limited, and may be a regular shape or an irregular shape, for example, a columnar groove of constant cross-section with a rectangular, elliptical, or track-shaped cross-section, a trapezoidal groove with a rectangular cross-section and a gradually changing cross-sectional dimension, a hemispherical groove with a circular cross-section and a gradually changing cross-sectional dimension, a semi-ellipsoidal groove with an elliptical cross-section and a gradually changing cross-sectional dimension, or the like. In the embodiments of the present application, the third accommodating groove **12140** may be configured into a shape in which the length of the cross-section is greater than the width, such as a rectangle, an ellipse, and a track shape, so as to facilitate the accommodation of the first gathering part **2212**.

[0340] In the third embodiment, the second gathering part **2213** is electrically connected to the first post terminal **12** directly or indirectly. For example, when the second gathering part **2213** is directly electrically connected to the first post terminal **12**, such as when the second gathering part **2213** is welded to the first post terminal **12**, the structure of the battery cell assembly **2** may be simplified, thereby reducing the number of components, simplifying the assembly process, and improving the assembly efficiency. The method and position of direct electrical connection between the second gathering part **2213** and the first post terminal **12** are not limited. For example, the electrical connection position of the second gathering part **2213** and the first post terminal **12** may be located on the second end wall **12121** and/or the second side wall **12123**. Further, the electrical connection position of the second gathering part **2213** and the second end wall **12121** may extend in the length direction or the width direction of the second end wall **12121**. Yet further, the second end wall **12121** is provided with the second recess **12122**, and the electrical connection position of the second gathering part **2213** and the second end wall **12121** may be located in the second recess **12122**, and the like. The corresponding technical effects can refer to the introduction of the above embodiments and will not be described in detail herein.

[0341] As an optional solution, the conductive part **22** may also be provided with an adapting piece **222** as required. In this case, the second gathering part **2213** is electrically connected to the first post terminal **12** indirectly. Specifically, FIG. **30** is a partial cross-sectional schematic diagram of the battery cell **10** according to some embodiments of the present application. Referring to FIG. **30**, when the conductive part **22** includes the adapting piece **222**, the adapting piece **222** is connected to the second gathering part **2213**, and the conductive part **22** is electrically connected to the first post terminal **12** via the adapting piece **222**. In this case, at least a part of the adapting piece **222** is accommodated in the second accommodating groove **12120**. In this example, at least a part of the second gathering part **2213** is also accommodated in the second accommodating groove **12120**.

[0342] In the above technical solution, the active substance-coated part **21** can be electrically connected to the first post terminal **12** through the first gathering part **2212**, the second gathering part **2213**, and the adapting piece **222** in sequence. The electrical connection position of the conductive part **22** and the first post terminal **12** is located on the adapting piece **222**. For example, the electrical connection can be achieved by welding (for example, laser welding) the adapting piece **222** to the first post terminal **12**. In addition, the adapting piece **222** and the tab plate **2211** are two separate components and are connected by welding (such as ultrasonic welding) or the like.

[0343] In the above technical solution, by accommodating at least a part of the second gathering part **2213** and at least a part of the adapting piece **222** in the second accommodating groove **12120**, the space in the first post terminal **12** can be more fully utilized, and the space occupied by the conductive part **22** in the housing **11** can be further reduced, so as to further improve the volumetric energy density of the battery cell **10**. Furthermore, by providing the adapting piece **222** of the plate structure, it is convenient for the adapting piece **222** to pass through the first perforation **12130** and extend into the second accommodating groove **12120**.

[0344] In addition, by using the adapting piece **222** to achieve an indirect electrical connection between the second gathering part **2213** and the first post terminal **12**, the adapting piece **222** can be welded to the first post terminal **12** at a part avoiding the second gathering part **2213**, such that the welding between the adapting piece **222** and the first post terminal **12** is secure, the risk of welding cracking is low, and the reliability and stability of the battery cell **10** can be further improved. Meanwhile, by electrically connecting the first post terminal **12** and the tab plates **2211** through the adapting piece **222**, the configuration of the tab plates **2211** can also be simplified.

[0345] The method and position of direct electrical connection between the adapting piece **222** and the first post terminal **12** are not limited. For example, the adapting piece **222** is electrically connected to the first post terminal **12** by welding. For example, the electrical connection position of the adapting piece **222** and the first post terminal **12** may be located on the second end wall **12121** and/or the second side wall **12123**. Further, the electrical connection position of the adapting piece **222** and the second end wall **12121** may extend in the length direction or the width direction of the second end wall **12121**. Yet further, the second end wall **12121** is provided with the second recess **12122**, and the electrical connection position of the adapting piece **222** and the second end wall **12121** may be located in the second recess **12122**, and the like. The corresponding technical effects can refer to the introduction of the above embodiments and will not be described in detail herein. When the electrical connection position of the adapting piece **222** and the first post terminal **12** is located on the second end wall **12121** and/or the second side wall **12123**, since at least a part of the adapting piece **222** is accommodated in the second accommodating groove **12120**, the structure of the adapting piece **222** can be simplified, thereby reducing the redundancy and lowering the costs.

[0346] Referring to FIG. **30**, in the fourth embodiment, the active substance-coated part **21** includes a current collector **211** and an active substance layer **212** arranged on the current collector **211**. The conductive part **22** includes a tab part **221** and an adapting piece **222**. The tab part **221** includes a plurality of tab plates **2211** electrically connected to the current collector **211**. Parts of the plurality of tab plates **2211** proximal to the current collector **211** converge to form a first gathering part **2212**, and parts of the plurality of tab plates **2211** distal to the current collector **211** converge and are connected to form a second gathering part **2213**. The adapting piece **222** is electrically connected to the second gathering part **2213**. When the accommodating part **121** is provided with the second accommodating groove **12120**, at least a part of the adapting piece **222** may be accommodated in the second accommodating groove **12120** and electrically connected to the first post terminal **12**.

[0347] In the above technical solution, compared with the solution including the adapting piece **222** in the third embodiment, in the fourth embodiment, at least a part of the adapting piece **222** is accommodated in the second accommodating groove **12120**, but the relative positions of the tab part **221** and the second accommodating groove **12120** are not limited, that is, at least a part of the tab part **221** may be accommodated in the second accommodating groove **12120**, and the tab part **221** may also be completely located outside the second accommodating groove **12120**, thereby meeting different structural design requirements.

[0348] In the above technical solution, by accommodating at least a part of the adapting piece **222** in the second accommodating groove **12120**, the adapting piece **222** can occupy the space in the first post terminal **12**, such that the space occupied by the adapting piece **222** in the housing **11** can be reduced, so as to accommodate a larger active substance-coated part **21**, thereby improving the energy density of the battery cell **10**. Moreover, the probability of short circuit between the adapting piece **222** and the active substance-coated part **21** can be reduced, and the risk of short circuit of the battery cell assembly **2** can be reduced, thereby improving the stability and reliability of the battery cell **10**.

[0349] In addition, by using the adapting piece **222** to achieve an indirect electrical connection between the second gathering part **2213** and the first post terminal **12**, the adapting piece **222** can be welded to the first post terminal **12** at a part avoiding the second gathering part **2213**, such that the welding between the adapting piece **222** and the first post terminal **12** is secure, the risk of welding cracking is low, and the reliability and stability of the battery cell **10** can be further improved. Meanwhile, by electrically connecting the first post terminal **12** and the tab plates **2211** through the adapting piece **222**, the configuration of the tab plates **2211** can also be simplified.

[0350] FIG. **31** is a partial cross-sectional schematic diagram of the battery cell **10** according to some embodiments of the present application. Referring to FIG. **31**, in other embodiments of the present application, the accommodating part **121** is provided with a fourth accommodating groove **12150**. The fourth accommodating groove **12150** is a groove body, and the groove body is a groove-shaped structure of a certain depth. A surface of the first post terminal **12** on a side distal to the active substance-coated part **21** is the post terminal outer end surface **123**, and the opening of the fourth accommodating groove **12150** is formed on the post terminal outer end surface **123**. The fourth accommodating groove **12150** is in communication with the interior of the housing **11** through the second perforation **12160**. The conductive part **22** may not be accommodated in the fourth accommodating groove **12150**. For example, the conductive part **22** may be provided in the second perforation **12160** in a penetrating manner, and the electrical connection position of the conductive part **22** and the first post terminal **12** is located at the wall of the second perforation **12160** formed on the first post terminal **12**.

[0351] In the above embodiments, by providing the fourth accommodating groove **12150**, the electrical connection between the conductive part **22** and the wall of the second perforation **12160** can be easily realized. Furthermore, in some cases, the sealing of the second perforation **12160** can be achieved by utilizing the electrical connection between the conductive part **22** and the first post terminal **12**. For example, the conductive part **22** may be welded to the wall of the second perforation **12160** at a position of the second perforation **12160** connected to the fourth accommodating groove **12150**, so as to facilitate the operation. In addition, by controlling a weld mark, the second perforation **12160** can be sealed by utilizing the weld mark and the conductive part **22**, so as to alleviate the problem of leakage of the electrolyte in the housing **11** from the second perforation **12160**.

[0352] It should be noted that the specific shape of the fourth accommodating groove **12150** is not limited, and may be a regular shape or an irregular shape, for example, a columnar groove of constant cross-section with a rectangular, elliptical, or track-shaped cross-section, a trapezoidal groove with a rectangular cross-section and a gradually changing cross-sectional dimension, a hemispherical groove with a circular cross-section and a gradually changing cross-sectional dimension, a semi-ellipsoidal groove with an elliptical cross-section and a gradually changing cross-sectional dimension, or the like.

[0353] In the embodiments of the present application, the shape of the second perforation **12160** may be a long strip shape, so as to match the shape of the plate-like part of the conductive part **22**, thereby facilitating the plate-like part of the conductive part **22** to pass through. Meanwhile, when the second perforation **12160** is long strip-shaped, the fourth accommodating groove **12150** may also be configured into a shape in which the length of the cross-section is greater than the width, such as a rectangle, an ellipse, and a track shape. In this case, the length direction of the second perforation **12160** may be set to be consistent with the length direction of the cross-section of the fourth accommodating groove **12150**, such that the space can be fully utilized.

[0354] It should be noted that the accommodating part **121** of the embodiments of the present application is not limited to the above forms in which at least one accommodating groove must be provided. For example, in other embodiments of the present application, FIG. **32** is a partial cross-sectional schematic diagram of the battery cell **10** according to some embodiments of the present application. Referring to FIG. **32**, the accommodating part **121** may be provided with only a third perforation **12170**, a surface of the first post terminal **12** on a side facing the active substance-coated part **21** is the post terminal inner end surface **122**, a surface of the first post terminal **12** on a side distal to the active substance-coated part **21** is the post terminal outer end surface **123**, the third perforation **12170** is in the form of a through hole and passes through the post terminal inner end surface **122** and the post terminal outer end surface **123**, and at least a part of the conductive part **22** is provided in the third perforation **12170** in a penetrating manner. The electrical connection position of the conductive part **22** and the first post terminal **12** is not limited. For example, the electrical connection position may be located on the wall of the third perforation **12170** formed on the first post terminal **12**, or the conductive part **22** may pass through the third perforation **12170**, such that the electrical connection position is located on the post terminal outer end surface **123** outside the third perforation **12170**. In addition, the shape of the third perforation **12170** is not limited. The third perforation may be a regular-shaped hole of constant cross-section, or a hole of variable cross-section. Furthermore, the cross-sectional shape of the third perforation **12170** is not limited, for example, a long strip shape, such as a rectangle, an ellipse, or a track shape, so as to match the shape of the plate-like part of the conductive part **22**, thereby facilitating the plate-shaped part of the conductive part **22** to be provided in the third perforation **12170** in a penetrating manner, which will not be described in detail herein.

[0355] FIG. **33** is a partial cross-sectional schematic diagram of a battery cell **10** according to some embodiments of the present application; FIG. **34** is a schematic diagram illustrating the fit between a battery cell assembly **2** and a support **3** according to some embodiments of the present application; and FIG. **35** is a cross-sectional view along the line C-C in FIG. **34**. Referring to FIGS. **33** to **35**, in the embodiments of the present application, the battery cell **10** further includes a support **3**. The support **3** is located in the housing **11**, and the support **3** is located on a side of the active substance-coated part **21** proximal to the first post terminal **12**. The support **3** is provided with a clearance hole **31** configured to provide clearance for the conductive part **22**. The conductive part **22** can pass through the clearance hole **31** to extend to the side of the support **3** distal to the active substance-coated part **21**, so as to be welded to the first post terminal **12**, thereby ensuring the normal charging and discharging operations of the battery cell **10**.

[0356] In the above technical solution, by providing the support **3** on the side of the active substance-coated part **21** proximal to the first post terminal **12**, the active substance-coated part **21** can be separated from the housing **11** by utilizing the support **3**, thereby improving the reliability of the battery cell **10**. Furthermore, by providing the clearance hole **31** on the support **3**, the conductive part **22** can be guided and constrained to be fitted to the first post terminal **12** by passing through the clearance hole **31**, thereby eliminating the need for the conductive part **22** to bypass the edge of the support **3** to approach the first post terminal **12**, which can not only simplify the arrangement of the conductive part **22**, save the material of the conductive part **22**, and reduce the cost, but also reduce the risk of short circuit connection between the conductive part **22** and the active substance-coated part **21** by the support **3** supporting and guiding the conductive part **22** to be fitted to the first post terminal **12**, so as to further improve the reliability of the battery cell **10**.

[0357] Referring to FIGS. **33-35** again, optionally, the support **3** is provided with a guiding part **32**, the guiding part **32** defines, in an enclosing manner, at least a part of the clearance hole **31**, and the guiding part **32** at least partially extends to the accommodating part **121**.

[0358] It is worth noting that the guiding part **32** protrudes from the support **3** and extends into the accommodating part **121**, and at least a part of the clearance hole **31** is formed in the guiding part **32**, such that when the conductive part **22** is provided in the clearance hole **31** in a penetrating manner, at least a part of the conductive part **22** can be easily accommodated in the accommodating part **121**, thereby improving the assembly efficiency of the conductive part **22**. Meanwhile, through the arrangement of the guiding part **32**, the fits between the support **3** and the first post terminal **12**, and between the support **3** and the conductive part **22** become tighter and more reliable, such that the structure of the battery cell **10** becomes more compact, which is more conducive to improving the energy density of the battery cell **10**.

[0359] Referring to FIGS. **33-35** again, optionally, the support **3** is provided with a third groove **38**, and at least a part of the first post terminal **12** located in the housing **11** is accommodated in the third groove **38**.

[0360] In the above technical solution, by arranging the third groove **38** on the support **3**, at least a part of the part of the first post terminal **12** located in the housing **11** is accommodated in the third groove **38** of the support **3**, such that the compactness of the structure can be improved, which is conducive to reducing the space occupied by the support **3** in the housing **11** and improving the volumetric energy density of the battery cell **10**.

[0361] In addition, in some embodiments, the guiding part **32** may be configured to participate in defining the third groove **38**, so as to simplify the structure of the support **3** and reduce the design and processing difficulties of the support **3**, which is conducive to increasing the wall thickness of the guiding part **32** and improving the guiding reliability of the guiding part **32**. Furthermore, it is also conducive to improving the stability and reliability of the first post terminal **12**, so as to ensure the reliability and stability of the electrical connection between the first post terminal **12** and the battery cell assembly **2**, and improve the reliability and stability of the charging and discharging operations of the battery cell **10**.

[0362] Referring to FIGS. **33-35** again, optionally, in the embodiments of the present application, the clearance hole **31** includes a first hole segment **311** and a second hole segment **312**. The second hole segment **312** is located on a side of the first hole segment **311** proximal to the active substance-coated part **21**, and the cross-sectional area of the second hole segment **312** gradually increases in a direction away from the first hole segment **311**. When the active substance-coated part **21** includes a current collector **211** and an active substance layer **212** arranged on the current collector **211**, the conductive part **22** includes a tab part **221** electrically connected to the current collector **211**, the tab part **221** includes a plurality of tab plates **2211**, parts of the plurality of tab plates **2211** proximal to the current collector **211** converge to form a first gathering part **2212**, parts of the plurality of tab plates **2211** distal to the current collector **211** converge and are connected to form a second gathering part **2213**, and the first gathering part **2212** connects the second gathering part **2213** and the active substance-coated part **21**, at least a part of the first gathering part **2212** may be accommodated in the second hole segment **312**, and the second gathering part **2213** may be provided in the first hole segment **311** in a penetrating manner.

[0363] In the above technical solution, arranging the clearance hole **31** to include the second hole segment **312** that gradually expands in the direction toward the active substance-coated part **21** facilitates the second hole segment **312** to accommodate more first gathering parts **2212**, so as to improve the compactness of the fit between the support **3** and the battery cell assembly **2**, such that the overall volume of the battery cell **10** is smaller, and the battery **100** can accommodate a greater number of battery cells **10**, thereby improving the volumetric energy density of the battery **100**. In addition, in the above technical solution, the specific explanations of the first gathering part **2212** and the second gathering part **2213** have been provided in the previous embodiments and will not be repeated herein.

[0364] In the embodiments of the present application, the support **3** may be of an integrated structure or a split-type structure. FIG. **36** is a schematic structural diagram of the integrated support **3** according to some embodiments of the present application. Referring to FIG. **36**, when the support **3** is of the integrated structure, the clearance hole **31** is provided in the form of a through hole penetrating through the support **3**. As such, the support **3** of the integrated structure is convenient to process, the support **3** has good reliability, and the support **3** can be conveniently assembled with the housing assembly **1**, thereby improving the assembly efficiency and fitting stability. It can be understood that how to process the support **3** can be specifically selected according to the material of the support **3**. For example, when the support **3** is an insulating plastic member, the support **3** of an integrated structure may be obtained by injection molding.

[0365] FIG. **37** is a schematic structural diagram of the split-type support **3** according to some embodiments of the present application. Referring to FIG. **37**, when the support **3** is of the split-type structure, the support **3** includes a first support **33** and a second support **34** that are separable. Both the first support **33** and the second support **34** are of a long-strip plate structure, and can be detachably connected, for example, in insertion fit or snap fit, to facilitate the assembly. Meanwhile, a half-hole structure is provided on a side of the first support **33** proximal to the second support **34**, and another half-hole structure of a matched shape is correspondingly provided on a side of the second support **34** proximal to the first support **33**. The half-hole structure of the first support **33** and the half-hole structure of the second support **34** together define, in an enclosing manner, the annular clearance hole **31**. That is, the clearance hole **31** is defined between the first support **33** and the second support **34**.

[0366] In the above technical solution, the clearance hole **31** is defined through the cooperation of the first support **33** and the second support **34**. When the support **3** and the battery cell assembly **2** are assembled, it is not necessary to pass the conductive part **22** from one end of the clearance hole **31** to the other end. Instead, the first support **33** and the second support **34** can be assembled at the position of the conductive part **22** to clamp the conductive part **22**, such that the clearance hole **31** surrounds the conductive part **22**, thereby facilitating the assembly of the support **3** and the battery cell assembly **2** and improving the assembly efficiency.

[0367] As an optional solution, when the cross-section of the clearance hole **31** is long strip-shaped, the first support **33** and the second support **34** are separately provided on two sides of the clearance hole **31** in the width direction. For example, if the width direction of the clearance hole **31** is a left-right direction, the first support **33** and the second support **34** are located on the left and right sides of the clearance hole **31**, so as to facilitate the fit of the first support **33**, the second support **34**, and the conductive part **22**.

[0368] FIG. **38** is a partial cross-sectional schematic diagram of the battery cell assembly **2** and the support **3** according to some embodiments of the present application, and FIG. **39** is an exploded view of the structure of the battery cell assembly **2**, the support **3**, and the housing assembly **1** according to some embodiments of the present application. Referring to FIGS. **33**, **38**, and **39**, in the embodiments of the present application, the structure of the support **3** is not limited to this. For example, the edge of the support **3** may also be provided with a housing entry guiding surface **35**. The housing entry guiding surface **35** may be an inclined surface or a curved surface. When the structure is orthographically projected in the axial direction R of the first post terminal **12**, the orthographic projection of the active substance-coated part **21** is completely located within the range of orthographic projection of the support **3**, and the range of orthographic projection of the support **3** is larger than the range of orthographic projection of the active substance-coated part **21**. During assembly, the support **3** and the battery cell assembly **2** may be pre-assembled together, and then the pre-assembled assembly is mounted into the housing **11**. During the mounting, the support **3** is located at the front end of the active substance-coated part **21**, that is, the support **3** enters the housing **11** before the active substance-coated part **21**, such that the difficulty of the support **3** entering the housing **11** can be reduced by utilizing the housing entry guiding surface **35**, and the active substance-coated part **21** can be protected by utilizing the relatively large projection area of the support **3**, thereby reducing the probability of scratching and damaging the active substance-coated part **21** and the housing **11**, and improving the assembly efficiency and the success rate. Moreover, the contact area between the support **3** and the active substance-coated part **21** can be increased, such that the stress concentration problem can be alleviated, and other structural members can be saved.

[0369] Referring to FIG. **38**, in the embodiments of the present application, the battery cell **10** may further include an inner insulating member **4**. The inner insulating member **4** is located in the housing **11** and enclosed outside the active substance-coated part **21**, and the inner insulating member **4** is connected to the support **3**. In the above embodiments, in one aspect, by wrapping the active substance-coated part **21** with the inner insulating member **4**, the insulation reliability between the active substance-coated part **21** and the housing **11** can be improved, the corrosion of the housing **11** caused by the contact between the active substance-coated part **21** and the housing **11** can be reduced or prevented, and the leakage of the electrolyte caused by the corrosion of the housing **11** can be reduced, thereby improving the reliability of the battery cell **10**. In another aspect, connecting the inner insulating member **4** to the support **3** can reduce the difficulty of fixing the inner insulating member **4** and improve the reliability of the inner insulating member **4** wrapped outside the active substance-coated part **21**.

[0370] Referring to FIG. **38**, in the embodiments of the present application, the support **3** includes a body part **36** and an extending part **37**. The body part **36** is located on a side of the active substance-coated part **21** proximal to the first post terminal **12**, and the extending part **37** is connected to the body part **36** and is located in the outer peripheral region of the active substance-coated part **21**. For example, the extending part **37** may be connected to the circumferential edge of the body part **36** to form an annular extending structure connected to the body part **36**, or may extend from a part of the circumference of the body part **36** to form a block structure protruding relative to the body part **36**. In one aspect, a limiting fit with the active substance-coated part **21** can be achieved through the arrangement of the extending part **37**, so as to mitigate the problem of corrosion caused by carbon powder falling off the edge of the active substance-coated part **21** and connecting to the housing **11**. In another aspect, the inner insulating member **4** can also be fixed through the arrangement of the extending part **37**, thereby improving the connection reliability between the inner insulating member **4** and the support **3** and achieving a better insulation effect.

[0371] Illustratively, referring to FIG. **38**, the body part **36** and the extending part **37** may define a positioning groove **39** located on a side of the extending part **37** distal to the active substance-coated part **21**, and an end part of the inner insulating member **4** is embedded in the positioning groove **39** to prevent the inner insulating member **4** from protruding from the edge of the body part **36**. In this way, when entering the housing, the body part **36** can be used to protect the inner insulating member **4**, thereby reducing the probability of scratching and damaging the inner insulating member **4** and the housing **11**.

[0372] Specifically, referring to FIG. **39**, the inner insulating member **4** may be an integrated film, and is provided with main body parts **41** located on both sides of the active substance-coated part **21** in the thickness direction and a connecting part **42** connecting the two main body parts **41**. The connecting part **42** is located on a side of the active substance-coated part **21** distal to the first post terminal **12**, and an edge of the main body part **41** on a side distal to the connecting part **42** extends to the extending part **37** and is connected to the extending part **37**, thereby realizing better insulation performance and facilitating the connection.

[0373] In the embodiments of the present application, the specific method for providing the first post terminal **12** on the housing **11** is not limited, for example, by riveting or by welding, which will be separately described below.

[0374] For example, in some embodiments, FIG. **40** is an exploded view of the structures of a first post terminal **12**, a housing **11**, and sealing gaskets according to some embodiments of the present application; FIG. **41** is an assembly view of the first post terminal **12**, the housing **11**, and the sealing gaskets shown in FIG. **40**; and FIG. **42** is a structural schematic diagram of the first post terminal **12** according to some embodiments of the present application. Referring to FIGS. **40** to **42**, the first post terminal **12** may be an integrated structure and riveted to the housing **11**, so as to improve the assembly efficiency of the first post terminal **12** and reduce the height of the first post terminal **12** protruding from the surface of the housing **11**, which is conducive to improving the energy density and improving the compactness.

[0375] Specifically, referring to FIG. **40** to **42**, before riveting, the first post terminal **12** may include a stopper part **1281** and a penetration part **1282**. During assembly, the stopper part **1281** is stopped inside the housing **11**, the penetration part **1282** is provided in the mounting hole **113** in a penetrating manner, the part of the penetration part **1282** located outside the housing **11** is riveted to form a flange part **1283**, and the flange part **1283** is stopped outside the housing **11**, thereby achieving the mounting of the first post terminal **12**.

[0376] Optionally, referring to FIGS. **40** and **41**, the housing assembly **1** may include several sealing gaskets fitted between the housing **11** and the first post terminal **12**, such as the first sealing gasket **191** and the second sealing gasket **192** shown in FIG. **40**. The sealing gaskets are assembled in place before riveting. After the first post terminal **12** is riveted, the first post terminal **12** presses the sealing gaskets to form a seal, such that the sealing performance of the fit between the first post terminal **12** and the housing **11** can be improved using the sealing gaskets. The number, position and material of the sealing gaskets are not limited. For example, the material may be silicone, plastic, etc., which is not limited here.

[0377] Optionally, referring to FIG. **41**, the length c of the flange part **1283** may be greater than or equal to 1 mm, and the thickness d may be greater than or equal to 2 mm, so as to improve the riveting strength of the first post terminal **12**. When the length c of the flange part **1283** is less than 1 mm and/or the thickness d is less than 2 mm, the reliability of the first post terminal **12** and the housing **11** is reduced under relatively strong vibration.

[0378] For another example, in other embodiments of the present application, FIG. **43** is an assembly diagram of a first post terminal **12**, a housing **11**, and sealing gaskets according to some embodiments of the present application; FIG. **44** is an exploded view of the structure of the first post terminal **12** shown in FIG. **43**. Referring to FIGS. **43** and **44**, the first post terminal **12** may be a split-type structure and connected by welding to be mounted in the housing **11**. For example, the first post terminal **12** includes a first part **1291** and a second part **1292**. At least a part of the first part **1291** is stopped outside the housing **11**, and at least a part of the second part **1292** is stopped inside the housing **11**. At least one of the first part **1291** and the second part **1292** is provided in the mounting hole **113** in a penetrating manner and connected to the other by welding (for example, laser welding), such that the mounting of the first post terminal **12** can be realized.

[0379] In some optional embodiments, FIG. **45** is an orthographic projection view of a battery cell **10** according to some embodiments of the present application; and FIG. **46** is an orthographic projection view of a battery cell **10** according to some embodiments of the present application. Referring to FIGS. **45** and **46**, a plurality of first post terminals **12** are provided and are all located on the surface of the housing **11** on the same side, thereby facilitating mounting and improving assembly efficiency.

[0380] It should be noted that the arrangement manner of the plurality of first post terminals **12** on the surface on the same side is not limited. For example, when the cross-section of the first post terminal **12** is a slender structure, for example, with the cross-sectional length greater than or equal to three times the cross-sectional width, such as an ellipse, a track shape, or a rectangle, the first post terminal can better match a thin and flat housing **11**. For example, the plurality of first post terminals **12** are all provided on a surface of the housing **11** on a side in the height direction (referred to as a first wall surface **110**). The length direction of each of the first post terminals **12** is consistent with the length direction of the first wall surface **110** of the housing **11**, and the plurality of first post terminals **12** are spaced apart in the length direction and/or width direction of the first wall surface **110**.

[0381] For example, in the example shown in FIG. **45**, when the first wall surface **110** is provided with two first post terminals **12**, the two first post terminals **12** are spaced apart in the length direction of the first wall surface **110**. Optionally, referring to FIG. **45**, the part of the first post terminal **12** located outside the housing **11** (referred to as the exterior of the post terminal) is annular. In the length direction of the first wall surface **110**, the inner ring length a1 of the exterior of the post terminal is greater than or equal to ⅓ of the length a0 of the first wall surface **110**, and in the width direction of the first wall surface **110**, the inner ring width b1 of the exterior of the post terminal is greater than or equal to ¾ of the width b0 of the first wall **110**. As such, the first post terminal **12** can provide a larger area for electrical connection to the busbar component, so as to further improve the current passage capacity of the first post terminal **12**. Illustratively, the inner ring length a1 of the exterior of the post terminal is greater than or equal to 50 mm, and the inner ring width b1 of the exterior of the post terminal is greater than or equal to 30 mm.

[0382] In addition, referring to FIG. **45**, when the first wall surface **110** is provided with two first post terminals **12** and the two first post terminals **12** are spaced apart in the length direction of the first wall surface **110**, in some optional embodiments, each first post terminal **12** includes a part located inside the housing **11** (referred to as the interior of the post terminal). In the length direction of the first wall surface **110**, the length of the interior of the post terminal is greater than or equal to ⅓ of the length of the first wall surface **110**, and in the width direction of the first wall surface **110**, the width of the interior of the post terminal is greater than or equal to ¾ of the width of the first wall surface **110**. As such, the first post terminal **12** can provide a larger area for electrical connection to the conductive part **22**, so as to further improve the current passage capacity of the first post terminal **12**. Illustratively, the length of the interior of the post terminal is greater than or equal to 50 mm, and the width of the interior of the post terminal is greater than or equal to 30 mm.

[0383] For another example, in the example shown in FIG. **46**, when the first wall surface **110** is provided with four first post terminals **12**, two of the first post terminals **12** are spaced apart in the width direction of the first wall surface **110** to form a group, and a total of two groups are spaced apart in the length direction of the first wall surface **110**. Optionally, referring to FIG. **46**, a part of the first post terminals **12** located outside the housing **11** (referred to as an exterior of the post terminal) is annular, and in the length direction of the first wall surface **110**, an inner ring length a2 of the exterior of the post terminal is greater than or equal to ⅓ of a length a0 of the first wall surface **110**, and in the width direction of the first wall surface **110**, an inner ring width b2 of the exterior of the post terminal is greater than or equal to ⅕ of a width b0 of the first wall surface **110**. As such, the first post terminal **12** can provide a larger area for electrical connection to the busbar component, so as to further improve the current passage capacity of the first post terminal **12**. Illustratively, the inner ring length a2 of the exterior of the post terminal is greater than or equal to 50 mm, and the inner ring width b2 of the exterior of the post terminal is greater than or equal to 8 mm.

[0384] In addition, referring to FIG. **46**, when the first wall surface **110** is provided with four first post terminals **12** where two first post terminals **12** are spaced apart in the width direction of the first wall surface **110** to form a group, and a total of two groups are spaced apart in the length direction of the first wall surface **110**, in some optional embodiments, each first post terminal **12** includes a part located inside the housing **11** (referred to as the interior of the post terminal). In the length direction of the first wall surface **110**, the length of the interior of the post terminal is greater than or equal to ⅓ of the length of the first wall surface **110**, and in the width direction of the first wall surface **110**, the width of the interior of the post terminal is greater than or equal to ⅕ of the width of the first wall surface **110**. As such, the first post terminal **12** can provide a larger area for electrical connection to the conductive part **22**, so as to further improve the current passage capacity of the first post terminal **12**. Illustratively, the length of the interior of the post terminal is greater than or equal to 50 mm, and the width of the interior of the post terminal is greater than or equal to 8 mm.

[0385] In some embodiments, referring to FIGS. **45** and **46**, a part of the first post terminal **12** is located inside the housing **11**, and a part of the first post terminal **12** is located outside the housing **11**. The orthographic projection area of the part of the first post terminal **12** located outside the housing **11** on the first wall surface **110** is greater than or equal to 5% of the area of the first wall surface **110**. For example, the orthographic projection area of the part of the first post terminal **12** located outside the housing **11** on the first wall surface **110** is greater than or equal to 5%, 6%, 7%, 8%, 9%, or 10% of the area of the first wall surface **110**. As such, the connection area between the first post terminal **12** and the busbar component can be increased, so as to increase the effective current passage area between the first post terminal **12** and the busbar component, which is conducive to improving the charging speed of the battery cell **10**.

[0386] In addition, in some embodiments, FIG. **47** is a cross-sectional view along the line D-D in FIG. **46**. Referring to FIG. **47**, the vertical height t1 of the part of the first post terminal **12** protruding from the outer surface of the first wall surface **110** (referred to as the exterior of the post terminal) from the first wall surface **110** may be less than or equal to 3.2 mm, and the vertical height t2 of the part of the first post terminal **12** protruding from the inner surface of the first wall surface **110** (referred to as the interior of the post terminal) from the first wall surface **110** may be less than or equal to 2 mm, so as to improve the volumetric energy density of the battery cell **10**.

[0387] FIG. **48** is a cross-sectional schematic diagram of the housing assembly **1** according to some embodiments of the present application; FIG. **49** is a cross-sectional schematic diagram of the housing assembly **1** according to some embodiments of the present application; FIG. **50** is an orthographic projection diagram of the battery cell **10** according to some embodiments of the present application; FIG. **51** is a cross-sectional view along the line E-E in FIG. **50**; FIG. **52** is a partial cross-sectional schematic diagram of the battery cell **10** according to some embodiments of the present application. Referring to FIGS. **48-52**, in the embodiments of the present application, the housing **11** specifically includes a housing body **111** and a housing cover **112**. The housing body **111** is of a square annular structure with one or both ends open. When one end of the housing body is open, the number of the housing cover **112** is one, and the housing cover lids an open position. When both ends of the housing body are open, the number of the housing cover **112** is two, and the housing covers lid both open ends of the housing body **111**, respectively.

[0388] Specifically, when the housing **11** includes the housing body **111** and the housing cover **112**, and one end of the housing body **111** is open, the housing body **111** is an integrally formed part, and may specifically be a square structure formed by stretching. In this case, the first post terminal **12** may be provided on at least one of the housing body **111** or the housing cover **112**. Illustratively, referring to FIG. **48**, the first post terminal **12** may be specifically arranged at an end of the housing body **111** distal to the housing cover **112**. When the battery cell **10** is used in a vibration environment, the vibration at the connection between the housing body **111** and the housing cover **112** is relatively small, and the connection position of the housing body **111** and the housing cover **112** is unlikely to crack, which can improve the reliability of the battery cell **10** and help reduce the wall thickness of the housing body **111**, thereby reducing costs and weight, and facilitating miniaturization of the battery cell **10**.

[0389] As an optional solution, referring to FIG. **48** again, when a plurality of first post terminals **12** are provided, all the first post terminals **12** are disposed at one end of the housing body **111** distal to the housing cover **112**. As such, when the battery cell **10** is used in a vibration environment, the vibration at the connection between the housing body **111** and the housing cover **112** is small, and the connection position of the housing body **111** and the housing cover **112** is unlikely to crack, which can improve the reliability of the battery cell **10**. Furthermore, the wall thickness e1 of an end wall of the housing body **111** distal to the housing cover **112** may be reduced to less than or equal to 2 mm, and the wall thickness e2 of the side wall of the housing body **111** connecting the above end wall and the housing cover **112** may be reduced to less than or equal to 0.8 mm, thereby reducing costs and weight, and facilitating miniaturization of the battery cell **10**.

[0390] When the first accommodating groove **12110** corresponding to the mounting hole **113** is formed on the first post terminal **12**, the wall thickness of a part of the first post terminal **12** located on a side of the first accommodating groove **12110** distal to the active substance-coated part **21** is small, such that the welding of the conductive part **22** and the first post terminal **12** can be achieved from the outside of the housing **11**. Referring to FIG. **48**, in the case where the housing **11** includes a housing body **111** and a housing cover **112**, and the housing cover **112** is arranged at an open end of the housing body **111**, even if the first post terminal **12** is arranged at a closed end of the housing body **111**, there is no need to worry about the difficulty in welding the conductive part **22** to the first post terminal **12** from the inside of the housing **11**, because the conductive part **22** and the first post terminal **12** can be welded from the outside of the housing **11**, such that the connection stability and reliability of the housing body **111** and the housing cover **112** can be improved by arranging the first post terminal **12** at the closed end of the housing body **111**.

[0391] When the above second accommodating groove **12120** is formed on the first post terminal **12**, since the welding of the conductive part **22** to the first post terminal **12** can be realized from the outside of the housing **11** through the opening of the second accommodating groove **12120**, referring to FIG. **48**, in the case where the housing **11** includes a housing body **111** and a housing cover **112**, and the housing cover **112** is arranged at an open end of the housing body **111**, even if the first post terminal **12** is arranged at a closed end of the housing body **111**, there is no need to worry about the difficulty in welding the conductive part **22** to the first post terminal **12** from the inside of the housing **11**, because the conductive part **22** and the first post terminal **12** can be welded from the outside of the housing **11**, such that the connection stability and reliability of the housing body **111** and the housing cover **112** can be improved by arranging the first post terminal **12** at the closed end of the housing body **111**.

[0392] Certainly, referring to FIG. **49**, in other embodiments of the present application, all the first post terminals **12** may be arranged on the housing cover **112** as required. As such, the assembly of the first post terminal **12** and the housing cover **112** is facilitated, which is not limited in the embodiments.

[0393] In addition, referring to FIGS. **50** to **52**, in the embodiments of the present application, when a plurality of first post terminals **12** are provided, the plurality of first post terminals **12** may also be separately disposed on two surfaces of the housing **11** on different sides. For example, a plurality of first post terminals **12** are provided and separately provided on surfaces of the housing **11** on two adjacent sides. For another example, a plurality of first post terminals **12** are provided and separately provided on surfaces of the housing **11** on two opposite sides.

[0394] When the first post terminals **12** are separately provided on the surfaces of the housing **11** on two opposite sides, the conductive part **22** can extend out of the active substance-coated part **21** proximal to the first post terminal **12** on each side, and the conductive part **22** is connected to the first post terminal **12** on the adjacent side in a fitting manner, such that the problem that the tab part **221** is pulled by the first post terminal **12** on the same side, resulting in breaking of the connection between the tab part **221** and the active substance-coated part **21**, thereby improving the reliability of the battery cell **10**. It is worth noting that the first post terminals **12** on two sides may be the same or different, and the connection modes between the first post terminals **12** on two sides and the conductive part **22** may be the same or different, which are not limited herein.

[0395] Certainly, in other embodiments of the present application, the first post terminal **12** may also be provided on a surface on a side of the largest area of the housing **11**. For another example, in some optional embodiments of the present application, the first post terminal **12** may be located on the top surface of the housing **11**. For another example, in some optional embodiments of the present application, the first post terminal **12** may be located on the bottom surface of the housing **11**, and the like. When the first post terminal **12** is located on the bottom surface of the housing **11**, the accommodating part **121** may be used to accommodate the electrolyte, thereby increasing the cycle life of the battery cell **10**. In addition, when the first post terminal **12** is located on the bottom surface of the housing **11** and the support **3** is located at the bottom of the active substance-coated part **21**, the contact area between the support **3** and the active substance-coated part **21** can be increased, thereby mitigating the stress concentration problem and saving other supporting structural members.

[0396] FIG. **53** is a schematic diagram of the structure of a housing cover **112** according to some embodiments of the present application. Referring to FIG. **53**, in the embodiments of the present application, the housing **11** is provided with a pressure relief part **16**. The specific structure of the pressure relief part **16** is not limited. For example, the pressure relief part may be an explosion-proof valve or a weak part, which can be used to relieve pressure when the pressure in the battery cell **10** is high, so as to improve the reliability of the battery cell **10**.

[0397] Optionally, the pressure relief part **16** and the first post terminal **12** may be located on a surface of the housing **11** on the same side. As such, processing and assembly can be facilitated. Optionally, the pressure relief part **16** and the first post terminal **12** may also be provided on surfaces on two opposite sides of the housing **11**. As such, the space can be saved, the volume of the first post terminal **12** can be increased, and the adverse effects on the first post terminal **12** when the pressure relief part **16** relieves pressure can be reduced.

[0398] Referring to FIGS. **48** and **53** again, in the embodiments of the present application, the pressure relief part **16** may also be provided on the housing cover **112** as required. Since the housing cover **112** does not need to assume the function of mounting the first post terminal **12**, and the connection position of the housing cover **112** and the housing **11** is less affected by vibration during the charging and discharging processes of the battery cell **10** and is unlikely to crack, the thickness of the housing cover **112** is set to be relatively small, which is more convenient for processing and manufacture of the pressure relief part **16**. Illustratively, the pressure relief part **16** can be formed directly on the housing cover **112** in an integrated engraving manner, so as to fully improve the manufacturability of the battery cell **10**. It is worth noting that in the embodiments, the first post terminal **12** may be provided on the housing body **111** or the housing cover **112**, which is not limited herein. For example, the pressure relief part **16** may be integrally formed with the housing cover **112**, thereby facilitating processing, simplifying assembly, improving production efficiency, and reducing costs.

[0399] Referring to FIGS. **3-5**, **18**, and **23-24** again, the battery cell **10** according to a specific embodiment of the present application is described.

[0400] In the embodiments of the present application, the battery cell **10** is in the shape of a rectangular parallelepiped, and the height direction of the battery cell **10** is the first direction Z, the length direction of the battery cell **10** is the second direction X, and the thickness direction of the battery cell **10** is the third direction Y The battery cell **10** includes a housing **11**. The housing **11** includes a housing body **111** and a housing cover **112**. The housing body **111** is of a square annular structure, one end of the housing body **111** in the first direction Z is open, and the other end in the first direction Z is closed. The housing cover **112** lids an open position of the housing body **111**.

[0401] Meanwhile, the closed end of the housing body **111** in the first direction Z is provided with two post terminals. The two post terminals are spaced apart from each other in the second direction X, and are separately a positive electrode post terminal and a negative electrode post terminal. The two post terminals are both first post terminals **12** provided with an accommodating part **121**, and the accommodating part **121** includes a second accommodating groove **12120**. Specifically, the first post terminal **12** includes a second end wall **12121** and a second side wall **12123**, the second end wall **12121** is located on a side of the second side wall **12123** proximal to the housing cover **112**, and the second end wall **12121** and the second side wall **12123** define, in an enclosing manner, a second accommodating groove **12120**. A surface of the first post terminal **12** on a side distal to the housing cover **112** is a post terminal outer end surface **123**, the opening of the second accommodating groove **12120** is formed on the post terminal outer end surface **123**, a first perforation **12130** is formed on the second end wall **12121**, and the first perforation **12130** is located at a position of the second end wall **12121** proximal to the second side wall **12123**.

[0402] In addition, the battery cell **10** includes a battery cell assembly **2**. The battery cell assembly **2** includes an active substance-coated part **21** and a tab part **221**. The active substance-coated part **21** is accommodated in the housing **11**, and the tab part **221** passes through the first perforation **12130** to extend into the second accommodating groove **12120** and is welded to the second end wall **12121**, such that the active substance-coated part **21** is electrically connected to the first post terminal **12** through the tab part **221**.

[0403] Moreover, a first cover plate **13** is embedded in the groove opening of the second accommodating groove **12120**, such that after the tab part **221** is welded to the second end wall **12121**, the groove opening of the second accommodating groove **12120** is closed by the fit between the first cover plate **13** and the first post terminal **12**. The first cover plate **13** is welded to the first post terminal **12** to form an electrical connection. Thereafter, when the battery cells **10** are electrically connected to each other by using a busbar component, the busbar component may be welded to the first cover plate **13** to achieve an electrical connection to the first post terminal **12**.

[0404] In the above technical solution, in one aspect, by providing the first post terminal **12** with the second accommodating groove **12120**, the weight of the first post terminal **12** can be reduced to a certain extent, so as to improve the gravimetric energy density of the battery cell **10** and the battery **100**. In another aspect, since the opening of the second accommodating groove **12120** is formed on the post terminal outer end surface **123** and the post terminal outer end surface **123** is a surface of the first post terminal **12** on a side distal to the active substance-coated part **21**, the second accommodating groove **12120** can be open in a direction away from the active substance-coated part **21**. In this way, when at least a part of the tab part **221** is accommodated in the second accommodating groove **12120**, the accommodation and arrangement of the tab part **221** can be easily realized through the opening of the second accommodating groove **12120**, and the welding operation between the tab part **221** and the first post terminal **12** can be easily realized through the opening of the second accommodating groove **12120**, thereby reducing the production difficulty of the battery cell **10** and improving the production efficiency of the battery cell **10**.

[0405] Furthermore, since the welding of the tab part **221** to the first post terminal **12** can be realized from the outside of the housing **11**, the first post terminal **12** can be provided on the closed end of the housing body **111**. In this way, when the battery cell **10** is used in a vibration environment, the vibration at the connection between the housing body **111** and the housing cover **112** is relatively small, and the connection position of the housing body **111** and the housing cover **112** is unlikely to crack, which can improve the reliability of the battery cell **10** and help reduce the wall thickness of the housing body **111**, thereby reducing costs and weight, and facilitating miniaturization of the battery cell **10**.

[0406] Meanwhile, since the second accommodating groove **12120** can be in communication with the interior of the housing **11** through the first perforation **12130**, the second accommodating groove **12120** can also serve as a buffering and temporary storage structure for the electrolyte, such that a larger number of electrolyte can be accommodated in the housing **11**. As the electrolyte will be consumed during the charging and discharging processes of the battery cell **10**, a greater amount of electrolyte can prolong the service life of the battery cell **10**. It is also due to the fact that the second accommodating groove **12120** can be in communication with the interior of the housing **11** through the first perforation **12130**, that the second accommodating groove **12120** can also serve as an accommodating and buffering structure for gas generated inside the battery cell assembly **2**, reducing the expansion of the battery cell **10** and improving the reliability and stability of the battery cell **10**.

[0407] According to some embodiments of the present application, the present application further provides a battery **100**, and the battery **100** includes the battery cell **10** according to any one of the above embodiments.

[0408] In the above technical solution, since the battery **100** is provided with the battery cell **10** described above and the first post terminal **12** of the battery cell **10** is provided with the accommodating part **121**, the weight of the first post terminal **12** can be reduced to a certain extent, thereby improving the gravimetric energy density of the battery cell **10** and the battery **100**. In another aspect, by accommodating at least a part of the conductive part **22** in the accommodating part **121** to occupy the space in the first post terminal **12**, it is therefore conducive to improving the volumetric energy density of the battery cell **10**, or the space occupied by the battery cell **10** can be reduced. This allows for the accommodation of a greater number of battery cells in the battery **100** of a given volume, thereby improving the volumetric energy density of the battery **100**. Moreover, the accommodation of at least a part of the conductive part **22** in the accommodating part **121** can reduce the redundancy of the conductive part **22** in the housing **11** to a certain extent, reduce the probability of short circuit between the conductive part **22** and the active substance-coated part **21**, thus further improving the working reliability and stability of the battery cell **10** and the battery **100**. According to some embodiments of the present application, the present application further provides an electric device **1000**. The electric device **1000** includes the battery **100** of the above embodiments, and the battery **100** is configured to provide electric energy for the electric device **1000**.

[0409] In the above technical solution, the electric device **1000** is provided with the battery **100** described above and the energy density of the battery **100** can be improved, which is therefore conducive to prolonging the service life of the electric device **1000**; and the working reliability and stability of the battery **100** can be improved, thereby enhancing the working reliability and stability of the electric device **1000**. It will be appreciated that, when the electric device **1000** is a vehicle, the service life of the battery **100** being prolonged is conducive to improving the vehicle's endurance mileage.

[0410] It should be noted that the embodiments and features of the embodiments in the present application may be combined with each other without conflict.

[0411] The above are only some embodiments of the present application, and are not intended to limit the present application. For those skilled in the art, the present application can be modified and varied. Any modification, equivalent substitution, improvement, and the like made within the spirit and principle of the present application shall all fall within the protection scope of the present application.

What is claimed is:

1. A battery cell, comprising:

a housing assembly, comprising a housing and a first post terminal disposed on the housing, wherein the first post terminal is provided with an accommodating part; and

a battery cell assembly, comprising an active substance-coated part and a conductive part, wherein the active substance-coated part is accommodated in the housing, and the conductive part is configured to electrically connect the active substance-coated part and the first post terminal;

wherein at least a part of the conductive part is accommodated in the accommodating part.

2. The battery cell according to claim **1**, wherein the accommodating part is provided with a first accommodating groove, a surface of the first post terminal on a side facing the active substance-coated part is a post terminal inner end surface, a groove opening of the first accommodating groove is formed on the post terminal inner end surface, and at least a part of the conductive part is accommodated in the first accommodating groove.

3. The battery cell according to claim **2**, wherein:

the housing is provided with a mounting hole, and the first post terminal is mounted in the mounting hole; and

in an axial direction of the first post terminal, a depth of the first accommodating groove is greater than or equal to a minimum distance from the post terminal inner end surface to the mounting hole.

4. The battery cell according to claim **2**, wherein the first post terminal comprises a first end wall and a first side wall, the first end wall is located on a side of the first side wall distal to the active substance-coated part, the first end wall and the first side wall define, in an enclosing manner, the first accommodating groove, and an electrical connection position between the conductive part and the first post terminal is located on the first end wall and/or the first side wall.

5. The battery cell according to claim **4**, wherein the first end wall is provided with a first recess, and at least a part of an electrical connection position of the conductive part and the first end wall is located in the first recess.

6. The battery cell according to claim **2**, wherein the first post terminal is provided with a first groove, a surface of the first post terminal on a side distal to the active substance-coated part is a post terminal outer end surface, and a groove opening of the first groove is formed on the post terminal outer end surface.

7. The battery cell according to claim **6**, wherein the housing assembly further comprises a groove cover, and the groove cover is arranged on the post terminal and closes the opening of the first groove.

8. The battery cell according to claim **2**, wherein the active substance-coated part comprises a current collector and an active substance layer disposed on the current collector, the conductive part comprises a tab part electrically connected to the current collector, the tab part comprises a plurality of tab plates, parts of the plurality of tab plates converge proximal to the current collector converge to form a first gathering part, parts of the plurality of tab plates distal to the current collector converge and are connected to form a second gathering part, the first gathering part connects the second gathering part and the active substance-coated part, and at least a part of the second gathering part is accommodated in the first accommodating groove.

9. The battery cell according to claim **8**, wherein at least a part of the first gathering part is accommodated in the first accommodating groove.

10. The battery cell according to claim **8**, wherein the conductive part further comprises an adapting piece, the adapting piece is connected to the second gathering part, the conductive part is electrically connected to the first post terminal through the adapting piece, and at least a part of the adapting piece is accommodated in the first accommodating groove.

11. The battery cell according to claim **2**, wherein the active substance-coated part comprises a current collector and an active substance layer disposed on the current collector, the conductive part comprises a tab part and an adapting piece, the tab part comprises a plurality of tab plates electrically collected to the current collector, parts of the plurality of tab plates proximal to the current collector converge to form a first gathering part, parts of the plurality of tab plates distal to the current collector converge and are connected to form a second gathering part, the adapting piece is electrically connected to the second gathering part, and at least a part of the adapting piece is accommodated in the first accommodating groove and electrically connected to the first post terminal.

12. The battery cell according to claim **1**, wherein the accommodating part is provided with a second accommodating groove, a surface of the first post terminal on a side distal to the active substance-coated part is a post terminal outer end surface, a groove opening of the second accommodating groove is formed on the post terminal outer end surface, the second accommodating groove is in communication with an interior of the housing through a first perforation, and the conductive part is provided in the first perforation in a penetrating manner and is at least partially accommodated in the second accommodating groove.

13. The battery cell according to claim **12**, wherein an electrical connection position between the conductive part and the first post terminal is located on a wall of the first perforation formed on the first post terminal.

14. The battery cell according to claim **12**, wherein the first post terminal comprises a second end wall and a second side wall, the second end wall is located on a side of the second side wall proximal to the active substance-coated part, the second end wall and the second side wall define, in an enclosing manner, the second accommodating groove, the first perforation is formed on the second end wall, and an electrical connection position between the conductive part and the first post terminal is located on the second end wall and/or the second side wall.

15. The battery cell according to claim **14**, wherein the second end wall is provided with a second recess, and at least a part of an electrical connection position of the conductive part and the second end wall is located in the second recess.

16. The battery cell according to claim **12**, wherein:
the housing is provided with a mounting hole, and the first post terminal is mounted in the mounting hole; and
in an axial direction of the first post terminal, a depth of the second accommodating groove is greater than or equal to a minimum distance from the post terminal outer end surface to the mounting hole.

17. The battery cell according to claim **12**, wherein the housing assembly further comprises a first cover plate, the first cover plate fits the first post terminal and closes the groove opening of the second accommodating groove, and the first cover plate is electrically connected to the first post terminal.

18. The battery cell according to claim **17**, wherein the first cover plate comprises a first conductive member and a second conductive member made of different materials, the first conductive member fits and is electrically connected to the first post terminal, and the second conductive member fits and is electrically connected to the first conductive member.

19. The battery cell according to claim **18**, wherein the first conductive member is provided with a second groove, the second conductive member is embedded in the second groove, and an opening of the second groove is formed on a surface of the first conductive member on a side distal to the second accommodating groove, such that the second conductive member is exposed from the opening of the second groove.

20. A battery, comprising the battery cell according to claim **1**.

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